

Basic Technology

Year 9

Pupils Text Book



Technical Vocational Education and Training
Ministry of Education
26th June, 2018

Preface

This text book has been written for the new Year 9 Integrated Basic Technology syllabus that was trialled in Fiji secondary schools in 2014. It is now the third edition of the Year 9 Basic Technology resource material.

It is designed to introduce students to the fundamental techniques of technical drawing, graphics and design, wood, metal and other common materials and processes with related knowledge on basic hand tools.

It is hoped that for beginners for Basic Technology this text book will be relevant for them and that it provides them the opportunity to pursue further in this field.

**MINISTRY OF EDUCATION,
SUVA.
26th June, 2018.**

Acknowledgements

This textbook for Year 9 Basic Technology has been produced by the Industrial Arts Section of the Technology & Employment Skills Training (TEST) Section of the Ministry of Education.

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Table of Contents



Safety

6

General Workshop Safety
Personal Safety
Hand Tools Safety
Electrical and Machine Safety
First Aid/OHS Regulations

Plane Geometry

12

Basic Technical Drawing Equipment
Preparing A4 Sheet and Paper Layout
Using the Tee & Set Squares
Types of Lines and their Uses
Styles of Lettering and Numbering
Symbols and Conventions
Sketching
Division of Lines, Angles, Arcs and Circles
Triangles
Quadrilaterals
Polygons
Ellipse

Design and Enterprizing

36

Design Cycle
Design Elements
Colors
Principles of Design
Design for Recycling
Green Design
Enterprising Skills

Hand Tools and Materials

57

Classification of Tools
Metals
Timber
Leather
Hardware
Finishes
Sharpening Tools
Abrasives

Geometrical Drawing

91

Orthographic Projection
Isometric Views and Drawings
Oblique Views and Drawings
Perspective Views and Drawings
Solid Geometry
Orthographic Projection of Geometrical Solids
Surface Development of Geometrical Solids

Joints & Processes

116

Wood Joints
Metal Joints

Chapter 1

SAFETY

Outcome

After studying this chapter the students will:

- Identify and follow general safety rules and signs.
- Practice good housekeeping in the workplace.
- Identify and use safe working procedures.
- Recognize hazardous situations in the workshop.
- Develop safe working attitudes and practices.



Introduction

Workplace safety rules are a means of keeping everyone safe and healthy in a protected environment.

Accidents to persons occur when handling and using hand tools, stepping and striking obstructions on the floor and the bench, lifting and moving materials around. These accidents can be avoided if proper precautions are taken to provide safe working conditions and techniques in a workplace.



Before using equipment and machines or attempting practical work in a workshop one must understand basic safety rules. These rules will help keep everyone safe in the workshop.

The following are general rules that apply to all workshop workers, whether attending a practical class or in the workshop for any other purpose.

Personal Safety

It is important to develop good attitude toward safety. This means that there is a need to have a strong feeling towards the importance of safety and willingness to give time and attention to learning the safest way to perform the work. A safe attitude will protect everyone not only in the workshop but also in activities outside school.

1. Wash hands before and after work.
 2. Walk – do not run at the workplace.
 3. Keep the workplace clean and tidy at all times.
 4. Never distract the attention of another co-worker.
 6. Concentrate on what you are doing.
 9. Report all hazards and unsafe conditions and work practices immediately
-
11. Dress properly, wear safety boots and avoid wearing loose fitting garments.
 12. Wear safety glasses or a face shield when doing any work that may endanger your eyes.
 13. Wear ear muffs while using machines with impact noise.
 14. Remove paint and oil from the floor immediately
 15. Two people should carry boards and long pieces of timber.
 16. Lift heavy objects with your arm and leg muscles and ensure to keep the back straight.



Hand Tool Safety

Workplaces contain many types of hand tools. It is essential that the correct method of handling and care of the tools is practiced. Inspect tools for any damage prior to its use and if a tool is defective, remove it from service.

IMPORTANT: Use the right tool to complete a job safely, quickly, and efficiently.

1. Carry tools securely in a tool box to protect yourself as well as the tool.
2. Do not carry too many tools at once.
3. Carry sharp edge or pointed tools close to the body and pointed down to the floor.
4. Never carry sharp or pointed tools such as a screwdriver and chisels in your pocket.
5. Pass a tool to another person by the handle; never toss it to them.
6. Store tools properly. Do not clutter tools all over the workplace (on the bench and projected over the edge of the bench).
7. Properly store tools being used in the bench well and in the tool box after use.
8. Use the right tool for the right job. Do not use your wrench as a hammer. Do not use a screwdriver as a chisel.
9. Always keep your tools in top condition. A dull blade or blunt point can lead to injury.
10. Cut in a direction away from your body, especially when paring with chisels.
11. Do not use broken or damaged tools, dull cutting tools, or screwdrivers with worn tips. Do not use hammers with loose heads or chipped faces and mushroomed or broken handles. Do not use files without handles.
12. Stay alert and make sure there is enough light. Talk to your instructor if you are uncertain how to use a tool.
13. Use a hammer for driving punches, nail, etc., and a mallet for chisels with wooden handles.
14. Start hand saw with a back stroke, guided by raised thumb of left hand.
15. Clean and return tools to proper place after using. Report any breaks and malfunctions to the instructor immediately.



First Aid

Workshops, by nature, contain items that can be hazardous. One of the most important things to plan for in a workshop is to carry out first aid. Encourage, stop, look and think before taking on any hazardous task or activity. The following are a few suggestions on first aid in the workshop:

1. All accidents, injuries, incidents, illnesses and hazards must be reported immediately.
2. All injuries require the completion of an accident report form by your instructor.
3. It is highly recommended that no one works alone in the workshop. You are to be supervised at all times by your instructor.
4. Be prepared for possible accidents. Consider these questions: - where is the nearest telephone, first aid kit and the nearest person who can help?



First Aid Kit

Workplace should have First Aid Kit, which should be visible to all the workers. Contents should be enclosed in the kit and the door of the Kit should be labeled with a red cross. The first aid box should contain the following items.



25 mm roller bandage	25mm adhesive plaster	Antibiotic cream
Triangular bandage	Scissors	Cold pack
Burn lotion	Absorbent cottons	Washing soap
Iodine lotion	Alcohol wipes	Gloves
Antiseptic cream	Safety pins	bandages



MACHINE TOOL SAFETY

NOTE: Exposure to high levels of noise produced by machines can induce deafness which is irreversible – operators, etc, **should wear ear plugs or ear muffs**. Machine tool and portable power tool accidents are generally caused by – (a) Electrical Faults (b) Mechanical Faults (c) Careless or incorrect working Techniques.

A. General Machine Shop Safety

1. Machine tool operators (pupils) should pass a safety test for each machine before being given permission to operate it.
2. Never interfere with (switch on or alter the controls) a machine without permission.
3. ALWAYS obtain PERMISSION to use a machine.
4. Observe correct workshop clothing rules
5. Wear goggles or eye shields when using machines which produce dust, chips or shavings – saws, sanders, grinders and lathes.
6. Observe classroom behaviour safety rules. No running – “no fooling around”.
7. Make sure there is adequate working space to operate the machine safely.
8. Do not talk to or distract the attention of the machine operator.
9. Remove scraps from around the machine and from the floor.
10. Examine material to be worked by the machine for faults which could be dangerous.

B. Electrical Equipment – Safety First

In most workshop the installation of electrical machines is carried out by trained electricians. Power points or outlets, switches, etc. being properly earthed. Where flexible leads or cables are used for machines or appliances take the following safety precautions.

1. Check the plugs frequently for fracture and to see that the leads are securely fixed to the terminals.
2. Don't use cracked or faulty plugs.
3. Don't use damp or faulty leads.
4. Store power tools and leads in a dry place when not in use.
5. Make sure your hands are dry and DON'T stand on a damp floor when plugging in, removing plugs or when operating the switch and the machine.



6. Switch off and disconnect leads after use.
7. Report any damaged leads, plugs or switches.

Occupational Health and Safety

Occupational health and safety applied in your situation is concerned with safe guarding your safety, health and welfare while in the workplace. The goal of all occupational health and safety programs is to foster a safe work environment. It also protects co-workers. The school you attend is required by the OHS Act 1996 and OHS Regulation to provide a workplace that is safe and minimizes risk to you, your fellow workers and teachers. Knowing and understanding the occupational health and safety regulations will help you avoid unnecessary injury, illness and damage to property. Under the OHS legislations the school is to provide the following:



- Safe premises.
- Safe tools, machines and materials.
- Safe systems of work.
- A suitable working environment and facilities

Activity

1. Name some safety device to be worn while sanding your projects.
2. Safety Calendar.
Make a safety calendar for a month with a safety message. It should encourage you to work safely in the workshop.
3. Safety Chart.
Make a safety chart showing important safety rules.
4. Why OHS should be observed at the work place?
5. List 4 essential items that should be kept in First Aid.



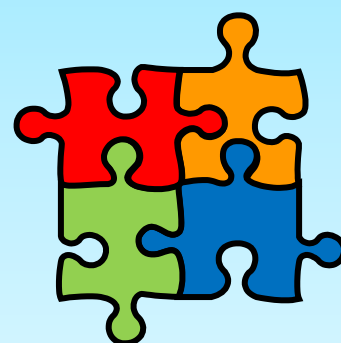
Chapter 2

STRAND: 2 PLANE GEOMETRY

Outcome

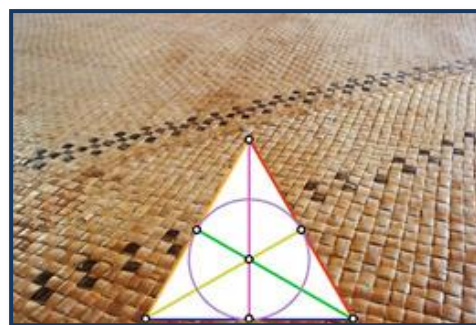
After studying this chapter, students will:

- Acquire knowledge and skills in the correct use of Technical Drawing Instruments through the clear, conventional and visual language of lines, forms of letters and shapes.
- Recognize and select correct technical drawing instruments to solve problems.
- Identify and use symbols and conventions correctly.
- Use correct lettering and numbering styles.
- Identify and develop basic skills in freehand sketching and rendering techniques.
- Identify and develop skills in geometrical construction methods utilized in making various types of lines, circles, arcs, angles, triangles, quadrilaterals, polygons and ellipse.



Introduction

Technical drawing is a means of clearly and concisely communicating all the information necessary to transform an idea or a concept into reality. Therefore technical drawing often contains various forms of lines, letters and figures instead of words. A common drawing can hold many purposes and meanings, while a technical drawing is intended to concisely and clearly communicate all needed specifications to transform an idea into information.



Geometry is all about lines, circles, triangles, quadrilaterals, polygons and ellipse that can be drawn on a flat surface is called a **Plane**. An understanding and mastering the skills in geometrical construction is useful in many careers such as architecture and carpentry.

BASIC TECHNICAL DRAWING EQUIPMENT

DRAWING BOARD

It is a drawing equipment made of wood or plastic on which the drawing paper is placed. Different instruments are used on it for drawing purposes.



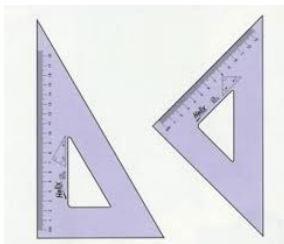
TEE SQUARE

A tee square is a technical drawing instrument used as a guide for drawing horizontal lines on a drawing board and also as a guide for set squares to draw vertical and diagonal lines.



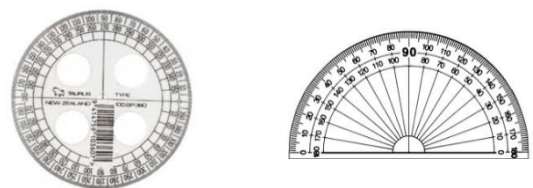
SET SQUARES

Set squares are used in technical drawing for providing straight edges at right angles to a base line. The two common types of set squares are $30^\circ/60^\circ/90^\circ$ and $45^\circ/45^\circ/90^\circ$ angles.



PROTECTOR

A protractor is a circular (360°) or a semicircular (180°) instrument used for measuring angles in degrees ($^\circ$).



DRAWING COMPASS

It is used for drawing arcs, circles and stepping off distances. Both the pencil and the needle of the compass should be sharp for good quality work.



DIVIDERS

They are used for stepping off and transferring distances and measurements accurately.



PENCILS

Good quality pencils are essential for neat and accurate work. Pencils are graded according to their hardness and blackness. 2H and HB.

Clutch pencils with acceptable lead refills can be used as alternatives to standard pencils.

ERASERS

Good quality erasers are essential for keeping the drawing papers neat in case of errors. Soft white erasers are preferred for all erasing works to be done on drawing papers and worksheets.



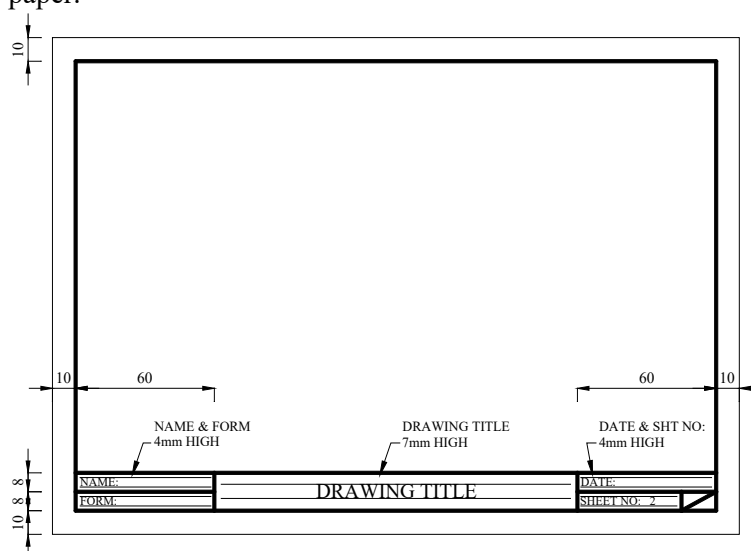
PREPARING AN A4 SHEET AND PAPER LAYOUT

Drawing Paper

Drawing paper is to be used for all the technical drawing exercises. Drawing papers are graded according to its weight, surface finish and size. The most common size of paper used in school is A2, A3, and A4. Folding a sheet of A2 paper will give two sheets of A3 paper and folding an A3 paper will give two sheets of A4 paper.

Paper Layout & Title Block

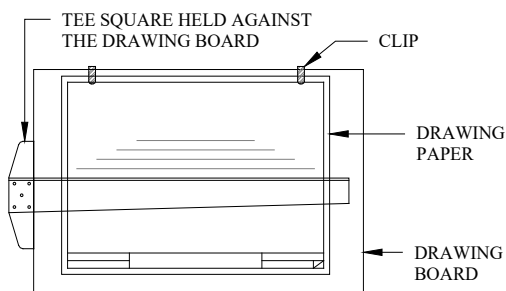
All drawing papers are to be set out as shown below using the given dimensions. The title block is drawn on the bottom of the paper as shown.



USING THE TEE & SET SQUARES

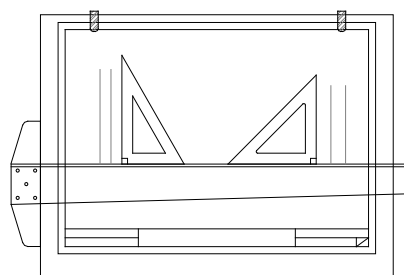
Drawing Horizontal Lines

Hold the T-square firmly against the drawing board and draw lines using the correct pencil for the appropriate line type.



Drawing Vertical Lines

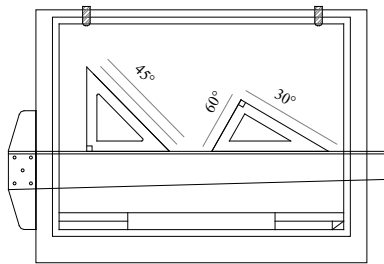
Place any one of the set squares against the T-square as the base line and draw lines using the correct pencil for the appropriate line type.



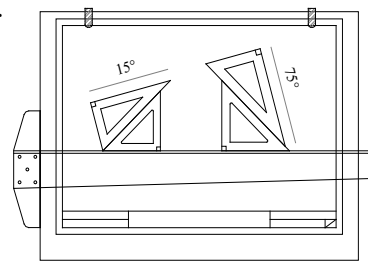
Drawing Lines At 45°, 60° & 30°

Drawing Lines At 15° & 75°

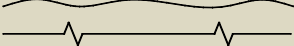
Place the required set square against the T-square as the base line. Draw lines using the correct pencil for the appropriate line type for the required angles.



Organise the two set squares as shown below. Practice other variations with the set squares by placing them against the T-square as the base line.



TYPES OF LINES AND THEIR USES

Line Type	Line	Application
Visible Outline		Thick continuous line used to represent features that can be seen like outlines, border lines, etc.
Construction Line		Very thin continuous line used for construction work in drawings
Dimension Line		Thin continuous line used to represent features like dimension, hatching/section lines and folded lines.
Hidden Line		Thin dashed used to represent features that cannot be seen in the current view.
Centre Line		Thin long and short dashed lines used to show the centre of symmetrical objects.
Cutting Plane Line		Thick long used to show the part which have been cut to reveal internal features through sectioning.
Break Line		Used to show where an object is broken to save space or to reveal internal features.

STYLES OF LETTERING AND NUMBERING

VERTICAL STYLE - UPPER CASE

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z & 1 2 3 4 5 6 7 8 9 0

VERTICAL STYLE - LOWER CASE

a b c d e f g h i j k l m n o p q r s t u v w x y z

SLOPING STYLE - UPPER CASE

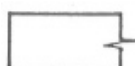
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z & 1 2 3 4 5 6 7 8 9 0

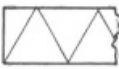
SLOPING STYLE - LOWER CASE

a b c d e f g h i j k l m n o p q r s t u v w x y z

SYMBOLS

A symbol is a simplified image or mark which stands for an object, idea or name to make communication quick and easy. The symbols and conventions used in this book are as follows:



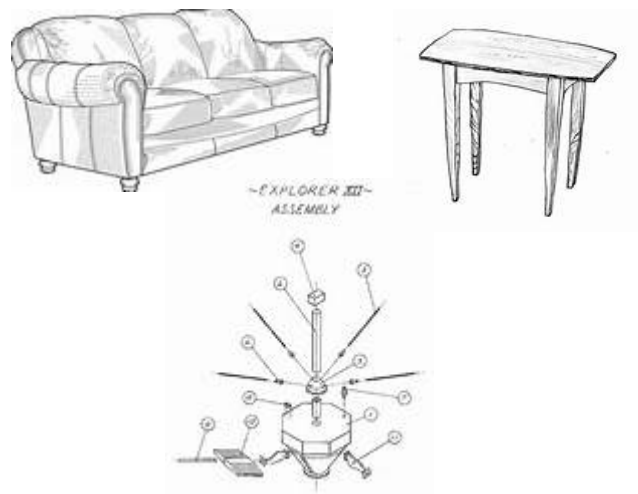
Sawn timber	Timber break	One way switch	Two way switch
			
Dressed timber	Concrete	Illuminating lamp	Socket outlet
			
Hardcore fill	Earth	Socket outlet with switch	Wall mounted lamp
			
Insulation	Distribution board	Fluorescent lamp	Point of entry
			

SKETCHING

A sketch is a quickly executed freehand drawing that is not intended as a finished work. In general, a sketch is a quick way to record an idea for later use.

"A picture is worth a thousand words". The ability to make freehand sketches is very important for a designer. The thought or an idea of a person can be easily seen through sketches produced on the spot quickly, neatly and clearly.

Sketches can be used to convey an idea as steps in preparing a working drawing, as sketches from which an object can be made, to explain or show details of an object, etc.

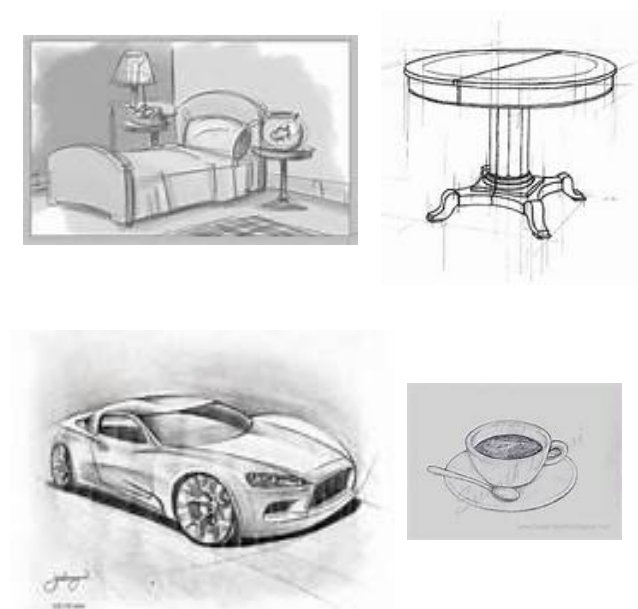


FREEHAND SKETCHING

A freehand sketch is a drawing made without the help of drawing instruments. The ability to produce a quick, clear and on the spot sketch using a pencil and paper is a valuable skill for most people.

Sometimes grids are used as an aid to sketching. The proportions of the sketch are judged by the eye and the line work is done freehand.

Objects can be sketched in any pictorial form depending on the nature of the object and the purpose of the sketch. The two common pictorial forms used in sketching are perspective and isometric.



METHODS OF SKETCHING

VP

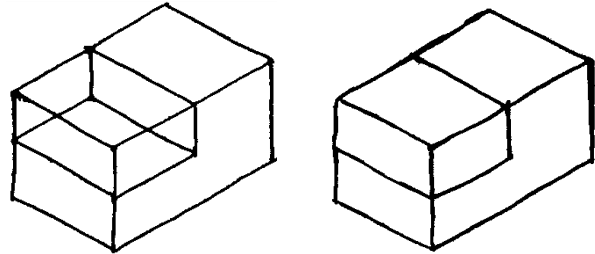
VP

Perspective Sketching

1. Draw a perspective box (to maintain the proportionality of the object) using construction lines.
2. Add details using light lines.
3. Clean up, show outline and shade if required.

Isometric Sketching

1. Draw the isometric crate (to maintain the proportionality of the object) using construction lines.
2. Add details using light lines.
3. Clean up, show outline and shade if required.



SKETCHING HORIZONTAL, VERTICAL AND DIAGONAL LINES

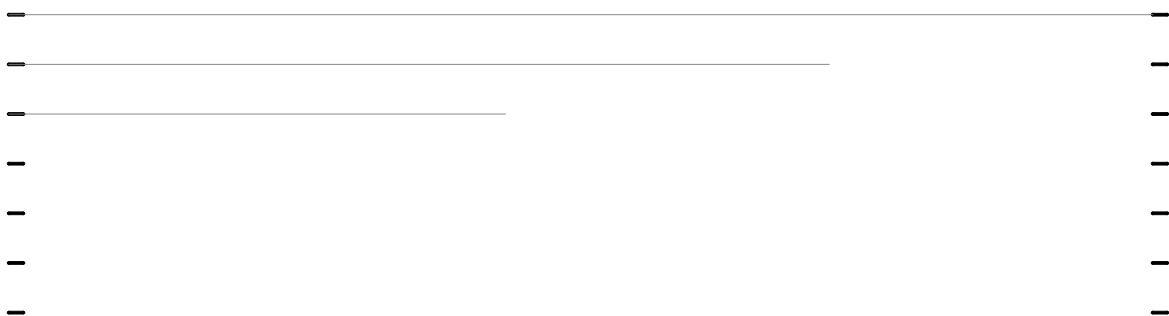
The skills in sketching a smooth, straight and continuous line is very important and needs to be developed at an early stage for producing quality freehand sketches.

The following steps are to be followed when attempting to make a smooth, straight and continuous line:

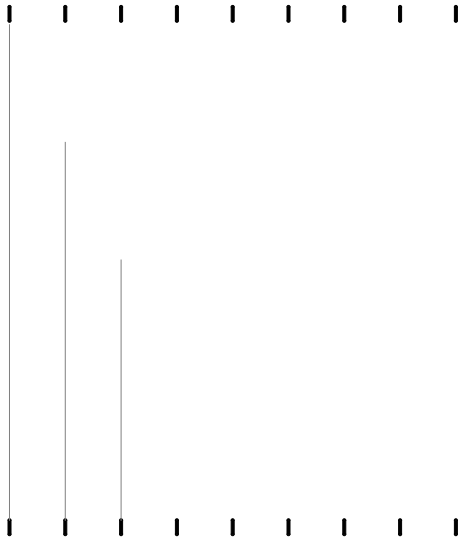
1. Mark the ends of the line with a small dash.
2. Hold pencil gently in normal writing position at right angles to the direction of the line.
3. Draw a couple of imaginary lines to get the feel of the lines.
4. Draw the line starting from the first dash while keeping your eye on the far dash.
5. Move hand and arm together. Keep hands free from the paper.
6. Do not attempt to draw the line in one continuous stroke. Sketch several strokes end to end.
7. Move your arm after each stroke. Strokes should not overlap.
8. For vertical and diagonal lines, turn the paper so that the line looks horizontal.

Given below are examples of producing smooth, straight and continuous horizontal, vertical and diagonal lines.

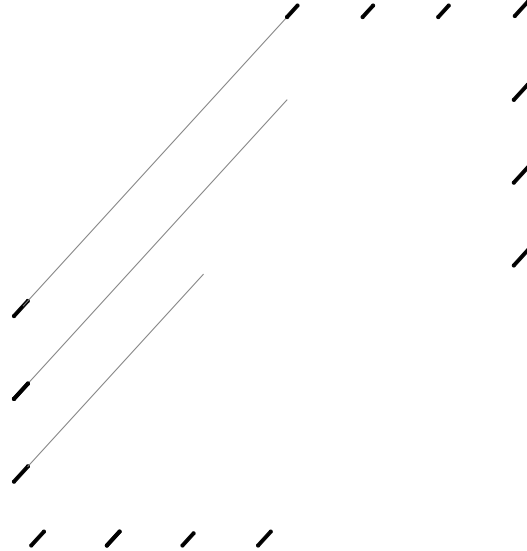
HORIZONTAL LINES



VERTICAL LINES



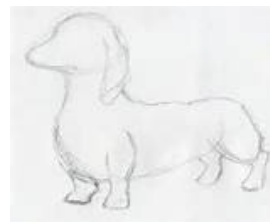
DIAGONAL LINES



RENDERING

Rendering or shading is useful in showing more clearly the shape and texture of objects. An impression of texture can be realized with the use of lines, shades or dots. Quite a number of methods of shading can be done using a pencil. Its use is limited to pictorial sketches. All rendering should be simple and must produce clear and easily read sketches.

The surface that is closest to the light source always has a lighter tone while the surface that is furthest has a darker tone. With some imagination almost any surface texture can be obtained.



Note the difference in the overall look of the sketches from outline sketching to light rendering.

RENDERING TECHNIQUES & TIPS



Note the difference in the overall look of the sketches from outline sketching to dark and sharp rendering showing the image clearly.

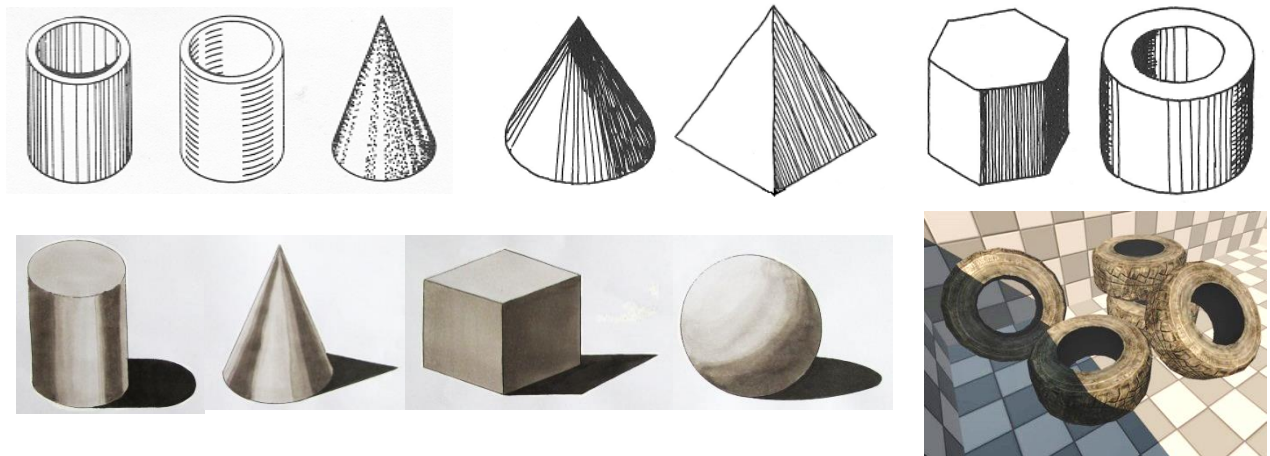
The technique of rendering is useful in showing more clearly the shape and texture of the objects.

A pencil sketch can serve as the basis for a work of art in another medium. It provides a rough idea of what the finished work will look like and need not be detailed.

A number of methods of shading can be carried out using pencil lines. They should be simple and limited to the requirement of producing clear and easily read drawings.

- Always use enough light shading.
- Avoid using heavy shading.
- Work from light to darker surfaces.
- Use the right amount of light, shade and texture.
- Practice as much as possible. "Practice makes perfect".

Shown below are some of the many surface textures that can be used for rendering to show clearly the texture of the objects being sketched.



DIVISION OF LINES, ARCS, ANGLES AND CIRCLES

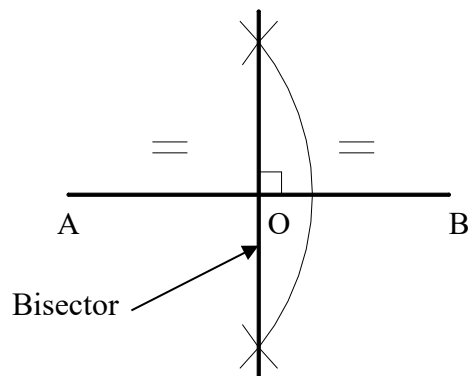
DIVISION OF LINES

1. To bisect a line AB

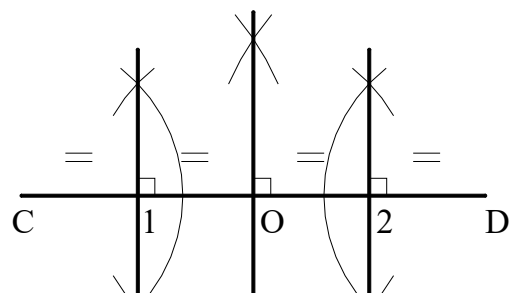
Set compass to more than half the length of AB and strike an arc from point A.

Using the same radius strike an arc from point B intersecting the first arc on the two sides.

Draw the bisector through the intersection of the arcs, label centre as 'O'. $AO = OB$.



2. To quadrisect a line CD



Bisect the line CD and label point O.

Bisect line CO and OD at 1 and 2.

Draw the bisector lines using the intersection of the arcs at 1 & 2.

$$C1 = 1O = O2 = 2D$$

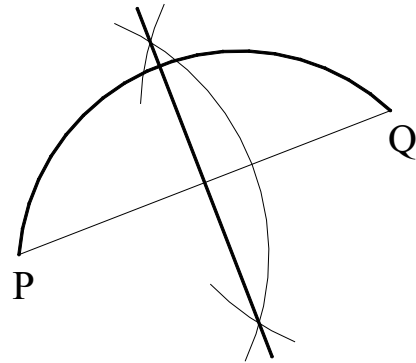
DIVISION OF ARCS

1. To bisect an arc PQ

Draw a straight line between P&Q.

Bisect the line PQ.

Produce the bisector that intersects the arc.



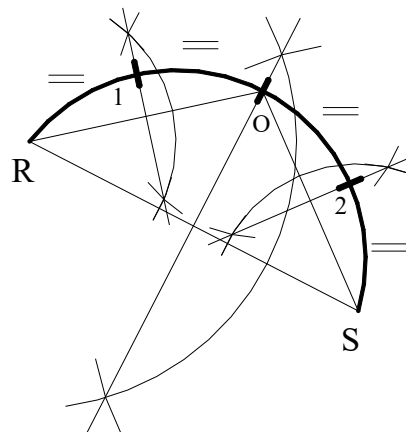
2. To quadrisect an arc RS

Bisect line RS.

Draw a line from R to O and another from O to S.

Bisect lines RO and OS.

Produce bisectors that intersect the arcs at points 1 and 2.



DIVISION OF LINES

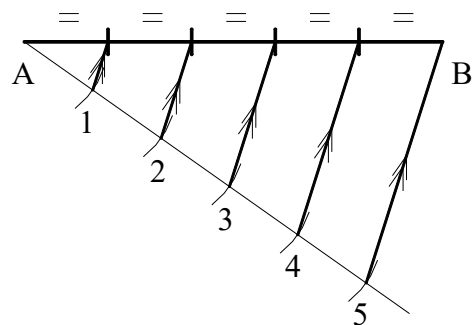
1. To divide a line into a number of equal parts. Example: 5 equal parts.

Draw a line at a convenient angle from point A.

Step off 5 equal divisions on this line with a compass.

Join 5 to B.

Draw lines parallel to B5 from 4, 3, 2 and 1.



2. To divide a line into a ratio.

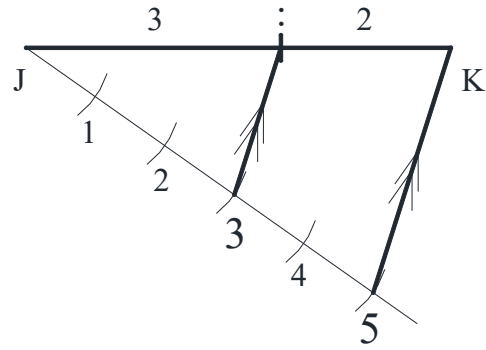
Example:3:2

Draw a line at a convenient angle from point J.

Step off divisions equal to the total of the ratio ($2 + 3 = 5$) on this line with a compass.

Join a line from point 5 to point K.

Draw a line parallel to 5K from point 3.



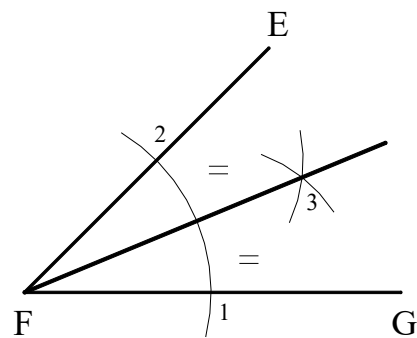
DIVISION OF ANGLES

1. To bisect an angle EFG.

With centre F and any radius draw an arc to intersect arms EF and FG at 1 and 2.

Place compass at 1 and 2 and draw arcs to intersect at 3.

Join F to 3. Line F3 bisects angle EFG.



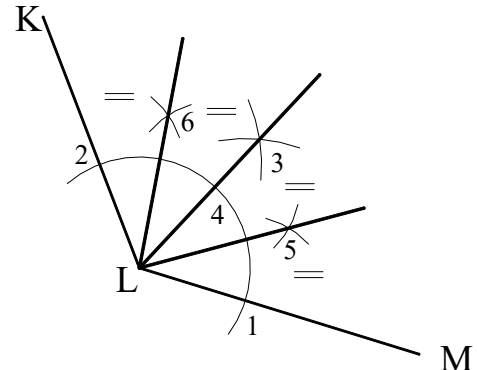
2. To quadrisect the given angle KLM

Bisect angle KLM as in above.

Place compass at 1 and 4 and draw arcs to meet at 5.

Place compass at 2 and 4 and draw arcs to meet at 6.

Join 5 to L and 6 to L.



CIRCLES

Circle is a plane figure bounded by continuously curved line, called circumference, every point which is equidistant from a point within a circle, called the centre.

Parts of a circle

Radius- is a straight line from the centre to the circumference.

Diameter - is the length of a line segment passing through the centre of a circle and touching the circumference. The diameter of a circle is twice its radius.

Chord – is a line segment whose endpoints lie on the circumference of the circle.

An arc - is any connected part of the circles circumference.

Tangent – a straight line that touches the circle at a single point.

Normal - a line drawn perpendicular to the tangent.

Segment – is an area bounded by a chord and an arc lying between the chords end points.

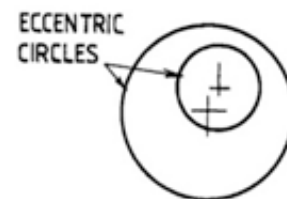
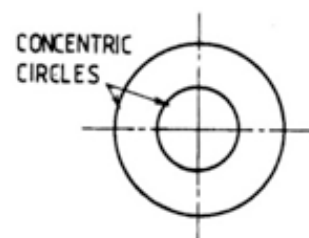
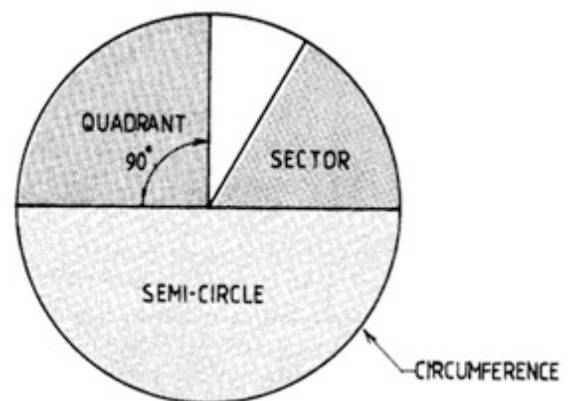
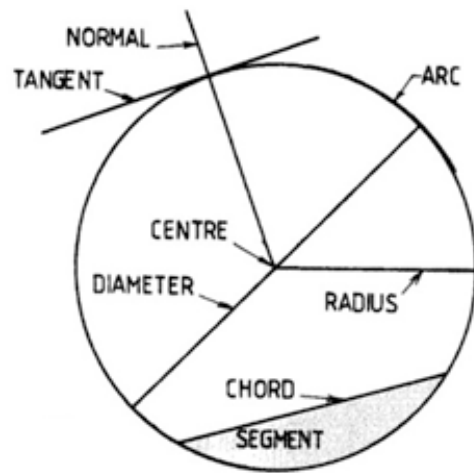
Sector – is a region bounded by two radii and an arc lying between the radii.

Quadrant – is a quarter of a circle.

Semi-circle – is half a circle. The diameter divides a circle into two semi-circles.

Concentric circles – are circles of different radius drawn from the same centre.

Eccentric circles – are circles not having a common centre point, although positioned one within the other.



DIVISION OF CIRCLES

1. To find the centre of a circle and divide it into four equal parts (Quadrants).



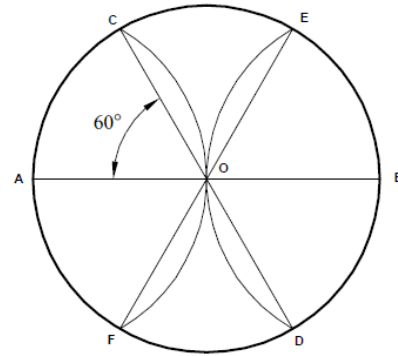
Draw a line to cut the circle at two points 1 and 2.
 Bisect line 1 2 and get points 3 and 4.
 Draw the bisector through 3 and 4 to get points 5 and 6.
 Bisect line 5 6 to get line 7 8.
 Line 7 8 passes through the centre.
 The circle is also divided into four equal parts (quadrants).

2. To divide a circle into six equal parts

Draw line AB, the diameter.

With A and B as the centres and radius AO, draw arcs to cut the circle at C, F, E and D.

The circle is divided into six equal parts (sextants).

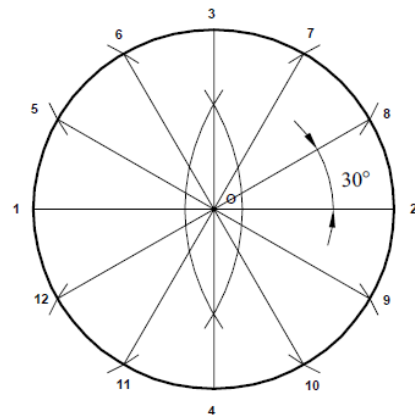


3. Divide a circle into twelve equal parts

Draw line 1 2 (diameter) and bisect at 3 4.

With centres 1, 2 and 3 and the radius of the circle, draw arcs to cut the circle at 5, 6, 7 and 8.

Join these through the centre to obtain the twelve divisions.



4. To divide a circle into five equal parts.

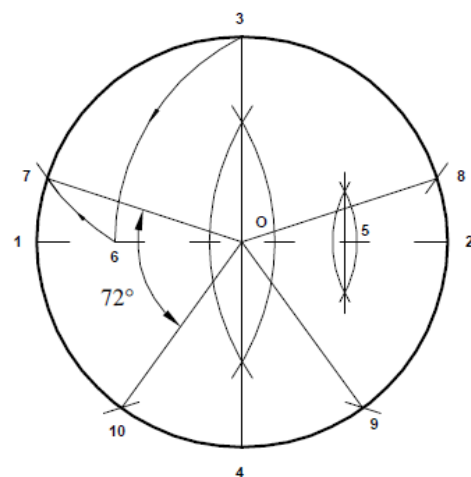
Bisect the diameter 1 2 to obtain line 3 4 which passes through centre O.

Bisect distance O 2 to obtain 5.

With centre 5 and radius 3 5, draw an arc to cut diameter at 6.

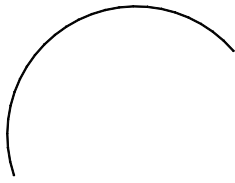
With centre 3 and radius 3 6, draw an arc to cut circle circumference at 7.

With radius 3 7, step off to obtain points 8, 9 and 10.
 The circle is divided into five equal parts.

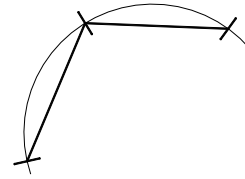


LOCATING CENTRE OF ARCS AND COMPLETING CIRCLES

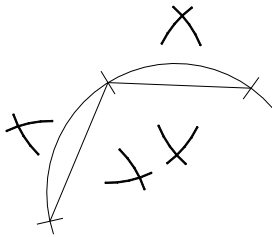
1. To locate the centre of the arc and complete the circle.



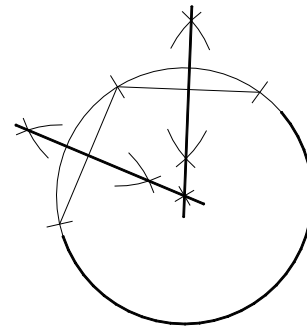
Identify and mark any 3 points. Join lines between the points marked.



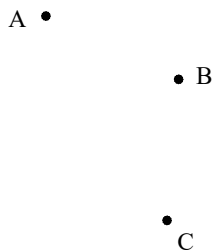
Bisect the two new lines produced.



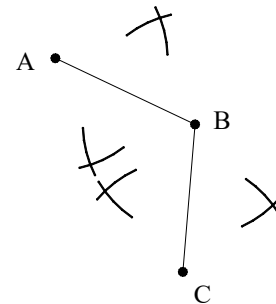
Draw the bisectors. From the intersection of the two bisectors and radius of the arc, construct the circle.



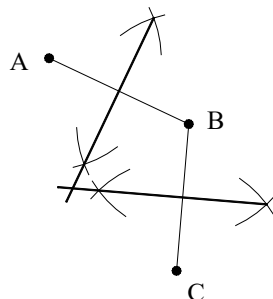
2. To circumscribe a circle through the three given points.



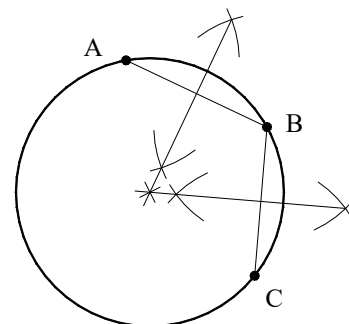
Join the three points. Bisect the two new lines produced.



Draw the bisectors of the two lines to intersect. Identify and mark the centre of the circle.



Using the centre and radius of the arc, construct a circle passing through the three points.



ANGLES

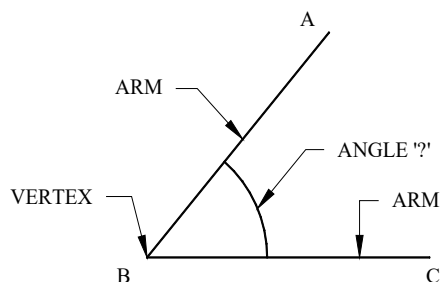
In geometry an **angle** is a figure formed by two rays or arms called the sides of an angle. The two arms meet at a common end point called the vertex of the angle. An angle can also be described as the amount of turn between two straight lines that meet at a point or the vertex.

Parts of an angle:

Arms: sides of an angle (AB & BC).

Angle: turn between any two lines.

Vertex: point where the two arms of an angle meet.

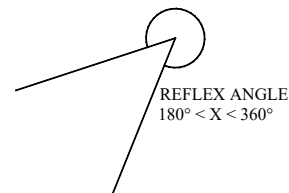
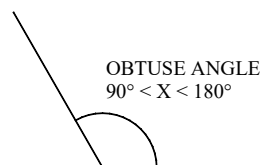
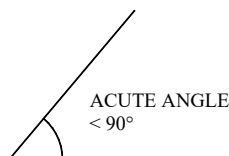
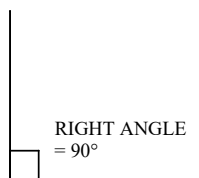


A right angle has a magnitude of 90° .

An acute angle has a magnitude less than 90° .

An obtuse angle has a magnitude greater than 90° but less than 180° .

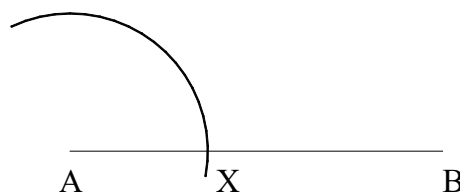
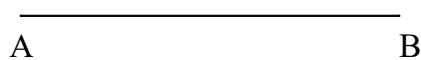
A reflex angle has a magnitude greater than 180° but less than 360° .



CONSTRUCTION OF ANGLES

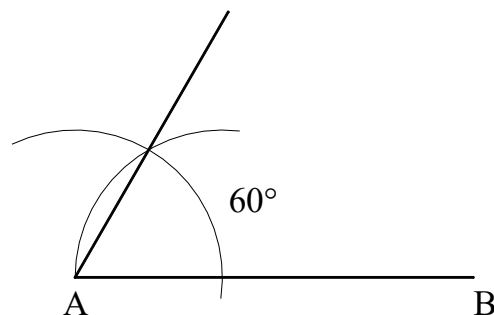
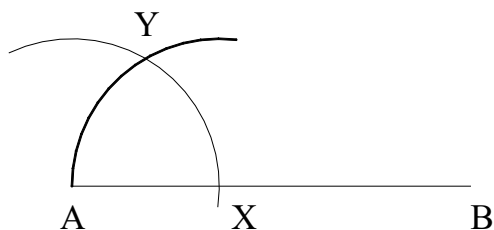
1. To construct an angle of 60° .

Set any radius on the compass and strike an arc from centre A to find point X along line AB.



Using the same radius as above strike another arc from centre X to intersect first arc at Y.

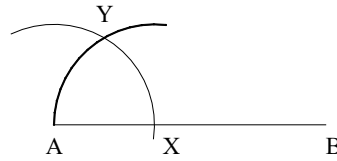
Draw a line through point Y producing an angle of 60° .



2. To construct an angle of 30° .

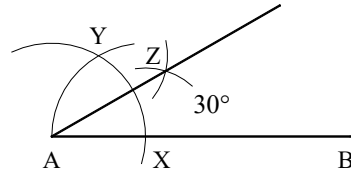
Set any radius on the compass and strike an arc from centre A to find point X along line AB.

Using the same radius as above strike another arc from centre X to intersect first arc at Y.



With centres X and Y draw arcs to intersect each other.

Draw a line from A through the intersecting arcs at Z to produce an angle of 30° .

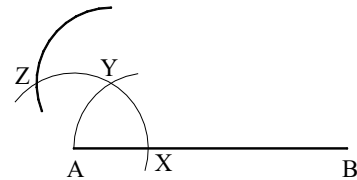


3. To construct an angle of 90° .

Using any radius, strike an arc from center A to find point X and extend the arc longer on the other side.

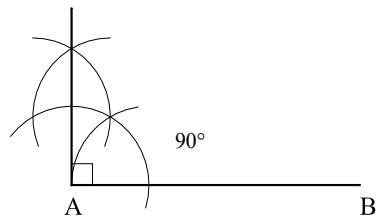
Using the same radius, strike another arc from centre X to intersect first arc at Y.

Use the same radius to strike an arc from centre Y at Z.



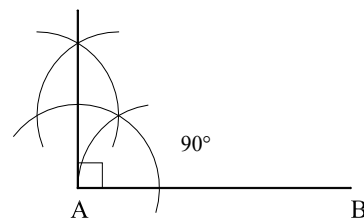
Using the same radius, strike another arc from centre Z to intersect the arc drawn from centre Y.

Draw a line from A through the intersection of the two arcs to produce an angle of 90° .



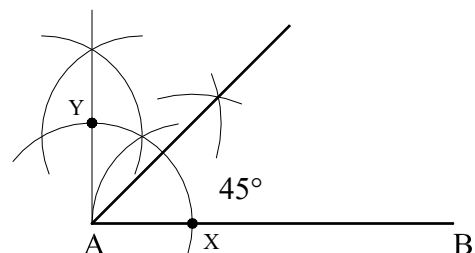
4. To construct an angle of 45° .

Construct an angle of 90° .



Bisect the 90° angle from points X and Y.

Draw a line passing through the intersection of the arcs to produce an angle of 45° .



TRIANGLES

A **triangle** is one of the basic shapes in geometry. It is a 2 dimensional plane figure bounded by three straight lines which meet at the three vertices. The sum of the interior angles of a triangle is equal to 180° .

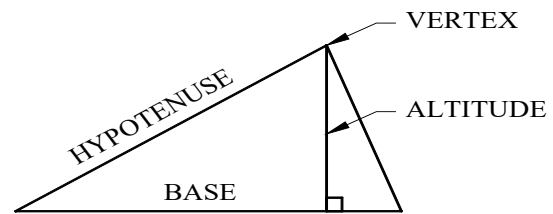
Parts of a triangle

Base: It is the side on which the triangle stands.

Vertex: It is the angle opposite the base.

Altitude: It is a straight line drawn from the vertex perpendicular to the base.

Hypotenuse: It is the longest side of a triangle.

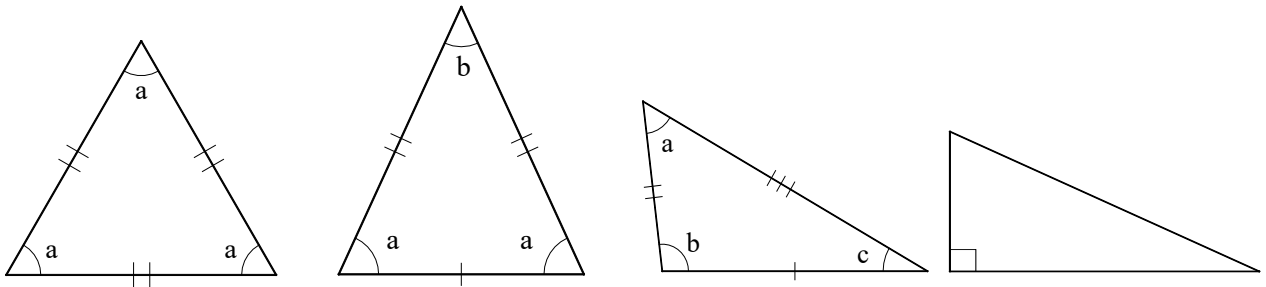


An **equilateral triangle** has three equal sides and three equal angles (60°).

An **isosceles triangle** has two equal sides and two equal angles.

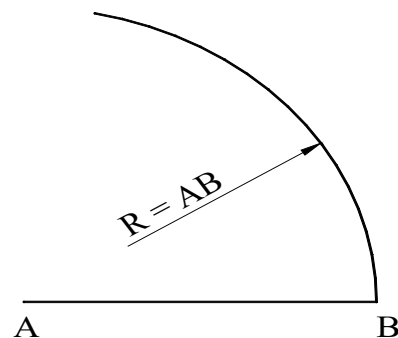
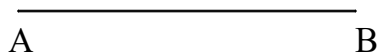
A **scalene triangle** has three unequal sides and angles.

A **right angled triangle** has one of its interior angles as 90° .



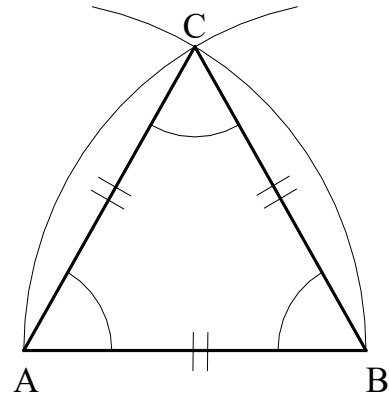
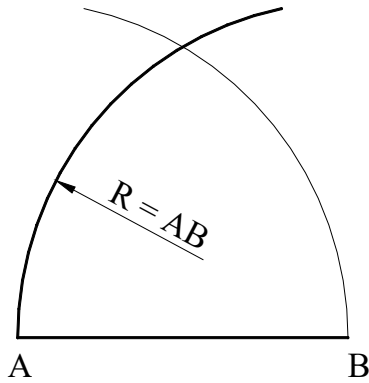
To construct an equilateral triangle ABC given the base AB.

1. With centre A and radius AB, draw an arc as shown.



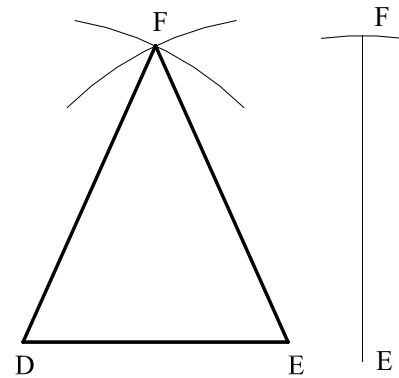
2. With centre B and radius AB, draw an arc to intersect at C.

3. Join the points to form the required triangle.



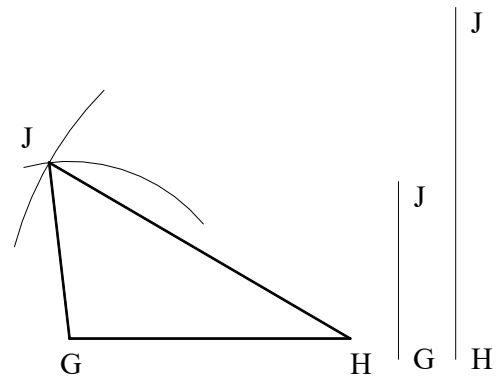
To construct an isosceles triangle DEF, given the base DE and side EF.

1. With centre D and radius EF, draw an arc as shown.
2. With centre E and radius EF, draw an arc to intersect at F.
3. Join the points to form the required isosceles triangle DEF.



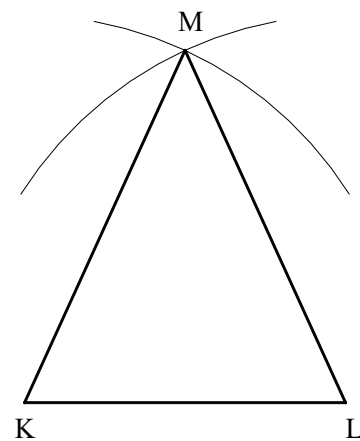
To construct a scalene triangle GHJ, given the base GH and sides GJ & HJ.

1. With centre G and radius GJ, draw an arc as shown.
2. With centre H and radius HJ, draw an arc to intersect at J.
3. Join the points to form the required scalene triangle GHJ.



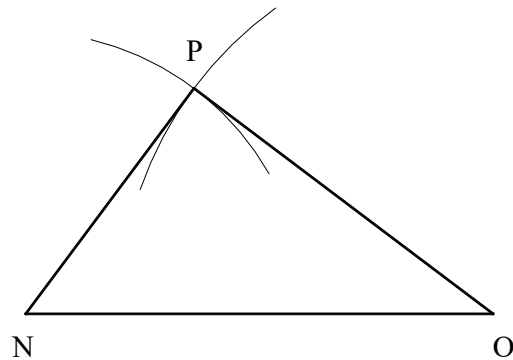
To construct an isosceles triangle KLM, given the base KL and the length of a side LM as 50 mm.

1. With centre K and radius 50 mm, draw an arc as shown.
2. With centre L and radius 50 mm, draw another arc to intersect at M.
3. Join the points to form the required isosceles triangle KLM.



To construct a scalene triangle NOP, given the base NO and the length of side NP = 40 mm and the length of side OP = 55 mm.

1. With centre N and radius 40 mm, draw an arc as shown.
2. With centre O and radius 55 mm, draw another arc to intersect at P.
3. Join the points to form the required scalene triangle NOP.



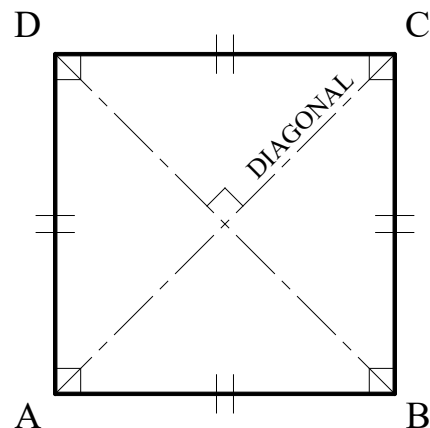
QUADRILATERALS

A **quadrilateral** is a 2-dimensional closed shape with four straight sides. E.g. The shape *ABCD* shown on the right is a quadrilateral.

A line segment drawn from one vertex of a quadrilateral to the opposite vertex is called a **diagonal** of the quadrilateral. *AC* is a diagonal of quadrilateral *ABCD*, as is *BD*.

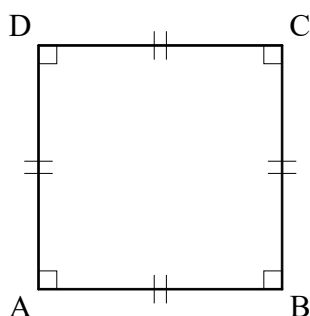
Quadrilateral just means “four sides” (*quad* means four, *lateral* means side).

Any four-sided shape is a Quadrilateral.



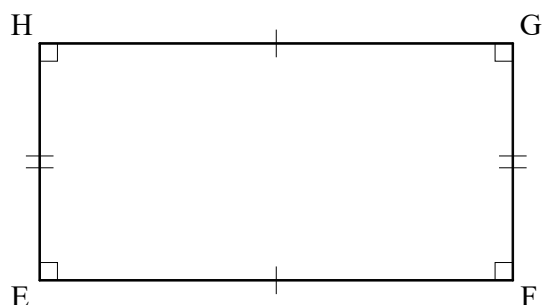
SQUARE

A square has equal sides and every angle is a right angle (90°). Also opposite sides are parallel and of equal length.



RECTANGLE

A rectangle is a four-sided shape where every angle is a right angle (90°). Also **opposite sides** are parallel and of equal length.

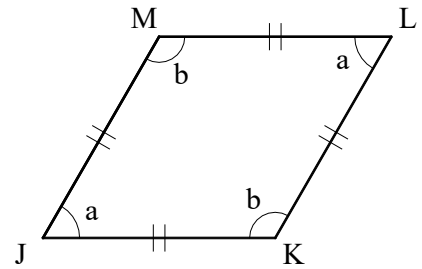
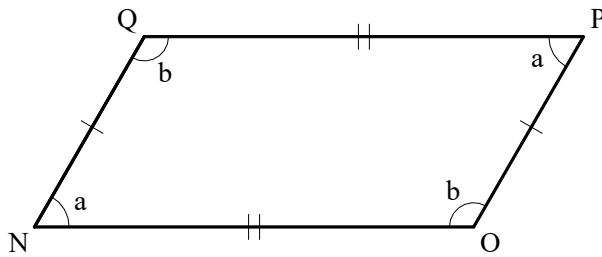


PARALLELOGRAM

A parallelogram has opposite sides parallel and equal in length. Also opposite angles are equal (angles “a” are the same, and angles “b” are the same).

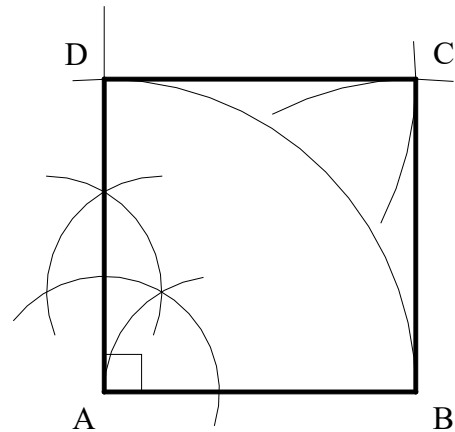
RHOMBUS

A rhombus is a four-sided shape where all sides have equal length and opposite sides are parallel. Also opposite angles are equal (angles “a” are the same, and angles “b” are the same).



To construct a square given side AB.

1. Construct a 90° line at point A on base AB.
2. With centre A and radius AB, draw an arc to intersect the perpendicular at D.
3. With centre's B and D, using radius AB draw arcs to intersect at C.
4. Join the points to form the required square.



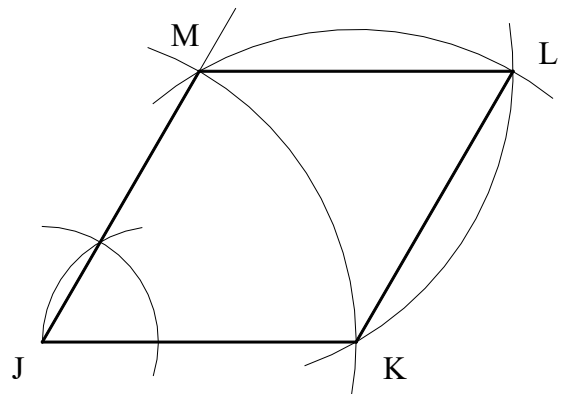
To construct a rectangle given sides EF and EH.

1. Construct a 90° line at point E on line EF.
2. Mark on the 90° line, the length of line EH from point E.
3. Using centre H and radius EF strike an arc.
4. Using centre F and radius EH strike an arc to intersect G.
5. Join the points to form the required rectangle.



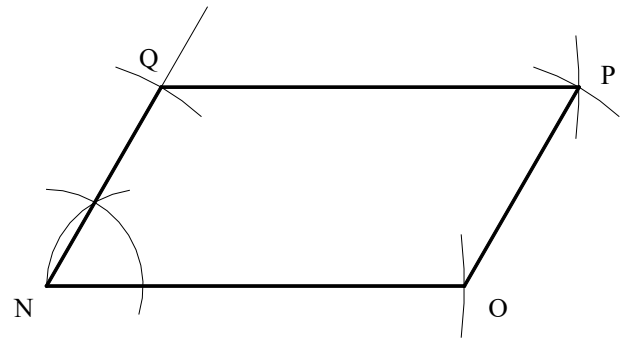
To construct a rhombus given base JK and an internal angle of 60°

1. Construct a 60° line at point J on line JK.
2. Mark on the 60° line, the length of line JM from point J.
3. Using centre M and radius JK, strike an arc.
4. Using centre K and radius JK, strike another arc to intersect at L.
5. Join the points to form the required rhombus.



To construct a parallelogram given base NO and an internal angle of 60°

1. Construct a 60° line at point N on line NO.
2. Mark on the 60° line, the length of line NQ from point N.
3. Using centre Q and radius NO strike an arc.
4. Using centre O and radius NQ strike an arc to intersect P.
5. Join the points to form the required parallelogram.

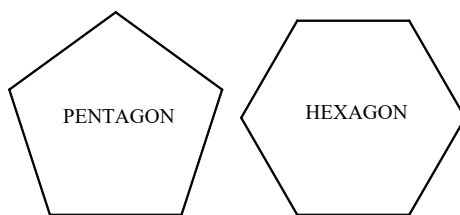


POLYGONS

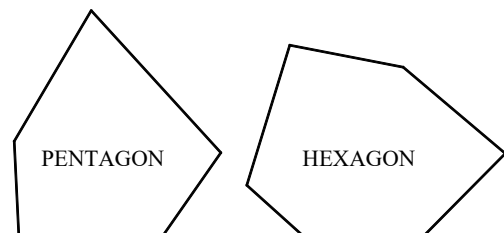
A **polygon** is a 2-dimensional closed plane figure bounded by several lines that are joined together. The sides do not cross each other. Polygons may be regular or irregular. The centre of a polygon is obtained by bisecting any two internal angles.

Regular polygons have all equal angles and all sides are the same length. Regular polygons are both equiangular and equilateral.

Irregular polygons have one or more sides unequal and also one or more angles unequal. Irregular polygons are neither equiangular nor equilateral.



REGULAR POLYGONS



IRREGULAR POLYGONS

Parts of a Polygon

Side - one of the line segments that make up the polygon.

Vertex - point where two sides meet. Two or more of these points are called vertices.

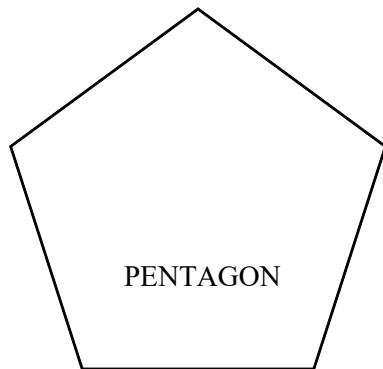
Diagonal - a line connecting two vertices that isn't a side.

Interior Angle - Angle formed by two adjacent sides inside the polygon.

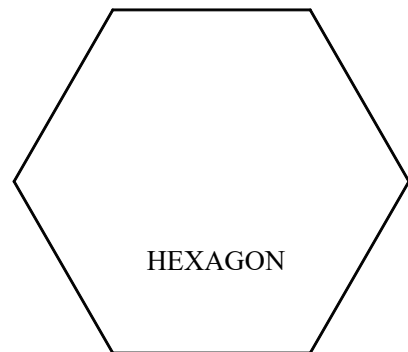
PENTAGON

HEXAGON

A regular pentagon is bounded by five equal sides. All the interior angles add up to 540° . All angles are the same - 108° .

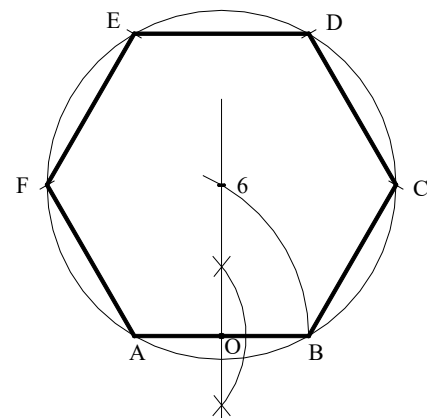


A regular hexagon has six equal sides. All the interior angles add up to 720° . All angles are the same - 120° .



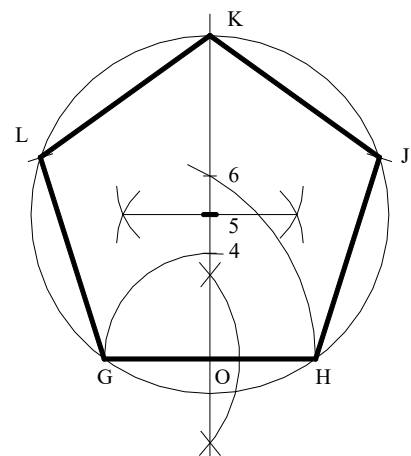
To construct a regular hexagon on the given base.

1. Bisect the base AB to find the bisector and point O.
2. Using radius AB from centre A, draw an arc to find point 6 on the bisector.
3. With centre 6 radius A6, draw an arc to pass through A and B.
4. Take side AB and step off on the circumference of the circle.
5. Join the points to form the regular hexagon.

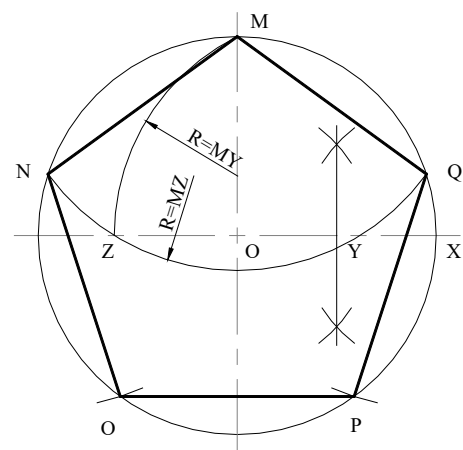


To construct any regular polygon, given the base, using the GENERAL METHOD e.g. Pentagon.

1. Bisect the base GH to find point O.
2. Using radius GH from centre G, draw an arc to find point 6 on the bisector.
3. Using radius OH from centre O, draw an arc to find point 4 on the bisector.
4. Bisect points 4 and 6 and label it as 5.
5. With centre 5 radius G5, draw a circle to pass through G and H.
6. Take side GH and step off on the circumference of the circle.
7. Join the points to form the regular pentagon.



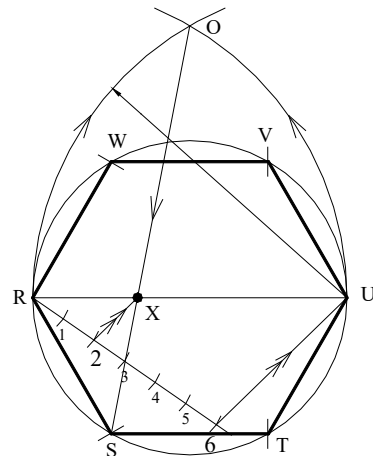
To construct a regular pentagon MNO PQ, within a given circle.



1. Bisect any one radius on the centre lines e.g. OX, to find point Y.
2. With centre Y and radius MY, draw an arc from M to intersect with the centre line on the opposite side at point Z.
3. With centre M and radius MZ draw arcs to either sides N & Q on the circumference of the circle.
4. Using the same radius (MZ) step off on the circumference to find points O & P.
5. Join the points to form the regular pentagon.

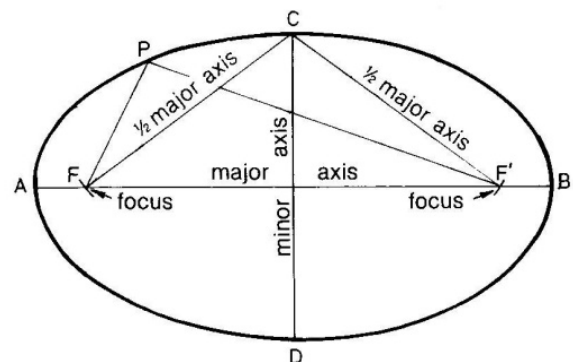
To construct any regular polygon within a given circle e.g. Hexagon.

1. Draw a line at any convenient angle from R and mark six points with equal radius.
2. Join point 6 to point U.
3. Draw a line parallel to line 6U to the diameter at point X..
4. With centre's R & U, draw arcs using radius RU to intersect at point O.
5. Draw a line from O through point X to intersect the circumference of the circle at S.
6. With length RS as first side, step off on the circumference to find points T, V & W.
7. Join the points to form the regular hexagon.



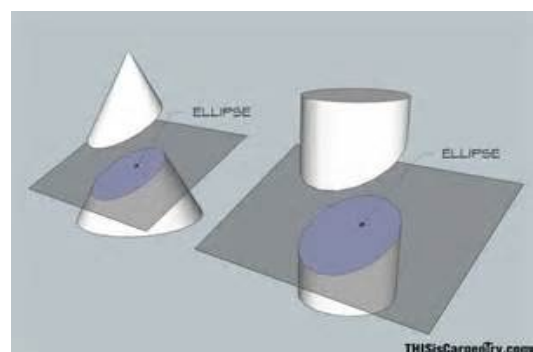
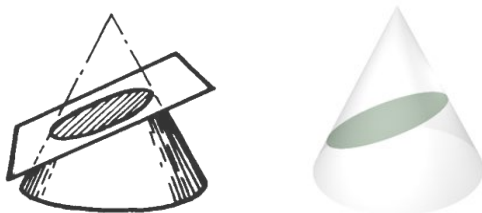
ELLIPSE

An ellipse is a closed symmetrical curve with a changing diameter which varies between the major axis and the minor axis. An ellipse may be defined geometrically as the curve traced out by a point (P) which moves so that the sum of its distances from two fixed points (F and F') is constant and equal to the major axis.



In the diagram shown on the right AB is the major axis; CD is the minor axis and F, F' are the focal points. To find the focal points, take half the major axis either from C or D then strike an arc to cut either side of the major axis.

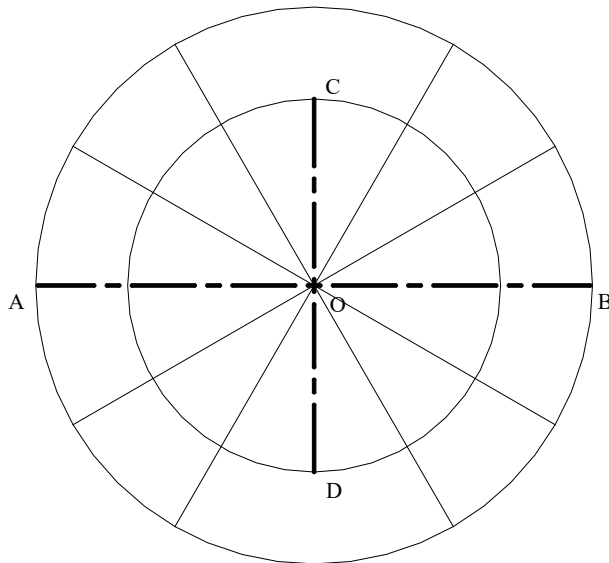
An ellipse is also the true shape formed by an inclined cutting plane passing through both the sides of a cone or a cylinder.



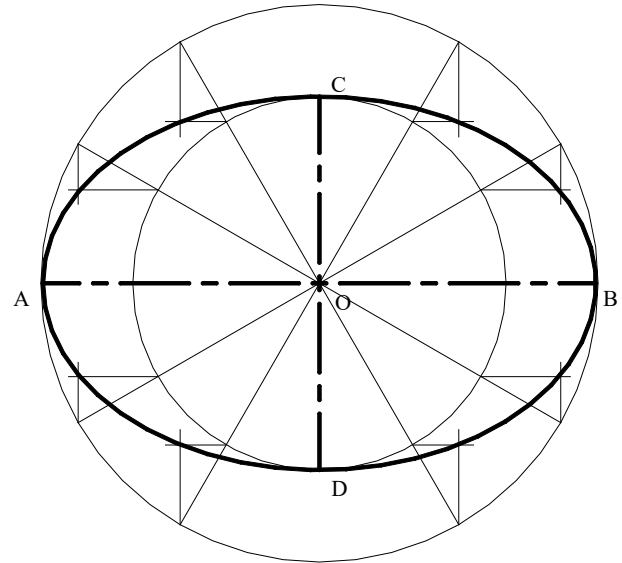
CONSTRUCTION OF ELLIPSE

Method 1: CONCENTRIC CIRCLES METHOD. Given the major axis AB and the minor axis CD.

1. Draw a circle from the centre 'O' using the radius OA of the major axis.
2. Draw a circle from the centre 'O' using the radius OB of the minor axis.
3. Divide the circles into 12 equal parts.

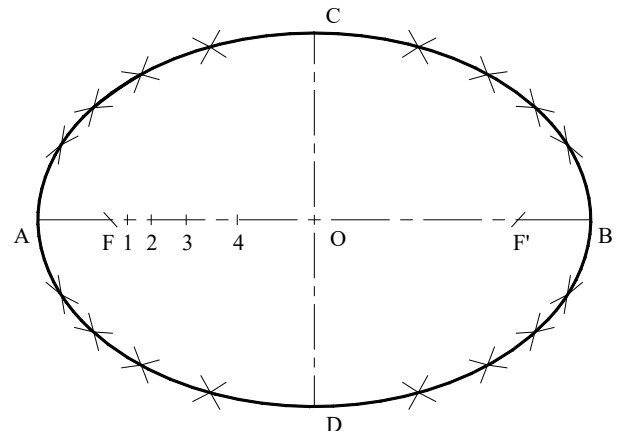
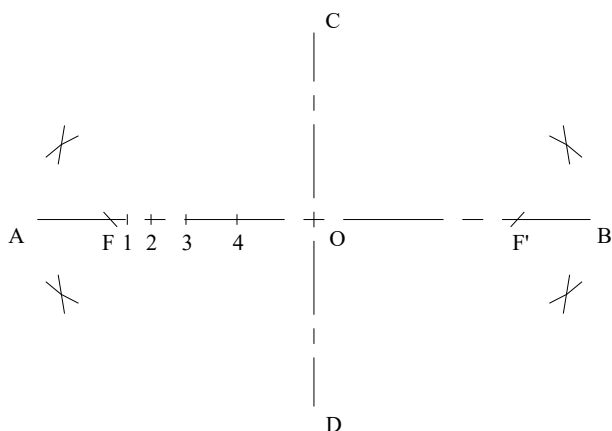


4. From the 8 points of the larger circle, draw lines perpendicular to the major axis AB inwards.
5. From the 8 points of the smaller circle, draw lines perpendicular to the minor axis CD outwards.
6. From the intersection of the two lines of each divisor, draw a smooth curve to get the ellipse.



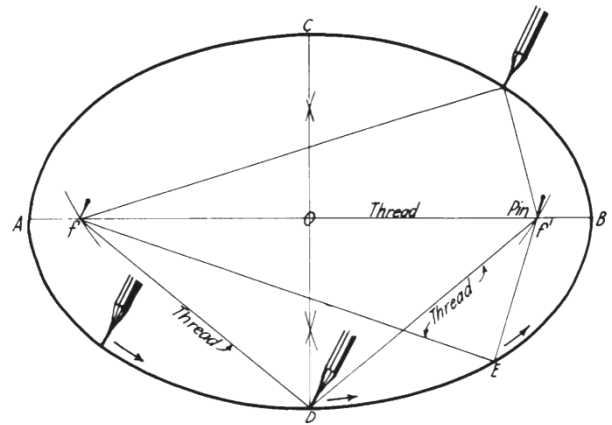
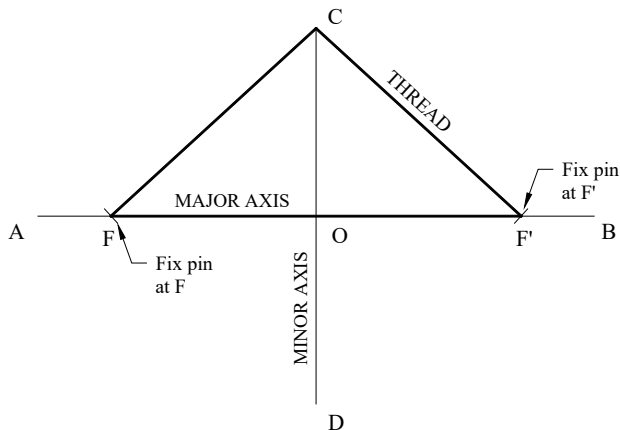
Method 2: INTERSECTING ARCS METHOD. Given the major axis AB and the minor axis CD.

1. Locate the two focal points F & F' by using OA as the radius, Then strike an arc on either side of the major axis from C.
2. Select points 1, 2, 3 & 4 between F & the centre.
3. Using radius A^1 and centre F & F', draw arcs in each of the four quadrants.
4. With radius B^1 and centre F & F', draw arcs to intersect each of the arcs drawn.
5. Using radius A^2 and centre F & F', draw arcs in each of the four quadrants.
6. With radius B^2 and centre F & F', draw arcs to intersect each of the arcs drawn.
7. Repeat the above steps for points 3 & 4 from A & B with centres F & F'.
8. From the intersection of the arcs, draw a smooth curve to get the ellipse.



Method 3: PIN AND THREAD METHOD. Given the major axis AB and the minor axis CD.

1. Locate the two focal points F & F' by using OA as the radius. Strike an arc on either side of the major axis from C.
2. Fix two pins on the two focal points F & F' .
3. Place a piece of thread round the two focal points and tie it to form a triangle on either point C or D.
4. Attach a pencil to the inside of the thread and hold it firmly.
5. Keeping the thread tight, move the pencil all around to trace the ellipse.



Chapter 3

STRAND: 3 DESIGN AND ENTERPRISING

Outcome

After studying this chapter you will:

- Identify, understand and interpret the relevance of the cycle in designing.
- Use basic design cycle and design elements to solve design problems.
- Know the effect of design on environment and how it is related to climate change.



Introduction

What is Design?

Design is about creating something to meet a need. This usually means that designers have to plan what they are going to do, consider the needs of their client and work within the constraints of time, money and safety. They also need to consider the social, cultural, ethical and environment issues that their designs might have.

Sometimes design is the invention of something new but sometimes it involves changing something that already exists to be different or better.



Meeting needs

Designing is about meeting needs. These needs might include:

- Something you need.
- Something your brother or mother needs.
- The needs in the school.
- Needs of pets.
- The needs in the religious worship area.
- Etc.

EXAMPLE OF NEEDS AT HOME



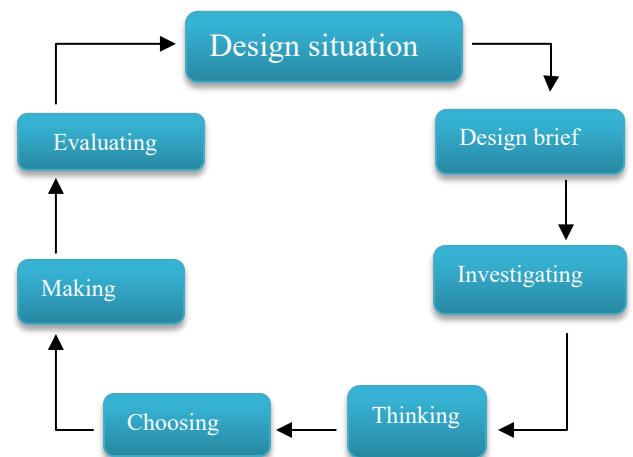
Fish scale remover and dust pan.

The Process of Design

The whole point of learning to 'design' is to learn how to work your way through a project to get the best possible solution. The first idea might not be the best idea.

Throughout the course you will be given projects /tasks and shown how to work through each using the process called **design process**.

Steps in a design process:

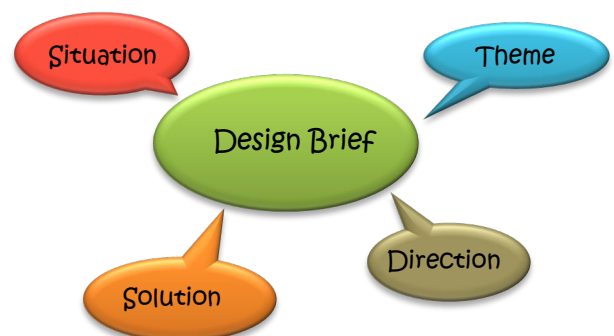


Problem Statement

Problem Statement gives the reason for designing. The problem (situation) will only occur when there is a need or opportunity to design.

Design Brief

A brief statement of what to design.



Investigating

The investigating stage is to find out what has been done by others on the project that you are designing.

Each design project will have activities. The students will need to:

- Collect information
- Look at similar design
- Consider issues and values
- Use appropriate design concepts.

Do not just copy and paste information in your report from someone else's work. Rewrite it in your own words to make it your own.

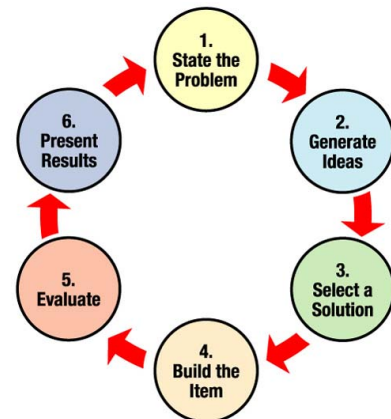
Collecting Information

You need to collect information on design materials related to your design project.



Concept Mapping

Results need to be presented in an interesting way and can be presented in written, oral, graphic, display and multimedia.



Thinking

Brainstorm on your own and then with others to think of as many ways as possible to meet the design brief.



Sketch at least five different ideas for your design. Label each one with your evaluation of its good, bad and interesting features. Use freehand sketches with labels and notes around them.

Choosing

Which of your ideas is your favorite? And more importantly which of your ideas best fits the design brief and your criteria for success? To answer the question, you need to evaluate your design ideas and list the strength and weaknesses.



Reason for Choice

- It looks good.
- It should be simple to construct.
- It will not cause any damage.
- I like it best.



Material and Resources

Make a list of materials you need, where you will get them and what they will cost.



Safety checklist

Before constructing the project, a checklist of the safety rules should be made

	Rule	What might happen	First aid in case of accident
Environment			
Attire			
Behavior			
Equipment			

Making

Before starting a design take some time and think whether you making a model or the real project?

If you are making a model, make sure that you can hold, mark out, measure, cut to shape, join and finish the materials you have chosen.

Technique Table

Prepare a technique table as shown below:

Process	Tools required	Skills to be learned	Skills learned

Evaluating

This is making decisions on which ideas you like best and things that you do not like. While working through your design project, a lot of decisions will come up, take some time to think about the decisions. Evaluation happens over and over again in order to keep checking on what you are doing. Ask questions like is it good as it could be? Or is it good as it needs to be?

Remember to always ask questions like: Can I do better? Can I do it another way? Why do other designers or manufacturers do it differently?

You can evaluate by using evaluating box as shown below:

How well did I:	Well	Not Well
- Use my time		
- Plan the project		
- Choose resource		
- Fulfill design brief		



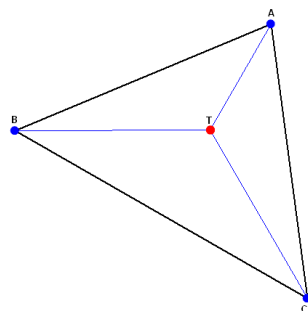
The Elements of Design are the language of the visual arts.

This introduction focuses on the elements that are most relevant to two-dimensional (flat) art works. Other elements include point, motion and elements related to three-dimensional art such as mass and volume.

Line	path of a point
Shape	perceivable area
Value	relative light and darkness
Color	color theory basics
Space	(2D) height, width and the illusion of depth
Texture	actual or simulated tactile quality

Point

A point is an element that has position, but no direction. It is a simple mark in space with a precise but limited location.



Line

Line - the path of a point.



Although the subject matter is the same in all three works, the differences in line quality have created works with very different impact. How you use line is one of the most important decisions to be made in creating a work of art

Shape



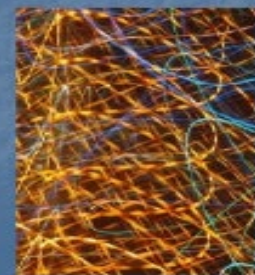
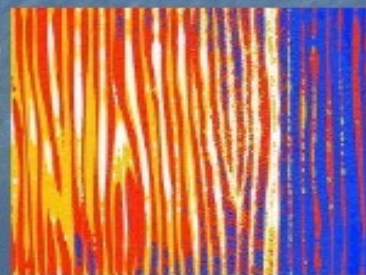
Shape - perceivable area. Shapes can be created by line, or by color and value changes which define their edges. As with line, the decisions you make concerning shape are important.

The shapes of the objects that you create or place in your images are positive shapes. The spaces around these shapes are the negative spaces. It is just as important to be attentive to the negative space as the positive shapes.

Texture



Texture - surface quality. We experience actual texture when we touch objects and feel their roughness, smoothness or patterns, which we can simulate or imply in digital imagery.



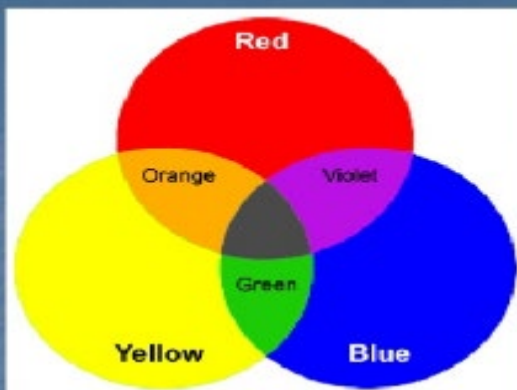
Color



Color - basic color theory. We respond to color on many levels. Color can be used simply to describe an object. It can also be used emotional (blue for sadness or spiritually, red for angry), symbolically (associated with a flag's color, corporation logo or sports team) and psychologically



Color harmony



Pigment Color (paint)

reflected light

Pigment color is created when a pigment absorb certain light wavelengths and reflects others. For example, a blue shirt absorbs all wavelengths except blue, which is reflected. The color wheel based on the three primary colors: red, yellow and blue, was developed in 1666 by Sir Isaac Newton.

Primary pigment colors - red, yellow and blue are the primary colors. All other colors are derived from these three hues.

Secondary pigment colors - green, orange and purple are created by mixing the primary colors.

Tertiary colors - yellow-orange, red-orange, red-purple, blue-purple, blue-green and yellow-green are the colors created by mixing the secondary colors.

Tone

Tone is light and dark. Light shows the world to us and shadow gives meaning to the things we see. Tone can give solidity, volume and weight to an image. It gives the impression of distance. Darker tones come forward and lighter tones go back to the image. Tone can give emotion to an image. Highly contrasting tones give life and energy; softer tones give gentle mood. Tone can create rhythm; with the eyes jumping from one dark tone to another. Tone is the property of color.

Direction

Direction is about how our eyes move around the art work. It can be horizontal, vertical, curved, sloped or straight. It can suggest movement by the speed at which it is changed. Direction can be balanced to give stability or imbalanced to give tension. It can have an emotional impact.



Size

Size is about the biggest or smallness of an area. It can give space, it can make closer objects appear larger and make distant object appear smaller.

Mass

Mass is the amount of material in any sculptural work. It can be suggested in a painting or drawing. Mass can be heavy or light in effects. Space or void refers to the lack of mass.



Traditional Designs

Indigenous people possess important traditional knowledge that has allowed them to sustainably live and make use of biological and genetic diversity within their natural environment for generations. Traditional Knowledge naturally includes a deep understanding of natural processes and the ability to sustainably extract useful products from the local habitat.



The traditional knowledge of indigenous people offers a set of proven practices and solutions for today's global dilemmas.

Most traditional knowledge is handed down through generations. These methods endure the test of time as the indigenous community works within the flow of nature to ensure a continued supply of the resources upon which their way of life depends. Industrial societies groping with health issues, species extinction, and habitat loss can learn much from indigenous people.



Design for Recycling

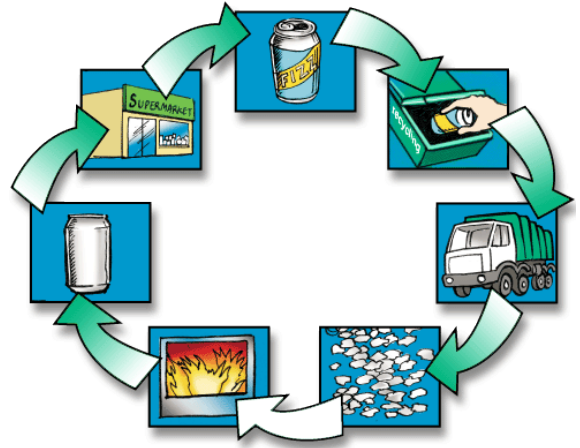
This is a process where pre-production planning for safe and efficient recycling by elimination as much as possible, of hazardous and non-recyclable materials from the production process.



Why Recycle?

Recycling is good for the environment

It takes less energy to create new items from recycled materials than it does to create new products from raw materials. Mining minerals and milling trees into lumber requires vast amounts of energy. Recycling allows us to reuse materials many times to conserve natural resources while creating the products we use in our everyday lives.



Recycling is good for Communities

Extracting materials from mines or forests is done far from the place where goods are consumed; however recycling starts in your own home. Gathering recyclables and reprocessing them into feedstock for future products is done locally by people who live, work, and spend money in their own communities.



Recycling Reduces Pollution

Burning garbage or throwing waste into landfills produces by-products that pollute the environment. Run-off from landfills and metals like mercury find their way into streams, rivers, oceans, fish, etc., and eventually into human beings, harming our health.



Waste water recycling reduces pollution

Green Design

What is green design?

Sustainable design is the philosophy of designing physical objects that will benefit many future generations without harming them, the environment and the natural resources in any way.

Boat made from waste plastic bottles.



Why Green Design is Important?

Site Evaluation

We study each site to understand the potential that site has to offer. Site features such as topography, predominant wind, solar exposure, views, watershed, and existing vegetation are all factored into the final design solution.

Energy Efficiency and Renewable Energy - Each project should be designed to meet specific performance expectations while minimizing the use of energy, saving money, and reducing greenhouse gas emissions. By specifying Energy Star appliances, solar hot water heaters, photo-voltaic systems, wind turbines, and geothermal heating systems, we are reducing our demand on traditional power and gas sources and reducing greenhouse gas emissions.

Waste Management

During construction, the contractor should minimize construction waste while maximizing the use of renewable building resources and material recycling. We encourage each builder to have a waste management plan and a goal of reducing waste. The building or space should also be designed in such a manner to encourage recycling and waste reduction.

Education

Upon completion, occupants are educated about the sustainable design, systems, required maintenance, and use of environmentally friendly materials. It is also important to educate the public about the importance of sustainable design and how every individual plays an important role in the future health of our environment.

How to Design with Sustainability in Mind

The key issues addressed by sustainability in product design are environmental issues, which include: carbon footprint, which refers to the production of greenhouse gases, the total energy consumed, and air and water pollution. Manufacturers trying to create sustainable products must first determine what their products are made out of and whether the materials have been assessed for their impact on human and environmental health.

Water Efficiency

By specifying fixtures and appliances that are low flow and rated Water Sense (an EPA program similar to Energy Star), we can reduce the amount of water consumption saving water and money. Native planting design and efficient irrigation systems are designed to minimize water used in the landscape. Implementing rainwater collection and storage as well as the use of grey water systems are other alternatives to consider.

Material Selection

Whenever possible, we select materials and products that are recyclable, renewable, non-toxic, and locally produced reducing CO2 emissions and promoting the local economy.



Effect of climate change in Pacific Island.

Assessing the Big Picture: Lifecycle Analysis

If products are to be designed effectively with little or no negative environmental impact, then the designer must consider the impact of the product throughout its entire lifecycle. What this means is that the designers and engineers need to look at everything that happens in the production, transportation, use and final disposal of the product.

Designing – You Try!

Let us start designing with a quick design project

- Design and make a set of bookmarks to use with this book. They should have your name, form and school on them.
- You will be able to use book mark to mark different section in this book. You will need to find pages quickly as your design.
- You will use hand design.



How would a Designer do it?

- Step 1: A designer finds out exactly what is needed by asking questions like: Why will I need bookmarks?
- Step 2: A designer makes sure that their idea of what is needed is the same as the idea of the person who is asking them to do it. Can it be one bookmark with four trails?
- Step 3: A designer looks for ideas and inspiration from already available and from lots of other places.
- Step 4: A designer thinks as many ideas as they can.
- Step 5: A designer chooses the idea their client like the best.
- Step 6: A designer sketches and then makes the chosen design for presentation.
- Step 7: A designer examines their work and determines its good and bad points. Then they think about how they might change it if they were to do it another time. These steps are part of the “**design process**”

Activity

1. What is the purpose of the design cycle in design activities
2. Write a design brief to a problem given by your teacher.
3. What is the purpose of investigation and research in the design process?
4. List and briefly explain the elements of design
5. What the abbreviation 3R stands for/
6. What is green design?
7. Why is it important to recycle waste materials?



Enterprising Skills

Learning directed towards the development of students skills, competencies, understandings and attributes, which equip them to be innovative, initiative and successfully manage personal, community, business and work opportunities.

The main aim of Enterprise Education is to provide learning opportunities where students identify real life projects and activities through which they develop and demonstrate transferable enterprising skills and attributes. It also introduces students to the world of business and work and promotes citizenship and sustainability. The projects can be related to social, personal, civics, community, business and work issues.

Enterprise education seeks to develop in people a range of skills to deal with change to assist them with their transition to post school life.

Enterprise Education is a way of helping deliver existing subject outcomes in a more different, meaningful and interesting way, through real life projects, which have meaning for the students and meet real needs?

Enterprising Education improves the development of enterprise capability – innovation, creativity, risk management and ‘can do’ attitude. The drive to make ideas happen is supported by:

1. Financial capability – the ability to manage one’s own finances and inform customers of financial services.
2. Business and economic understanding – the ability to understand the business content and make informed choices between alternative uses of scarce resources.

Enterprising skills and attributes are a part of the curriculum and assessment regime for all learning areas.



Enterprising

Enterprise in education is about taking an enterprising approach to teaching and learning. Enterprise encourages all young people to learn and develop in a way that meets their needs and develops skills for learning, skills for life and skills for work.



Young people need to be prepared for a world which is changing rapidly. Many of the jobs they will do when they leave school do not yet exist and they will probably have several jobs during their lifetime. They need to have the skills and attitudes to cope with an unpredictable future, to be able to deal with setbacks and disappointments in a positive way, and to continue to learn for the rest of their lives.

Taking an Enterprise Approach

Good enterprising should:

- provide opportunities for learners to think and act in enterprising ways.
- provide a clear focus on core and employability skills, and the ability to transfer these two different contexts, in particular the world of work.
- provide opportunities for work-related experiences, both in and out with the classroom.
- adopt an enterprising approach to learning and teaching.
- promote positive attitudes.
- provide opportunities for learners to develop skills such as problem solving, decision making and evaluating risks.
- provide entrepreneurial experiences.

The contribution that enterprise education makes to the personal growth of children and young people can enhance their life chances and choices. It can help them to become successful learners, confident individuals, responsible citizens and effective contributors to society and at work, with a clear understanding of their roles in the world.



How to Estimate Cost

Estimating the total cost of a project is one of the first puzzling challenges you'll face when turning your project into business. There are so many things to consider and it's easy to get bogged down in the details before you really get started.

But there is no avoiding it. You can't even price your work without knowing the cost. In fact, many workers figure their final price by adding a profit margin onto their cost estimate. And, even if you price your work based on market factors (you do, right), you'll still want to check that price against your costs. If you don't, you could be losing money without knowing it.

Formula for Estimating Cost

Total cost = materials cost + labour costs + overhead.

Materials Cost

Anything you purchase to construct or build the project goes into the "Materials cost" category.

Labour Cost

The labour estimate is the least exact piece of the puzzle. The easiest technique is to figure your labour cost as percent of your materials cost. It may also depend on the things like style of joinery to be used and the amount of custom design work required.

Overhead

The overhead category includes all the other things you spend money on, even if you don't build anything. Things like utilities (electricity, water, telephone, etc), rent, insurance, office expenses, software, advertising and anything else not directly part of the project cost.

Profit margin

If you're basing your price directly on the cost estimate, you'll want to include a profit. Remember, you've already build in labour costs (even if the labour is your own), so the extra amount you deserve for taking the entrepreneurial risks. This is usually calculated as percentage of other costs.

A Sample Cutting List

A sample cutting list is given below which you can use to work out the required materials for a particular project.

Cutting list									
Project no					Date:				
Project									
Item description									
	member	No off	Finished size			Total Length	Remarks cross section material		
			L	W	T				
1									
2									
3									
4									

A Sample Costing List

A sample costing list is given below which you can use to work out the cost for particular projects.

Costing list								
Project no				Date:				
Project								
Item description								
	member	No off	Finished size			Total Length	Unit cost	Total cost
			L	W	T			
1								
2								
3								

Quality Control and Quality Assurance

Quality control is the process employed to ensure a certain level of quality in a product or service. Quality is determined by the product users, clients, or customers, not by society in general. In relation to the projects you will make the inspection of the project throughout the entire production cycle that will reduce defects and cost.

Another aspect is known as **the Quality Assurance** or **QA** for short, which is the systematic monitoring and evaluation of the various aspects of projects, to maximize standards of quality attained by the production process.

There are two principles included in QA: "Fit for purpose- the product should be suitable for intended purpose; and "Right first time"- mistake should be eliminated. QA for your project work will basically include the assurance of quality of raw materials, assemblies, production and inspection processes.

The basic goal of quality control would be to ensure that the project made meet specific requirements, satisfactory and fiscally sound. Inspections for quality can be conducted at any point throughout the production process, with the maximum benefit observed when strategically employed at the beginning (first-stage), in-process (30%-50% complete) and pre-completion (100%). The idea is to identify, contain and resolve issues as quickly as possible. A key component and initial step of the inspection process is the development of an inspection plan which can be done in consultation with your teacher.

Folio Production

1. Front Cover

The front cover of your folio should include your name, and what it is the folio is about. It could also include the course you are sitting; BASIC TECHNOLOGY and possibly a sketch of your finished model. The front cover should be fairly plain so that the information stands out.

It is a good idea to make the front cover the last part of the folio that you make.

3. Overall guidelines

When constructing the folio, it is important to make it look neat and tidy. There should be a corporate image, all the pages should have the same layout, use the same type of text, color sequence etc. so that the pages look as if they fit together.

2. Content Page

It is a good idea to include a content page in your folio. It makes sure that you have all the relevant pages in it and keeps them all in the correct order. It also helps to give your folio a sense of continuity.

4. Contents

- Front cover
- Contents
- Design brief
- Specification
- Research-sizes
- Research- materials
- Research- joints
- Research- finishes
- Initial ideas
- Developed idea
- Working drawing
- Cutting list
- Sequence of operation
- Presentation drawing
- Evaluation

What is a DESIGN BRIEF?

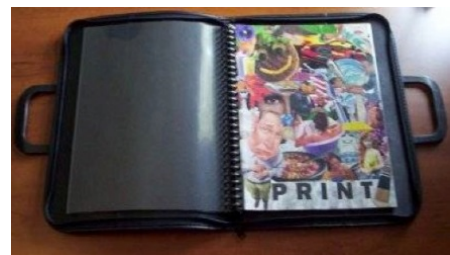
A **DESIGN BRIEF** is the starting point of the Design Process. It is a short statement of what the problem is. It should give enough information to state the problem clearly-identify a need. The brief should clearly give a designer a 'lead-in' but not restrict the design options by being too specific.

What is ANALYSIS?

ANALYSIS is a way of breaking the problem down into all the different things you must consider. **MIND MAPPING** or **BRAIN STORMING** is the most common.

What is a SPECIFICATION?

A **SPECIFICATION** is a list of things that your design should do or be. It should be written with reference to your ANALYSIS. It should start with the **PRIMARY FUNCTION** of your design, the most important thing that your design must do then list the rest with bullet-points.



Worked Example

Enterprising Project: CD & DVD Racks

DESIGN BRIEF

My bedroom is becoming more untidy by the day and my mum is always on at me about the mess. There are CD's and DVD's lying all over the place. The problem gets worse at night when I take off my jewelry and take loose change out of my pocket. If I could design something that could store these items then my bedroom would be much tidier.

ANALYSIS [Storage Device]

Aesthetic:

- Proportion
- Shape
- Color

Function:

- To store CD's and DVD's
- To keep room tidy
- To look good

Ergonomics & Anthropometrics

- Who
- Users
- Hands

The Mess

- CD's
- DVD's
- Coins
- Jewelry

Materials

- Metal
- Plywood
- MDF
- Plastic
- Wood

Location

- On the wall
- My bedroom
- Chest of drawers
- On top the cupboard

SPECIFICATION

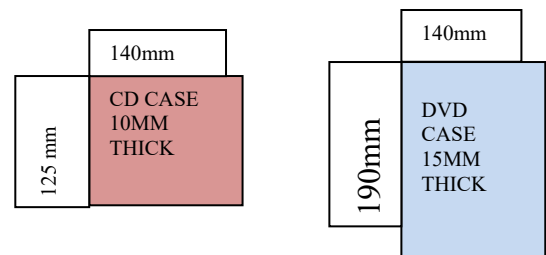
From my analysis, my design should:

- Have the primary function of storing CD's and DVD's.
- Have the secondary function of storing jewelry and loose change.
- Be good looking and fit in well with the decoration of my room.
- Be suitable in size and proportion to sit on top of my chest of drawers or fix to the wall.
- Be suitable in size to allow easy access.
- Use materials that are cheap and available but still durable.

RESEARCH AND INVESTIGATION

RESEARCH should continue on from the **ANALYSIS** and **SPECIFICATION**. It should cover all the important aspects from your analysis and make sure it has covered the specification. It should look at:

- Important sizes (CD cases and DVD cases)
- Ergonomics and Anthropometrics (human sizes and reach etc.)
- Materials (different types of material and joining processes)
- Aesthetics (finishes, proportion)
- Similar products already on the market



Important Sizes

My design will be used to store CD's and DVD's so the sizes of these items will be important.

Ergonomics

Ergonomics is taking into consideration the user of the design. The product will be used by people of all ages. To ensure the safety of all the user the design must have no sharp corners and for comfort and convenience the design must be easily accessible.

Anthropometry

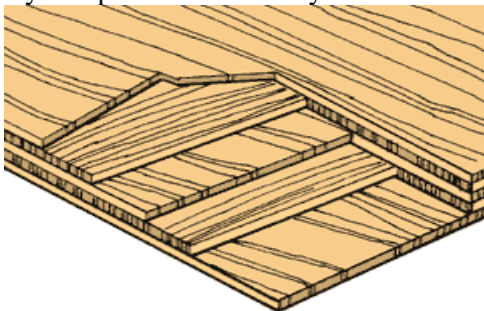
Anthropometric Data is found in tables and lists all aspects of human dimensions. The data is available for both men and women over different age groups.

RESEARCH & INVESTIGATION

Materials: Plywood is a man-made board made from thin layers called veneers that are glued together. The layers are glued with the grain direction at right angles to each other. This method of cross bonding keeps shrinkage to a minimum as movement across the grain is restricted by the glue and the cross-ply.

Red Pine

Red Pine or 'Redwood' is softwood. It is light in colour and has reddish grain and knots. It is relatively inexpensive and is easy to work with.



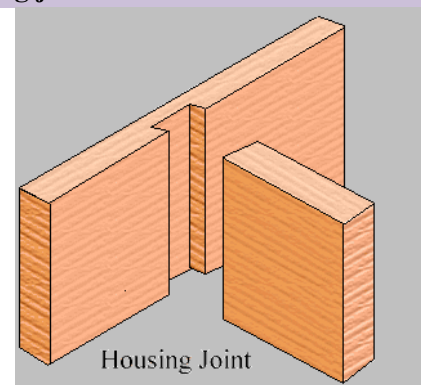
Mahogany

Mahogany is a hardwood. It is red-brownish in colour and is more expensive than pine or plywood. It is a reasonably durable material and is commonly used to make furniture.

Joining Methods

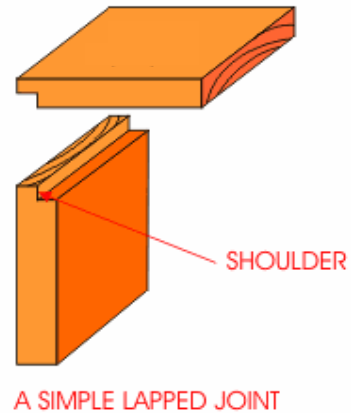
Through housing joints are used when fitting shelves or partitions. It involves cutting a groove that has the same thickness as the shelf so the shelf can sit in.

Housing joints



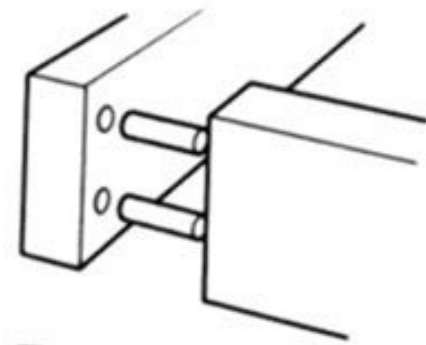
Lap Joints

Lap joints could be used for the corners of the frames or boxes. A rebate is cut with the same thickness as the end of the adjoining wall.



Dowel Joint

Dowel joints can be used for joining two pieces of wood at right angles. Small round pieces of wood called dowel pins are used and saw cuts are made on these to allow any excess glue and air to escape from the holes.



Finishes

Gloss Paints

Painting my model would give me a range of colours to choose from and would also protect the wood. Paint would be applied in three stages: PRIMER COAT - to seal the grain of the wood; UNDERCOAT - to complete the sealing and cover the grain; GLOSS COAT - to add a shiny finish. Each coat should be brushed on both with and across the grain, and the product should be sanded down between coats.



Stains

Stains of different colours can be applied to woods. They are applied by cloth or brush change the colour of the wood. When the stain is dry, the surface should be smoothed down before varnishing.

Polyurethane Varnish

Varnish is a clear, hard wearing finish which shows the natural grain of the wood. It is applied in the same way as paint, although the final coat should be rubbed down with very fine glass paper.

INITIAL IDEAS

- **INITIAL IDEAS** are the first steps to creating your design. They should be quick freehand sketches of any ideas that you have.
- Do not rub out bad ideas, work on them and if they don't improve, ignore them.
- If there are any parts of your ideas that are not obvious in the drawings, add **ANNOTATION** or little notes to give more information about ideas.
- Don't worry at this stage about how you might build your design, so be as imaginative as you can.

DEVELOPED IDEAS

DEVELOPED IDEAS are ideas that build on your initial ideas. Select the best ideas from your previous designs and work on them. Improve them by making small changes or combining different parts of your initial ideas together to make new designs.

This is the stage where your designs should show how they might be used, with some information about the different parts of your designs.

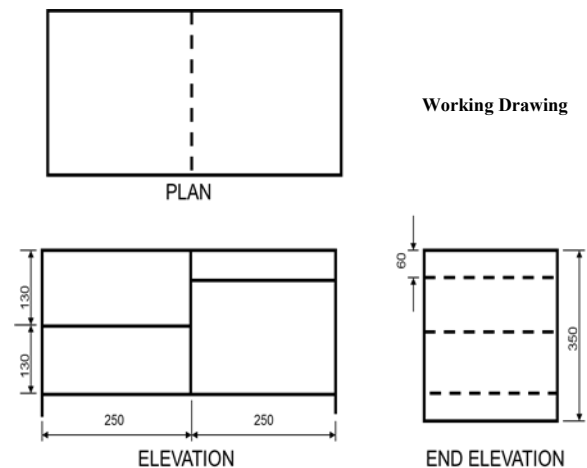
You might also want to show specific parts of your designs in more detail, giving examples of how different parts might be joined together.

PLANNING FOR MANUFACTURE

This section of the folio should contain all the information that is needed to build your model. There should be:

WORKING DRAWING

A WORKING DRAWING This should be an orthographic drawing with sizes. This drawing should show all the parts of your design.



CUTTING LIST

A CUTTING LIST This should be in the form of a table and contain all the different parts, dimensions and materials.

Parts	Dimensions Length x breadth (mm)	Materials	No off
Sides	350x150x12	Plywood	2
Bottom	510x150x12	Plywood	1
Shelves	250x150x12	plywood	2
Partition	280x150x12	Plywood	2
Top	250x165x12	Plywood	1
Back	350x520x3	Hardboard	1

SEQUENCE OF OPERATIONS

A SEQUENCE OF OPERATIONS This is a detailed list of all the steps needed to make the model. It should cover four main stages: marking out, cutting, drilling and shaping, assembling and then finishing. Each main stage should have three or four steps. Each step should be numbered clearly and the tools used should be listed. Sketches can be used to show important steps but should be kept fairly simple.

PRESENTATION DRAWING

A PRESENTATION DRAWING This is a 3-Dimensional drawing of your finished product with colour. It is normally a Perspective, but oblique and Isometric are also acceptable.

CUTTING LIST

SEQUENCE OF OPERATIONS

1. Mark and cut housing joints on sides and partition. Tools used: Rule, try square, tenon saw, chisel, mallet, hand router.
2. Mark and cut lap joint on top. Tools used: rule, marking gauge, chisel, mallet.
3. Assemble cabinet and check angles and sizes. Tools used: Rule, try square.
4. Sand down all surfaces and apply iron-on edging to front facing edges. Tools used: Sand paper, iron, craft knife.
5. Glue bottom into housing joints on sides and leave to dry. Tools/Materials used: PVA glue, sash cramps.
6. Glue partition into position. Then glue shelves into sides and leave to dry. Tools/Materials used: PVA Glue, sash cramps.
7. Glue top into position and leave to dry. Tools/Materials used: PVA glue, sash cramps.
8. Cut chamfer on edges of back. Tools used: jack plane.
9. Glue and nail hardboard into position. Tools used: ball peen hammer, panel pins.
10. Sand down all surfaces with 80grade, then 120 grades, then 240 grade sandpaper and brush dust off. Materials used: sand paper, brush
11. Apply thin layer of varnish and leave to dry. Materials used: paint brush, polyurethane varnish.
12. Rub down with steel wool.
13. Apply second coat of varnish and leave to dry.
14. Rub down with steel wool and apply polish.

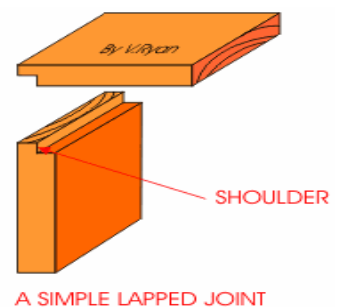
What is Evaluation?

EVALUATION is the last part of the folio. It should be done after construction and should check the quality of what has been designed. Each item in the **SPECIFICATION** should be considered and the answers should state how well or badly the model satisfies each item.

Being critical is not a bad thing and improvements can be made.

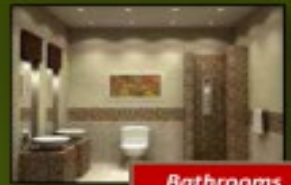
Example:

My design successfully stores 50 CD's and 25 DVD's. It also has a space to store my loose change and jewellery. It looks good and having the natural grain that fits in with the rest of the furniture in my bedroom. It is slightly too big to sit on my chest of drawers and would be difficult to attach to the wall but it is useful as a bedside cabinet.





Bedrooms



Bathrooms

Home Painting Guide

A step on step guide to Wall & Room Painting



Kitchen



Drawing Room

Important Home Painting Factors

- ✓ Home Painting Overview.
- ✓ How to Start Painting.
- ✓ Painting Process.
- ✓ Internal Wall Painting.
- ✓ External Wall Painting
- ✓ Post Painting Care & Common Defects.
- ✓ Different Rooms.



Home Painting Overview



Exterior Home Painting



Interior Home Painting

- [Home Painting](#) is more than simple wall protection.
- Your Painting radiates your personality. They add life to your home.
- Home Painting can broadly be classified into (a) Exterior Home Painting & (b) Interior Home Painting.

Home Painting Process



Pre Painting Work



Surface Preparation



Painting

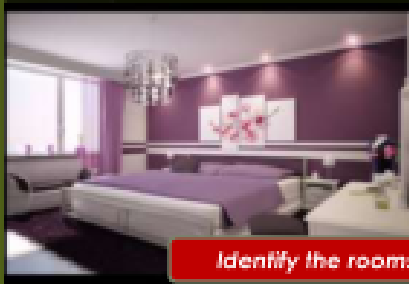
- The [process of Home Painting](#) is divided in 3 parts:

- **Pre-painting work** : Includes preparation work like masonry work, crack filling, water proofing etc. These have to be undertaken to ensure maximum life for the final paint film.

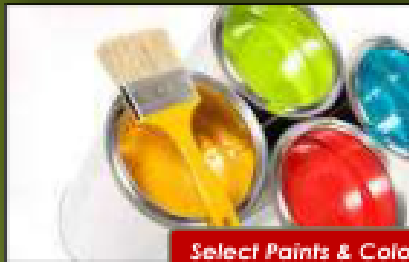
- **Surface Preparation** : This process of wall painting involves making the surface fit for paint application. For example, if you are painting your interior walls you would need to make them smooth by applying Plaster of Paris (POP) or Stucco/ Putty.

- **Painting**: This includes getting the final paint on the surface.

How to Start Painting



Identify the rooms.



Select Paints & Colours.



Remove dust from walls.

- Before you start painting, identify the rooms to be painted.
- Choose the shades/ scheme you want to be used in each room. Darker colours makes a space seem smaller while lighter shades brighten a space, making it appear more expansive.
- Measure your rooms to find out the area you need to paint. Choose the most appropriate paint that meets your needs.
- Remove door knobs, handles and other decorative accessories from doors and windows, so as to prevent damage due to paint spillage.
- Use a dry mop to thoroughly remove the dust from the walls. Wash bathroom and kitchen walls with water and allow them to dry.

Interior Wall Painting



- Painting the [Interior Walls](#) of your home requires you to follow the same procedures:

- **Pre-painting work :** Check surface for dampness or water seepage before painting. Remove any loose plaster from the wall. Ensure proper curing time (7 to 21 days depending on extent of plastering). Allow re-plastered surfaces to dry thoroughly.

- **Surface Preparation :** Newly plastered surfaces should be allowed to cure for a period of at least 6 months. Remove loose particles and paint flakes. Scrape with sandpaper to ensure that the surface is dry and free from dirt, dust, or grease.

- **Painting:** It is always recommended that new surfaces be primed. Avoid application of surface or joint compound while painting exterior surfaces. Fill cracks with a 1:3 (by volume) cement and sand mixture. Always stir the paint or stain thoroughly before use, as pigment has a tendency to settle at the bottom of the can. Stir periodically during application to ensure colour consistency.

Exterior Wall Painting



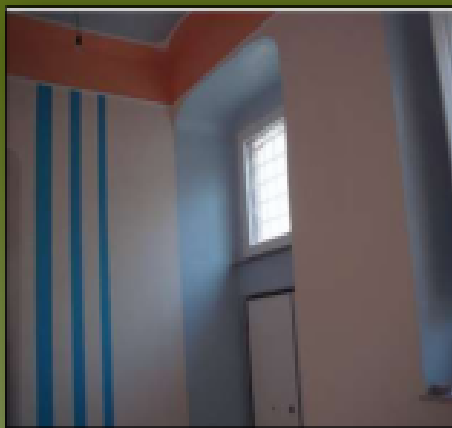
• Painting the Exterior Walls of your home requires you to follow the same procedures:

• **Pre-painting work** : For exterior surfaces it is very important to undertake repairs of any wall imperfections before commencing on painting job. The most common problem with exterior wall painting is cracks in the wall. Avoid applying putty or filling compound while painting exterior surfaces. All external drainage pipes must be checked for rusting or leakage. Any faulty plumbing must be rectified.

• **Surface Preparation** : Newly plastered surfaces must be allowed to cure for at least 35 to 45 days before wall painting. Previously oil painted surfaces should be sanded thoroughly to remove loose particles and made dull and matt for better adhesion. Exterior rough surfaces, previously coated with cement paint, should be wire brushed and washed with water thoroughly and dried completely. Before wall painting, ensure that the surface is free from chalking.

• **Painting**: It is always recommended that new surfaces be primed. Apply Exterior Wall Primer. Apply two coats of top coat paint on surface. It is recommended that horizontal surfaces like the tops of awnings and parapets must be given an additional coat for greater protection. Start Painting.

Post Painting Care & Common Defects



- Most wall paints (like emulsions and washable distempers) are washable. This means they can be periodically cleaned with mild soap water solution. However, stubborn and very old stains may be more difficult to remove from the film.

- The newly painted surface should not be cleaned in less than 2 weeks after application of finished coat.

- Special care should be taken for the passages, kitchens and children's rooms when home painting.

- To minimize permanent or difficult-to-remove stains, clean walls immediately before the stains dry up.

- Walls should be checked periodically for loose, flaking or peeling paint. Once signs of moisture are detected, they should be checked and resolved immediately, so as to minimize damage to the area.

- It is important to protect your painting job. Proper [post painting care](#) ensures that your painting job lasts for a long time & your walls stays protected.

Wall Painting for Various Rooms



Oil Paints for Bed Rooms



Plastic Emulsions on Kitchen Walls



Acrylic Paints for Bath Rooms



Gloss Enamel for Living Rooms

Activity

1. Who is an entrepreneur? Explain.
2. List the important characteristics and skills required in enterprising.
3. Draw a costing list for a project that you have completed.
4. What is a folio?
5. What is the function of the folio?



Chapter 4

STRAND: 4 HAND TOOLS AND MATERIALS

Outcome

After studying this chapter, students will:

- Identify the basic woodwork and metalwork hand tools, their parts and uses.
- Develop skills in use of the hand tools and processes.
- Acquire practical skills based on sound theoretical principles and knowledge of the hand tools.



Introduction

In most jobs and projects, the correct hand tools are essential to complete the task. Whether you are making a small task, project or big furniture, you will need the right tools. After all, it is the hand tools and the skills in using them the correct way which is important. This plays huge satisfaction and enjoyment when the expected results achieve at the work place. There are many types of tools available and used for various kinds of tasks. You may not be able to learn and use all the tools but only basic and essential hand tools that you may use for your tasks and projects.



CLASSIFICATION OF TOOLS

All the hand tools may be generally divided into classes or groups according to their uses. The main groups of tools which you will be introduced to are classified as follows:

1. **Holding and supporting tools** - These tools are used to hold or support the job while working on it.
2. **Geometrical tools** - These tools are used for measuring, marking, testing and setting out the work.
3. **Abrading and Cutting tools** - These tools are used for reducing the timber to required sizes and shapes.
4. **Percussion and impelling tools** - These tools are used to drive nails, screws, chisels and take out nails and screws.
5. **Boring tools** - These tools are used to bore holes.

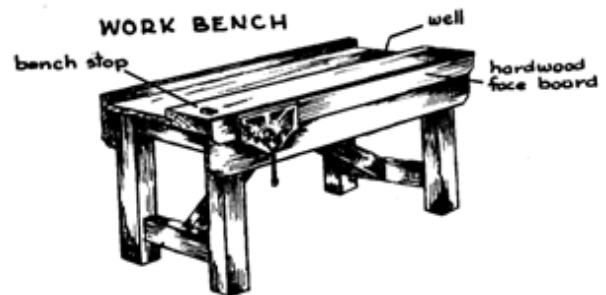
In this book, the hand tools have been split into their different categories. This is done for ease of reference and for you to be able to understand and use these tools effectively in completing your tasks and projects.

1. HOLDING AND SUPPORTING TOOLS

Woodworking Bench

A very essential equipment for any wood workshop. Take care of the bench at all times so that the surface remains smooth and level.

The work bench is used for supporting your practical work piece or project while you work on it and during various other operations in completing your tasks and projects.



Bench Appliance

Bench Well

It is the middle part of the bench top which forms a storage space called the 'well'.

It is used for storing the tools which are in use.

Bench Stop

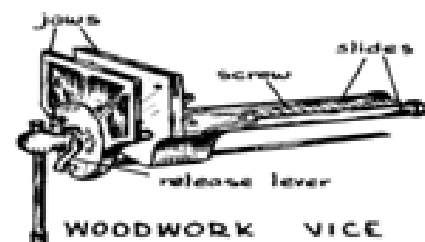
It is an adjustable wooden support slotted and fitted on the end of the bench usually on both sides. It is set slightly lower than the thickness of the timber being planned.

It is used to block and prevent the timber from sliding away while planning on the bench.

Bench Vice

is made of metal and fitted with wooden guards and checked and assembled to the bench usually one on each side.

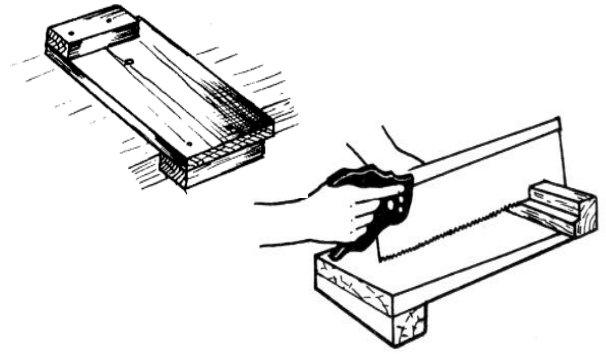
The vice is used to hold your job piece while you work on it; cutting, chiseling, sanding, assembling, etc.



Bench Hook

It is made of wood and designed to hook over the bench top.

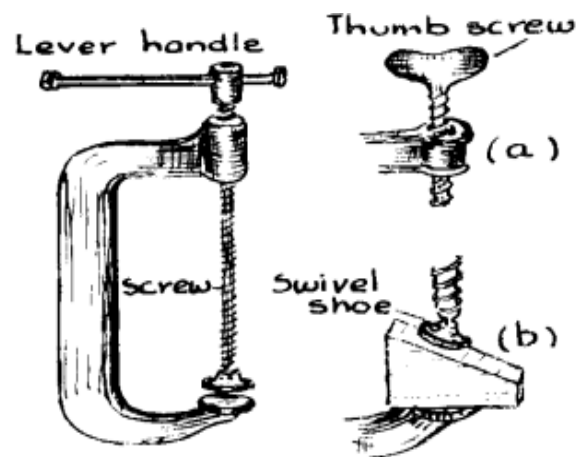
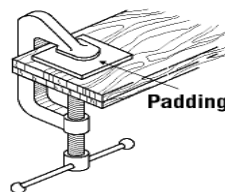
It is used for holding the timber while you saw or chisel on the bench and at the same time preventing the bench top from damage.



G-Cramp

It is an essential holding device made of metal which is shaped in the form of a 'G'.

It is used for holding down the piece of timber on the bench while you work on it and also used to hold together glued pieces of timber and projects left for drying.



2. GEOMETRICAL TOOLS

Measuring and marking tools for basic woodworking are those that are used for measuring, marking out, setting out and testing a job. It is constantly used for measuring and marking. It is essential that measurements are correct and pieces are worked on to finished sizes. If they are not, the project might not fit properly and will not be as good. Remember to follow one simple rule "*measure twice and cut once*" and you will not go wrong.

Steel Rule

It is a handy measuring tool to have when making precise measurements on wood.

It is used to measure pieces to length, measure the side and end dimensions and for checking edges and surfaces for flatness.



Pencil

It is an important marking tool. Pencils are useful in marking where appearance is important as it can be easily rubbed off or sanded.

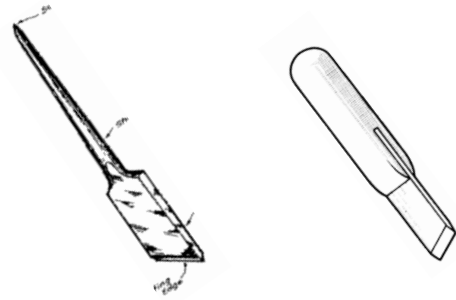
It is used for marking out on timber for lengths, face marks, waste wood and numbering of pieces for assembly.



Marking Knife

It is a simple tool which makes a more permanent line than what a pencil does.

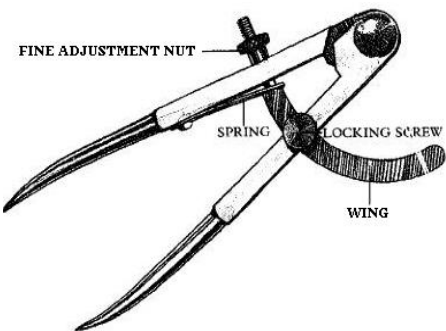
It is used where a fine and accurate cut line is required for marking joints and other more detailed markings on timber. It also cuts veneer, leather and card board.



Wing Compass

Has two steel points in the form of a divider and opened by the means of a wing.

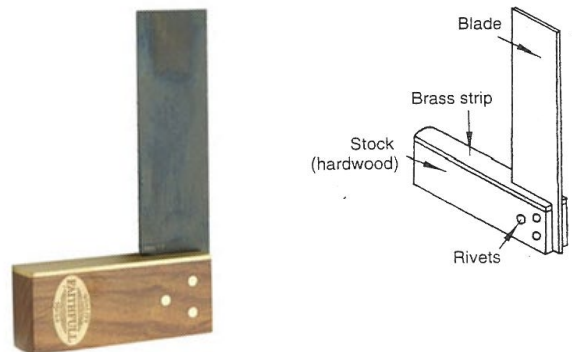
It is used for setting out and marking arcs and circles for designed work pieces.



Try Square

The blade of all the squares are made of steel and secured to the stock by means of rivets. The inside edge of the wooden stock has a brass plate fitted to prevent wearing and splitting.

It is used for marking lines at right angles on a piece of timber, testing timber surfaces for square and marking out square lines on the work piece.



Sliding Bevel

Has a wooden stock fitted with an adjustable blade through a slot in the stock and held by means of wing nuts for locking the blade at a certain angle.

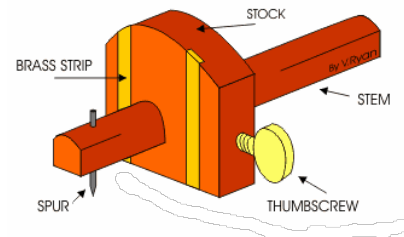
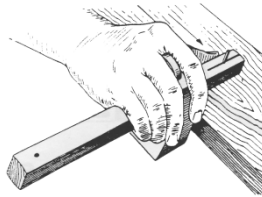
It is used for marking lines at any angle on a work piece and setting out, testing bevels and mitres.



Marking Gauge

It comprises of a wooden stock slotted with a stem which is held by means of a thumbscrew.

It is used for making lines parallel to the face or working side or edge of timber.



3. ABRADING AND CUTTING TOOLS

Saws are the most heavily used of all woodworking tools which take away small pieces or particles of the timber at a time. It is important to hold the saw properly and the best method is described as 3-1-1 which represents the placing of the fingers on the handle. The cut made by the saw is called a **kerf**. Saws can be divided into 3 main saw groups based on their type, sizes and uses:

Back saws - these are the saws used for general bench and fine cutting work. A stiffening rib is fitted over the blade to prevent the blade from bucking while cutting.

Hand saws - are saws used for larger cuts.

Curve cutting saws - are narrow bladed saws used for cutting curved shapes.

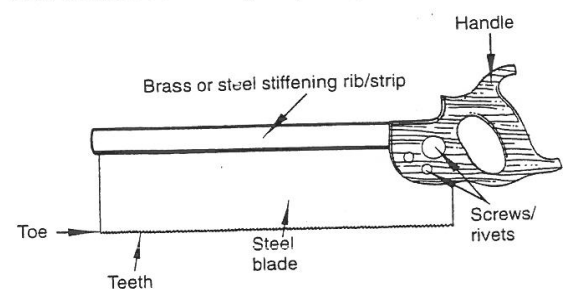
Back Saws

These are small saws with stiff thin blades used for making accurate cuts in small pieces of timber.

Tenon saw

It is considered the best general purpose saw for a beginner. The handle of a tenon saw is made from either plastic or wood and is fixed onto a steel blade with screws/rivets. The brass or steel rib/strip strengthens the blade and enables fine straight cutting.

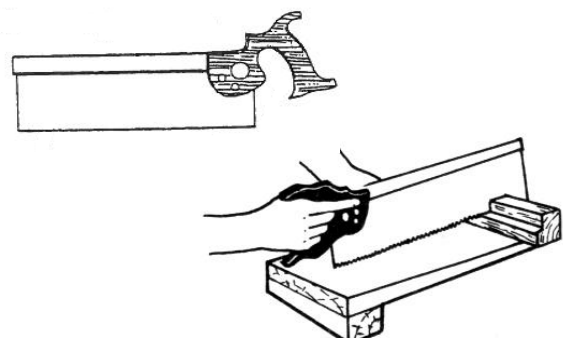
It is used for general bench work such as cutting of joints, angles, with and across the grain.



Dovetail saw

It is smaller than the tenon saw and has an open handle.

It is used for fine accurate work and for cutting dovetails.



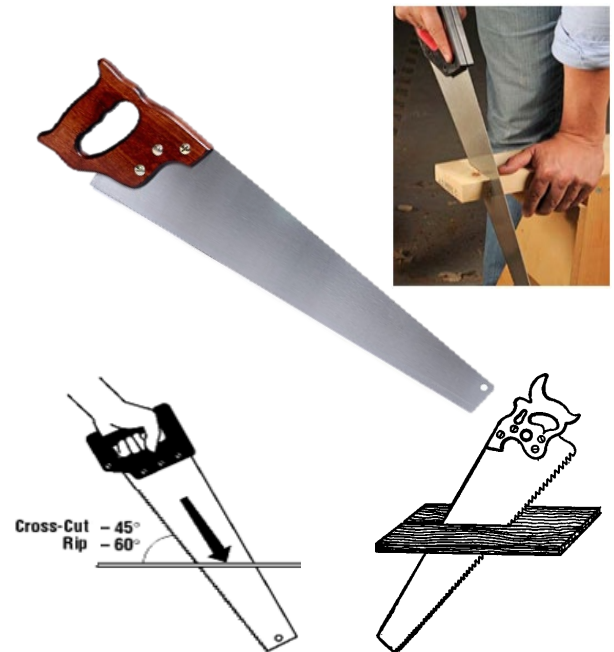
Hand Saws

These saws have long, flexible blades ranging from 500 to 660 mm in length. Unlike a backsaw, there is no stiffening back thus, allows the blade to cut right through a wide board.

Cross cut saw

It is not suitable for cutting small pieces of wood. To get the best from a cross cut saw, hold it at about 45° to the wood.

It is specially designed to cut timber across the grain of timber, especially cutting large pieces of timber.



Rip Saw

It is the largest hand saw and is not suitable for sawing small pieces. Use the saw at a low angle to start the cut and continue at about 60° to the wood using the full blade length.

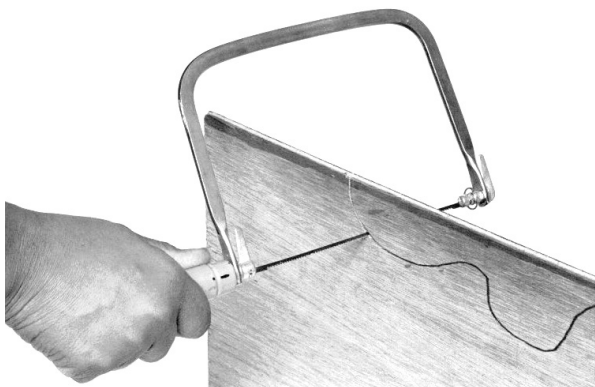
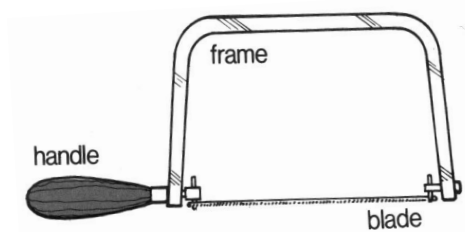
It is used for cutting timber along the grain or ripping lengthwise, especially cutting large pieces of timber.

Curve Cutting saws

Are narrow bladed saws and available in various shapes and sizes. This book will only cover the most commonly used curve cutting saw which is called the coping saw.

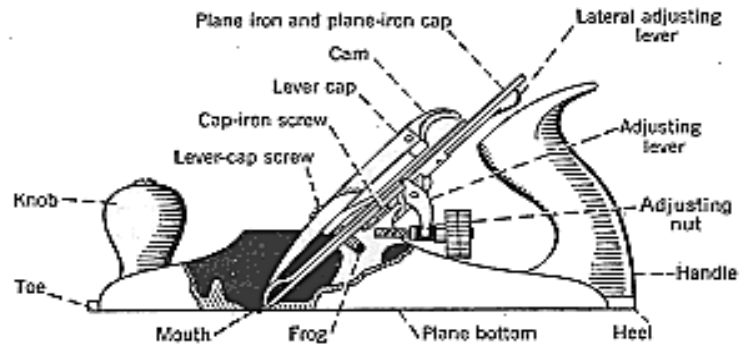
Coping Saw

Consists of a narrow, flexible steel blade held in tension to a 'C' shaped frame to which a handle is fixed. The coping saw cuts best on the pull stroke so the blade is fixed with the teeth facing the handle. It is used to cut small curves, slots and difficult shapes in fairly thin timber, plastic, glass fibre and mild steel.



Bench Plane

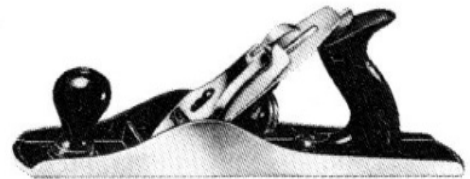
It is a traditional tool that is used to level off wood and for finishing prior to sanding, painting and sealing. A range of planes have been developed and they each have a different but specific use. When working with planes remember it is best to work with the grain as this allows for easier use.



Jackplane

It is designed to take off heavy shavings.

It is used to square up rough timber to correct size and quickly removes waste wood.



Smoothing plane

It is used with a finely adjustable blade to skim the surface of wood that is already flat. It tends to ride up and down uneven surfaces so it cannot be used for squaring timber.

It is used to produce a final smooth planed surface on the wood.



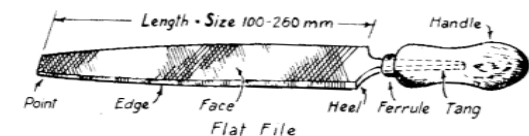
Files

Files are used in woodwork for two purpose:

1. Tool sharpening e.g. a saw.
2. Smoothing edges and small curves in wood which are difficult to reach with other tools.

Files have three distinguishing features:

1. *Length* is measured from heel to shoulder.
2. *Kind, shape or style* may be flat, mill half-round, circular, or triangular.
3. *Cut* having set of furrows; single or double cut.



Rasp

It is similar in appearance to a file except it has teeth instead of furrows.

It removes the wood more rapidly than a file but leaves a coarse finish which must be smoothed with a file or glass paper.



4. PERCUSSION AND IMPELLING TOOLS

Chisels are among the handiest of all woodworking tools and for many jobs. Chisels are available in wide range of shapes and sizes. These useful tools need to be handled with care and skills for the best results. *Before using a chisel it is important to ensure it is sharp.*

Chisels are tools with a long flat beveled blade with a square cut edge attached to a handle. They are used to remove unwanted parts from the wood or to carve and shape. The chisel is held by one hand, while the other hand is used in striking a mallet at the chisel's handle.

A variety of sizes of chisels are used in woodworking. The large chisels are usually used for larger work while the small chisels are saved for detailed tasks.

Beveled-edge firmer chisel

Has a beveled-edge and slim blade making it suitable for light work.

It is used for joint making; reaching into awkward corners in paring trenches, dovetails and chamfering.



Paring chisel

Is the same as a beveled-edged chisel except that it has a longer blade and is used for pairing work.



Mortise chisel

Is specifically designed for heavy work. The blade is thicker than most other chisels and the handle is stronger.

It is used for heavy work, cutting out mortises and slots.



Firmer chisel

It is the most common general purpose chisel that has a strong, thick rectangular section blade.

It is used for a wide range of tasks such as paring, trenching and light chopping of waste wood.



Parts of a chisel and uses



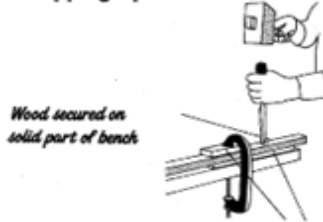
- | | |
|------------|----------------------------|
| 1 Blade | 4 Tang |
| 2 Shank | 5 Grinding bevel |
| 3 Shoulder | 6 Sharpening bevel |
| 7 Ferrule | 8 Handle - wood or plastic |

Vertical Paring:



Timber supported, note position of hands

Chopping Operations



Wood secured on solid part of bench

Horizontal Paring:



Note position of elbow and position of hands!

N.B. Keep all parts of the body behind cutting edge.

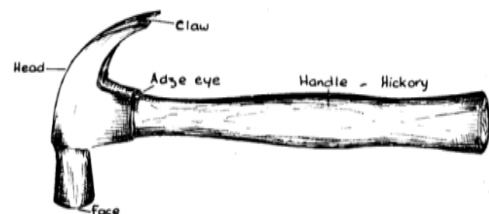


Hammers

Hammers come in variety of head weights and handle lengths. The head of a hammer is made of forged steel and the handle is made of either wood or steel. Always choose the hammer that fits your hand and is designed for the work to be carried out. You should acquire the habit of grasping the handle of the hammer at the end, as this will give greater force to the blow. Upon light work, the hand will naturally slip a little toward the head.

Claw hammer

Is the commonly used hammer in woodworking and is mostly used for heavy work. It is mainly used for heavy work in driving large nails. As the name suggests, it has a claw used for extracting nails



Warrington hammer

Is a light hammer with a cross pein.

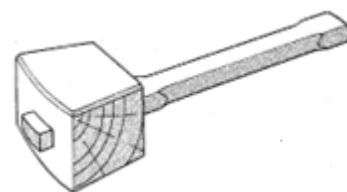
It is used for light hammering and general use in cabinet work. The cross pein is used for starting small nails and also for pressing small inlays and veneers into position when gluing.



Mallet

It is similar to a hammer made of wood with a large rectangular head fitted with a tapered handle.

It is used for driving chisels and for knocking pieces in assembling jobs.



Pincers

Is a tool made from drop forged steel. A block or scrap piece of timber is used with the pincer and the claw hammer while extracting nails to prevent damage to the surface and provide greater leverage.

It is used for extracting nails.



Nail punch

A small tool made of steel.

It is used with the hammer for driving nails below the surface of timber.



Screwdriver

Has a blade made of alloy steel which may be flat, round or square. One end of the blade is ground to form a tip(s) to fit the slots of screws, and the other end shaped to a tang which is fitted into a handle made of tough hardwood or plastic. These are available in a variety of sizes and patterns.

The screwdriver is a driving or impelling tool used for inserting or removing screws for general work.



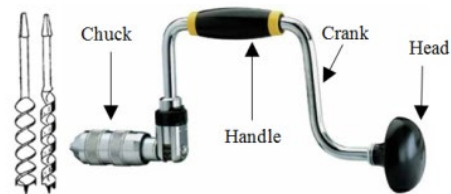
5. BORING TOOLS

These are tools that are used for boring or cutting holes in timber.

Ratchet brace

It is fitted with a *ratchet* so that holes can be bored in confined spaces, where it is impossible to turn or sweep the crank for complete turn.

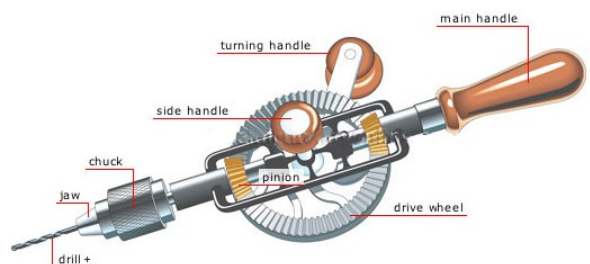
The brace is used with screwdriver bit for driving screws.



Hand drill

It is useful for woodwork because the drills cut quickly and do not split the wood.

It is used mainly for boring holes for screws or nails.



Hand Drill Bits

Many varieties of bit are designed for boring holes. Bits have a round shank to fit into the chuck of the hand drill. The term boring means cutting holes in materials.



Metalworking Bench

It is a very essential equipment for any metal workshop. It comes in many designs and shapes and should be strong and firm. Care of the bench must be taken at all times so that the surface remains smooth and level.

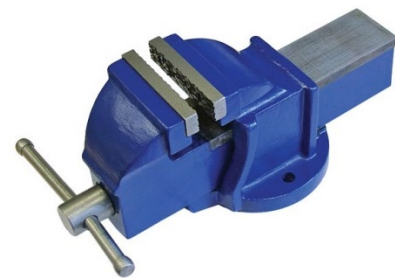
It is used for supporting your practical work piece or project while you work on it and during various other operations in completing your tasks and projects



Bench Vice

It is made from malleable cast steel. The parts are machined to slide together, or fabricated from mild steel. They are fixed to a work bench and are a common feature in any metal shop.

They are used for holding and supporting the work while cutting, grinding, drilling and welding.



Marking Out

It is the process of transferring measurements onto a work piece as the first step in the design process. It consists of transferring the dimensions from the work plan to the work piece then constructing or manufacturing the item.

The use of tools to measure and mark the work play an important role with the aid of rulers, gauges, squares, dividers, pencils, etc. This will determine the accuracy of measuring and marking during practical work.

Marking off

It is the process of laying out needed information in the form of center lines, circles, outlines, to show the position and area of work to be done and in setting out the work piece.

Surface Preparation

Surfaces are usually prepared to assist in marking and to ensure that the lines are clearly seen. First, oil or grease should be cleaned from machined surfaces, rust or scale brushed from steel plate, and castings dressed to remove sand or any irregularities.

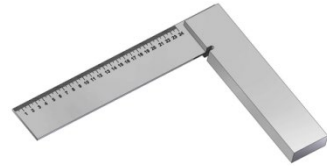
Steel rule

It is used for setting out straight lines and distances, setting caliper and dividers and for checking edges and surfaces for flatness.



Engineer's square

Has a metal blade fitted to a metal stock.
It is used for setting out and testing angles of 90°.



Centre punch

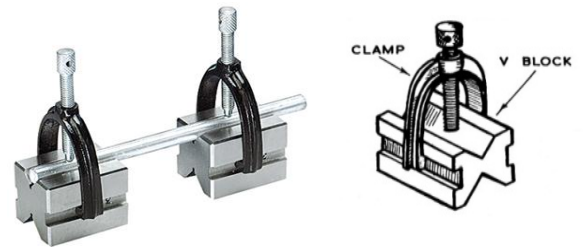
Has a thicker section and the point is 90°.
It is used for marking centers for drilling holes.



Vee Blocks

Normally come in pairs and are usually fixed with fasteners to hold the job piece.

It is used to hold up cylindrical work during marking.

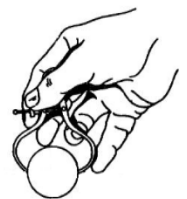
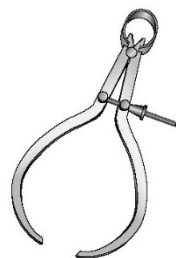


Calipers

Are used for measuring work, checking and transferring dimensions.

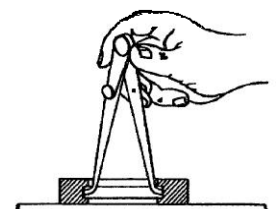
Outside calipers

Are used for checking and testing external diameters.



Inside calipers

Are used for checking and testing internal diameters.



Shaping Using Hand Tools

These are important processes, such as shaping with files or hacksaws. The correct position of shaping tools is one which is comfortable to the operator and permits the tool to be used effectively.

Files

Are made from carbon tool steel forged to shape. The distinguishing features of a file are:

- *Name* - is determined from its cross sectional shape (triangular, square, round, half round).
- *Length* - is measured from the shoulder to the point.
- *Cut* - refers to the way the teeth are cut, whether single or double.
- *Grade* - commonly used are: bastard, second cut and smooth.

When ordering a file the following description can be given length, type, and grade e.g. 300 mm flat bastard file.

Using files

When filing, the vice should be at elbow height but for work needing heavy filing, it should be lower.

The file must be held horizontally and "rocking" motion should be avoided to prevent any convex surface.

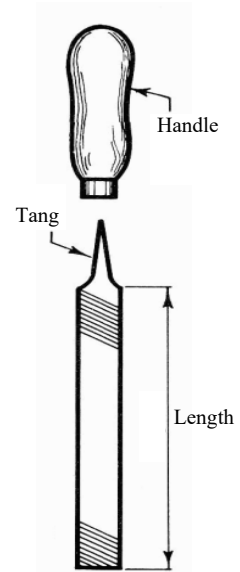
For heavy filing, a forward straight through stroke is best. The position of the stroke should be changed to prevent "grooving" by the teeth on the surface of the work.

Apply sufficient pressure on the cutting stroke to prevent the teeth from sliding over the work.

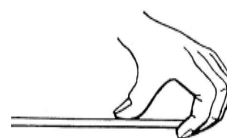
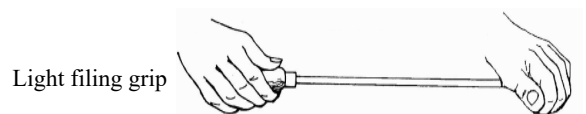
Hacksaws

Are used with the job piece held securely on the vice with the saw held in both hands. The saw must start at the far side at a low angle, and the angle slowly reduced while cutting to keep as many teeth as possible in contact with the work. The whole length of the blade must be used with pressure eased on the return stroke. Cutting strokes should be steady at a pace which is comfortable and effective. The speed is reduced when cutting harder metal. For effective cutting, the correct blade for the material should be used.

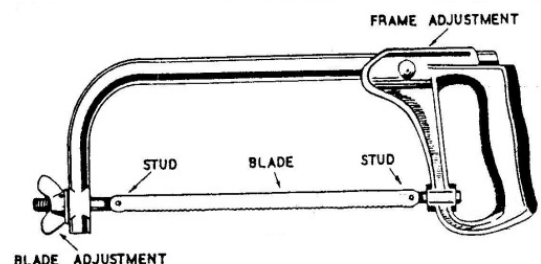
It is used for cutting rods, bars, angle plates to required lengths and thick sheet metals to shape.



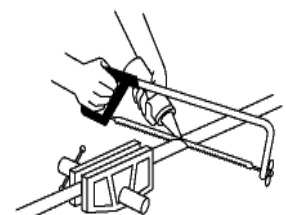
Grip for filing



File handle for heavy filing



While sawing, use lubricant to reduce friction.



Spanners

Are available in a variety of types according to the screws and nuts and the awkward positions in which they may be used.

They are used in turning set screws and nuts in the assembly or removing of structural, motor or machine parts.

Set spanners

Are made single or double ended with the jaws offset 15 degrees to allow further movement in confined spaces when the spanner is turned over.



Ring spanners

Are safer and have better control since nut faces or the corners will not slip. They are made as six or twelve point spanners, the twelve points being the most common and giving a better range of movement in difficult positions.



Tube and box spanners

Are six-sided and safe to use since the side grip on the nut lessens the chances of slipping. They are turned by a bar through holes in the body and used for work on nuts in awkward places e.g. spark plugs.



Adjustable wrench or adjustable spanner

It is a wrench with a "jaw" of adjustable width, allowing it to be used with different sizes of fastener head (nut, bolt, etc.) rather than just one fastener, as with a conventional fixed spanner.



Hammers

Are drop forged cast steel with the head and pein shaped and smooth. The face are hardened to withstand the heavy blows, and tempered to prevent the edges chipping.

Ball pein hammer

It is used for the assembly of parts, and the pein for riveting or hollowing and dishing sheet metals.



Cross pein hammer

It is used for forming seams in corners or for hammering metals to a grooved shape.



Soft-faced hammer

It is used to assemble finished work. Soft faced hammer prevents finished work from being damaged by hammer blows.



Mallet

It is used in place of hammers as they do not damage the metal.



Tinmans mallets

Are used for working on flat surfaces, straight and curved edges and for folding seams and edges.



Bossing mallets

Are used for shaping sheet metal into concave shapes by beating into sand bag mould.



Chisels

Should be used with the job piece held in the vice at elbow height. They are used for cutting or where the amount of metal to be removed is too much for filing.

Flat chisel

It is the common chisel used for general work such as chipping flat surfaces, removing waste metal or cutting thin sheet metal.



The soldering bit

May be heated electrically or by flame to sufficiently melt the solder for joining two pieces of metal.

It is used to apply solder to a joint.



Tin-snips

Are made for cutting thin sheet metal by hand. When using snips, the blades should not be closed fully or a ridged surface will result.

Straight snips

Are used for straight cutting or for curves with waste on the outside.



Curved snips

Are used for curve cutting with the waste on the inside of the cut.



Flat Nose and Long Nose Pliers

Are used for securing flat pieces of metal, or bending wire.



Combination pliers

It is the most commonly used pliers and is suitable for holding flat or small cylindrical work, bending and cutting wire.



Pop rivet gun

Is a tool with pliers-like handles and nose piece to insert rivets.

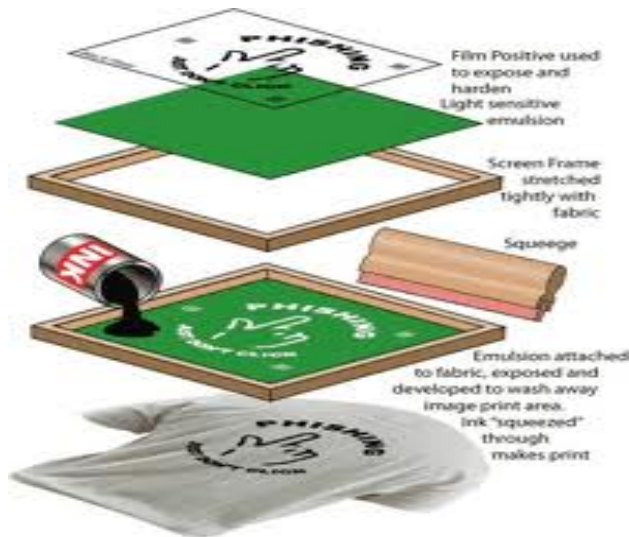
It is used to insert rivets to join sheet metal material together.



Activity

1. Name the bench appliances and state the purpose of each appliance.
2. State the purpose of marking tools. List the important marking tools.
3. Explain the purpose of a pencil and a marking knife.
4. Draw a neat sketch of a try square and label all the parts.
5. Name the different types of back saws and state its uses.
6. Name and sketch the tool for cutting a curve in thin plywood.
7. Name the most frequently used plane in the workshop.
8. State the tools classified under impelling tools. Give two examples of impelling tools and state their main purpose.
9. State the purpose of a metalworking bench vice.
10. Explain the terms *marking out* and *marking off*.
11. State the use of a hacksaw.
12. Name the curve cutting tool used for metalworking.
13. Name and state the use of the common metalworking chisel.

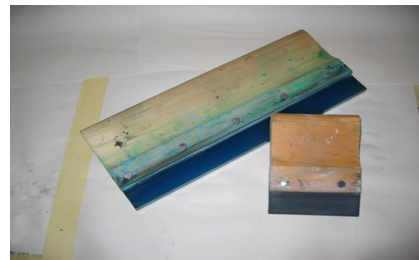
Graphic Arts and Painting Tools



Screen printing is a printing technique that uses a [woven mesh](#) to support an ink-blocking [stencil](#). The attached stencil forms open areas of mesh that transfer ink or other printable materials which can be pressed through the mesh as a sharp-edged image onto a material. A [roller](#) or [squeegee](#) is moved across the screen stencil, forcing or pumping ink past the threads of the woven mesh in the open areas.

Squeegee is a wooden framed tool with a rubber blade in the front. It is also available in different sizes and can be purchased depending on the size of the screen frame.

It is used to control the flow of ink when doing screen or stencil printing.



Screen frames are normally made from wood. Simple corner joints such as mitre, simple butt or rebated butt are used for its construction.

The screen is placed on the frame by the help of a staple gun. It must be ensured that the screen is stretched firmly to ensure that there are no creases formed.

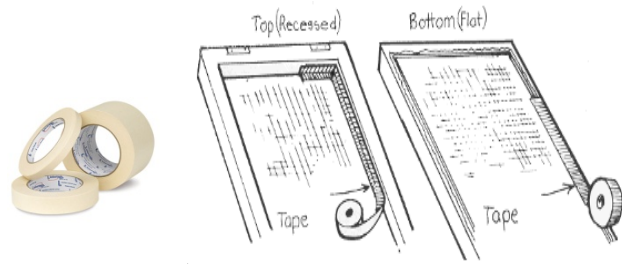


Screen Printing Inks are available in various forms for screen/stencil printing.

Water based inks are normally used for stencil printing as the stencil can be washed after use and stored for use with a different stencil at a later date.



Masking tapes help maintain a "tight" screen, and prevents ink from leaking under the screen frame during printing and will keep the edges of your prints clean. To get maximum adhesion of the tape, rub it with a spoon or wooden stirrer.



Screen Printing Stencils

A stencil is what the [art](#) on a silk screen is called when it's ready for printing.

Paper stencil

This is the best method for a beginner. It is the fastest, least expensive, and simplest way to prepare a screen. It is done by cutting the message or illustration from paper. Keep the paper flat and not wrinkled. For more accurate and durable cut paper stencils, use waxed paper. Designs can be cut with scissors or stencil knife or they can be "torn" to create a textured appearance.



Carving Tools – carving tools are simply the various types of chisel and gouges used to shape wood. There are two main types of carving: chip and relief carving.

Chip carving is a style of carving in which [knives](#) or [chisels](#) are used to remove small chips of the material from a flat surface in a single piece.

Relief carving is a sculptural form in which figures are carved in a flat panel of wood. The figures project only slightly from the background rather than standing freely. Depending on the degree of projection, reliefs may also be classified as high or medium relief. Relief carving can be described as "carving pictures in wood". The process of relief carving involves removing wood from a flat wood panel in such a way that an object appears to rise out of the wood. Relief carving begins with a design idea, usually put to paper in the form of a master pattern which is then transferred to the wood surface. Most relief carving is done with hand tools - [chisels](#) and [gouges](#) - which sometimes require a [mallet](#) to drive them through the wood.



Activity

1. Name the common tools used for graphic arts.
2. What is the purpose of the carving tools.
3. Explain how the screen printing stencils are used.

MATERIALS AND HARDWARE

Outcome

After studying this chapter, students will:

- Identify and follow general safety rules and signs.
- Practice good housekeeping in the workplace.
- Identify and use systematic working procedures.
- Recognize hazardous situations in the workshop.
- Develop safe working attitudes and practices.



Introduction

Materials come in many forms and are used for various purposes. Materials are refined and processed first before being used and are rarely used in its original form. The most significant feature is the alteration of the properties of the material to suit the purpose.

There are many kinds of materials used for engineering applications. Most materials fall into one of three classes that are based on the atomic bonding forces of a particular material. These three classifications are metallic, ceramic and polymeric. Additionally, different materials can be combined to create a composite material. Composite materials are often grouped by the types of materials combined or the way the materials are arranged together.



MATERIALS

All the materials in this book are classified under the following groups:

- metals –ferrous and non-ferrous.
- non-metals –wood, ceramics, polymers and leather.
- composites – concrete, fibreglass and manufactured boards.

These classes can be further broken into sub-groups, each with different applications. Every material we come across belongs to one or combinations of these classes.

Metals

Metals have been extensively used by humans since the early Bronze Age. Metals are normally used to make solid products from cabinets to bridges. Due to the good electric conductivity of metals, all electronic materials rely on metals or its alloys to transmit electricity. Metals generally have good electrical and thermal conductivity with high strength, hardness and have good ductility. Some metals, such as iron are magnetic.



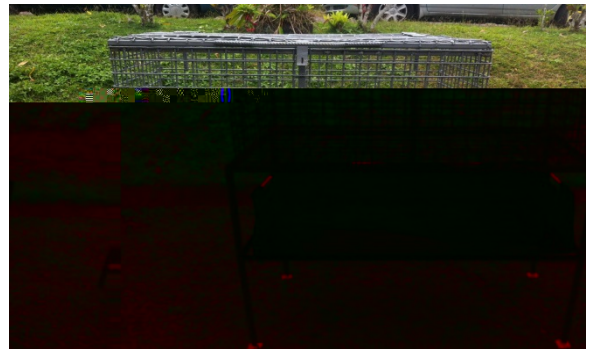
Ferrous metals

Ferrous metals are metals whose main element is iron. Wrought iron can be grouped as pure metal because it contains pure iron or very little carbon up to 0.2%.

Iron

There is some indication that man-made iron was available as early as 2500 B.C. Today Iron is used in most of our machinery and homes. Iron has greatly developed our manufacturing and building industries.

Metal Projects



Steel

These are a few of the many types of steel:

Name	Carbon content	Used	Properties
Dead mild steel	0.1% to 0.15%	Car bodies, wires	High ductility and soft malleability
Mild steel	0.15% to 0.3%	Bars, rods, tubing, wire sets	Ductile and malleable
Medium carbon steel	0.31% to 0.8%	crankshaft, springs and cutting tools	increased toughness and hardness
High carbon steel	0.81% to 1.4%	chisels ,files, taps and dies, knives etc	high hardness, tempered to reduce the hardness and increase the toughness.
Carbon steel	Above 3.2% carbon	Cast irons	high compression strength, high fluidity, low tensile strength and are best used for machine base.

Non-ferrous metals

Non-ferrous metals are metals which have no iron and exist in the earth as ores. They are referred to as pure metals; they are mined and processed in metal factories to extract the metal from the ore while the impurities are removed.

Name	Properties	Use	Diagram
Copper	reddish in color, ductile, malleable and a good conductor of heat and electricity	Used in electrical wirings, tips of soldering iron and some decorative purposes	
Lead	is highly ductile, malleable and non-corrosive	Used for soldering electrical circuits and sheet metal joints for containers and pipes.	
Tin	highly malleable and ductile and is resistant to corrosion	used as coating on food cans	
Aluminum	abundant metal ore found on the earth's crust, strong and light	Used in aircraft industry and in many engineering applications, window frames, trusses, radio and television parts and also for food preparation such as wrappers known as aluminum foil.	

Metal Alloys

Metal Alloys contain more than one metallic element. Their properties can be changed by changing the elements present in the alloy. Examples of metal alloys include bronze which is an alloy of copper, zinc and tin.

Non Ferrous alloys

Brass is an alloy of copper and zinc. It has a golden appearance, is harder than copper and costs more than zinc. Brass is used for musical instruments, ornaments, electrical fittings, screws, taps, hinges and door knobs.

Timber

Timber is converted to wood which is suitable for building and constructional purposes. The timbers used in woodwork are obtained from two botanical classes of trees, but this does not always mean that the timber is hard or soft.

- Hardwood – seeds in fruit and has broad leaves.
- Softwood - naked seeds in cone and usually has narrow leaves.



Timber projects



Softwood

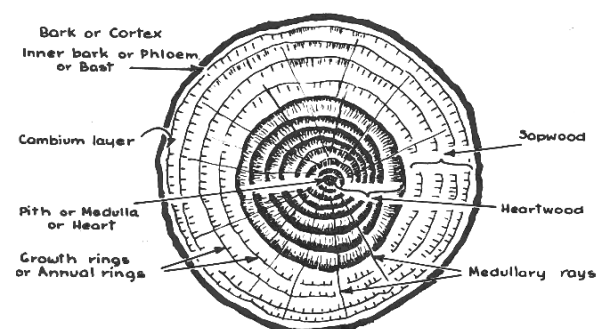
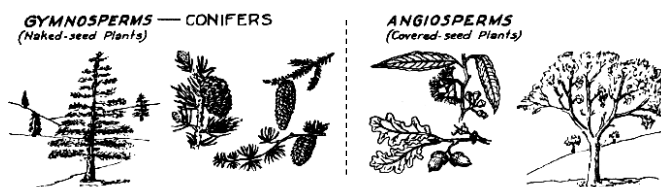
Softwoods or coniferous timber are obtained from cone-bearing trees (conifers) such as pines. The structure of softwood is simple. They have no vessels or pores; as a result, they are often referred to as 'non-pored' wood.

Hardwood

These are obtained from broad-leave trees. The structure of hardwoods is more complex than that of softwoods for they have two types of cell: 'vessels' or 'pores' and 'fibers'. Hardwoods are often referred to as 'pored'.

Difference between softwood and hardwood

Softwoods	Hardwoods
Needle-like leaves, cone-bearing pines and firs Usually evergreen	Broad leaves. Usually but not necessarily deciduous i.e. lose leaves in winter.
A branch usually grows in whorls, i.e. more than two at the same level.	Branches usually grow at different levels with one or two at the same time.
Medullary rays are narrow and very light in appearance.	Medullary rays vary from narrow to wide and may be also very light in appearance.
One type of cell (tracheid's) which serves both functions; conduct sap and support the tree.	Two types of cells, vessels or pores conduct sap and fibers support the tree.
Non-porous i.e. without vessels or pores.	Porous i.e. contains vessels or pores.
Produce uncovered or naked seeds .e.g. pine cone seeds	Produce covered seeds e.g. acorns (oak)
Timber is usually fairly soft and easily worked.	Timber is usually heavy and hard.



Cross section of a tree trunk

Function of each part

Name	Functions
Sapwood	carries water and mineral salts from the roots of the tree to leaves and other parts of the tree.
Medullary rays	cells that radiate from the center of the tree outwards and act as storage cells and transport food and water horizontally in the tree.
Bark or cortex	helps to act as a protective layer for the inner cells from animal, insects and weather.
Inner bark	passes the food in the form of sap down the tree to the cambium layer. (phloem vessels).
Cambium layer	the growth of new cells takes place here.
Growth rings	determine the age of the tree. The growth is more rapid in favorable conditions such as spring or summer. The differing summer winter growth rates each year produce the growth rings.
Truewood	dry sapwood becomes heartwood ; darker and stronger wood in the center of the tree. The main function is to strengthen the tree. The best timber for furniture and building comes from the heartwood of the tree.
Pith	dead cells present in the middle of the tree.

Leather

Leather is a durable and flexible material created via the tanning of animal rawhide and skin, primarily cattle hide. Usually available with the hair removed. These animal skins and hides are treated to preserve and make them suitable for use for a longer time. By the process of tanning, these skins are converted into a stable and non-decaying material



Today, most leather is made of cattle skin, but many exceptions exist. Lamb and deer skin are used for soft leather in more expensive apparels. Deer and elk skin are widely used in work gloves and indoor shoes. Pig skin is used in apparel and on seats of saddles.

Buffalo, goats, alligators, dogs, snakes, ostriches, kangaroos, oxen, and yaks may also be used for leather.

Leather is a product with high environmental impact, most notably due to:

- the impact of livestock.
- the heavy use of polluting chemicals in the tanning process.
- air pollution resulting from the transformation process (hydrogen sulfide during de-hairing, ammonia during deliming and the solvent vapour).

Uses and Benefits of Leather

Leather is a naturally versatile material which is warm in winter and cool in summer. There is no other natural fabric that has the insulation characteristics that allows both ventilation and evaporation to take place. Genuine leather provides the service and durability.

Leather is used to prepare leather apparel and leather garments like leather jackets, leather coats, leather lingerie, leather pants and leather undergarments. Leather shoes are also commonly used all over the world. Leather backpack, leather briefcase, leather bags, seat covers, pencil case, wallet and leather purses also have very important significance in our daily life.



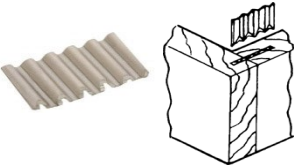
HARDWARE

Nails

These are used to hold wood parts together. Nails should generally be driven in a slanting direction, as they hold better than if driven straight. To do a satisfactory job, the nails must be the correct type.



Application of Hardware

Name	Description	Use	Diagrams
Box or Clout nail	They have a smaller diameter body and a large thin head.	They are used for box and crate construction.	
Spring Head (Roofing nail)	A galvanized nail with a cylindrical or square twisted body and large beveled or dome-shaped head.	Used for fixing corrugated roofing and wall cladding.	
Panel Pins	Thin nail with an inverted cone shape head which enters wood cleanly.	Fixing thin timber and manufactured boards.	
Wiggle nail or Corrugated Fastener	Made of corrugated plate irons with parallel corrugations that have plain or saw like edges.	Used for strengthening edge to edge joints and also for holding frames of cheap flush panel doors and carcase constructions.	
Rivets	The head is performed by stamping. Small rivets are closed by hammer and set.	Is a permanent method of fastening two or more pieces of sheet metal together.	



Activity

14. Name the bench appliances and state the purpose of each appliance.
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25. Name the curve cutting tool used for metalworking.
26. Name and state the use of the common metalworking chisel.

HARDWARE

Screws

These are available in a wide variety of sizes, types, metals and finishes.

The parts of a screw are:

The head gives the screw its name, such as round head, countersunk and raised head.

The shank is the body of the screw which is threaded to a point for about two thirds of its length. The shank determines the thickness of the screw.

The thread is the spiral groove which draws the screw into the timber and provides its holding power. Three types of screws are in general use:

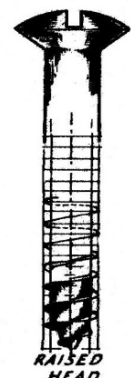
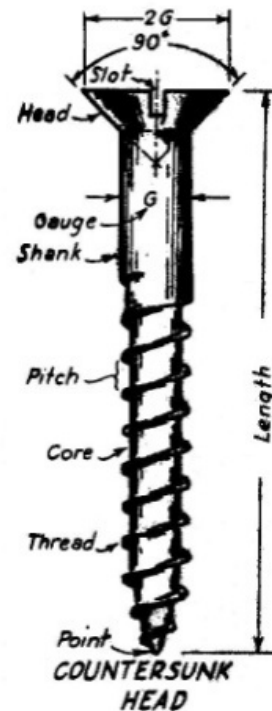
- Countersunk head
- Round head
- Raised head

Countersunk and round head screws are usually made of brass or steel finished. Raised head screws are generally nickel or chromium-plated, but can be obtained in brass or steel.

Countersunk head – is used when the head is required to be flush with or below the surface of the work.

Round head- mainly used for fastening metal fittings to work.

Raised head –They are very decorative and can be easily removed easily without damaging the surface surrounding the screw. They are often used to fix metal fittings to work.



Activity

1. Name the groups of materials and list the materials in each group.
2. Name and explain the properties of metals.
3. Explain the difference between softwood and hardwood.
4. Draw the cross-section of a tree trunk and label all the parts.
5. Name the common nails and screws.
6. Which nail is used for fixing corrugated roofing irons.



Finishes

Finishes are materials which are applied to give an eggshell gloss. They improve with age. The finished surface has high resistance to wet, dry heat, stains and normal wear. To clean the surface, wipe with a damp piece of cloth.

Varnish

Varnish is a transparent coating material. When spread on the surface of the material, they dry to form a glossy or shiny film. They are generally applied with brush, spray gun or rubber.



Uses:

- To preserve and protect the surface, from moisture, dirt, stain and wear.
- To protect the color and the grain of the timber.
- To beautify by adding luster or gloss to the surface.
- To produce a surface film that is easy to clean.



Applying Varnish

Varnishing should be carried out under dust free condition by sprinkling water on the floor around the work to keep the dust down.

Steps:

- a) Seal the surface using shellac or sanding sealer with a small amount of turpentine/thinner/etc and varnish (optional).
- b) Allow the sealer to dry and rub down with very fine abrasive paper.
- c) Dust-free the work carefully. Wipe with cloth moistened varnish so it is tacky.
- d) Dip the brush in varnish work on a clean piece of paper back and forward to evenly distribute the varnish in the brush.
- e) Dip the brush again and scrape against the side of the container.
- f) Brush the edges first then fill in the centre.
- g) Finish with light stroke with the tip of the brush.

Allow 24 to 48 hours or more for drying then rub down with very fine abrasive before applying the second coat.

Stains

Timber may be stained for the following reasons:

- a) To colour the wood.
- b) To imitate a more costly wood.
- c) To enhance the beauty of the grain.

Once staining has been carried out, it is usually protected with a transparent coating (varnish).



Types of Stains

Water stain, spirit and oil stains are available. Oil stains are water proof where water and spirit stains need a transparent coating.



Applying Stains

Once the surface is prepared for applying water stain, the surface must be dampened once or twice and sanded when dry. Water stain is best applied with soft brush which is then wiped off with soft cloth. Spirit stains are more difficult to apply as it does not raise the grains. They dry up very quickly and overlapping strokes are normally visible. They are ideal for small objects. Oil stains are easy to apply as they are water proof and also protects timber but are slower in drying. They are best applied by brush.

Sharpening Tools

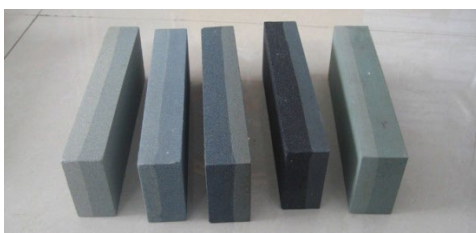
Oilstones

Natural oilstones

Available in two grades: hard and soft, with the coarser stone used for softer honing operations. To prevent overheating during sharpening, a lubricant is necessary. A non-drying oil is used on finer stone and water for coarse stone.

Artificial oilstones

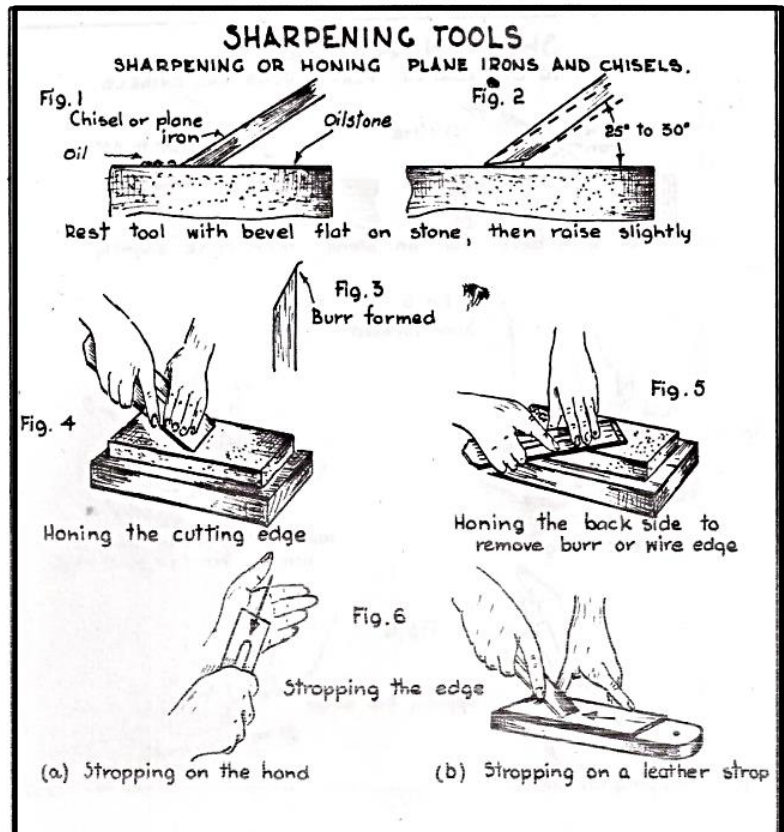
These are made from silicon carbide or aluminum oxide. Very fine machine oil or a mixture of machine oil and kerosene is used on the artificial stones as lubricant to prevent overheating.



Sharpening or Honing

The process of sharpening or honing must be carried out carefully to obtain a fine keen edge. Sprinkle the surface with the lubricant. This will reduce friction and carry away fine metal particles and prevent the stone being worn away during the sharpening process. Place the tool with the grinding bevel flat on the stone and lift slightly to make an angle of about 25 to 30 degrees. Move the tool steadily back and forth along the full length of the stone holding at the same angle until a slight burr appears on the face of the blade. To remove this, turn the blade over and keeping it absolutely flat, rub back and forth on the same stone until the burr is gone. If the sharpening bevel is blunt or damaged, it should be grounded at an angle of 20° to 25° before sharpening on the oilstone.

Sharpening Operation



Sharpening Saws

(a) Jointing and Topping

If the teeth are found to be uneven, it is necessary to joint the saw i.e. to run a file along the tops of the teeth until they are all of even height.

(b) Reshaping

Shaping becomes necessary if the teeth are irregular but not necessary after every time the saw is sharpened. Place the saw in the saw vice and file all the teeth, working from one side of the saw. File straight across with the file at right angles to the blade.



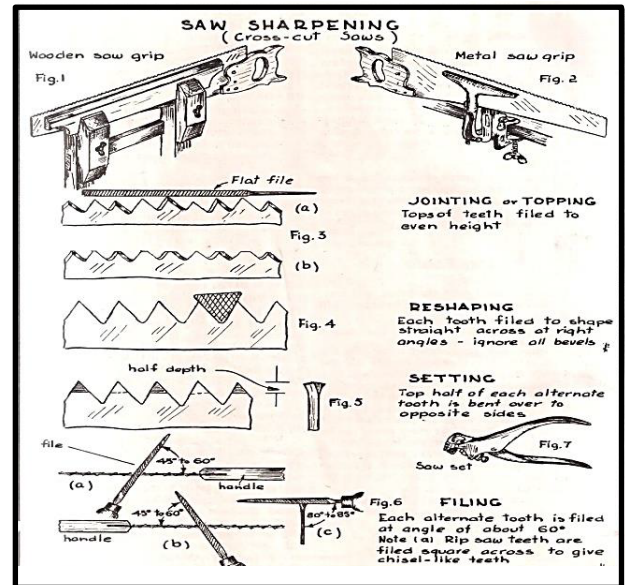
(c) Setting

Setting is always necessary after jointing and shaping.. It is the process of bending the adjacent teeth to the opposite sides so that the cut or the kerf made by the saw is slightly wider than the blade. Sawing gives clearance and prevents jamming when sawing. Only top third to a half of each tooth is to be bent.



Sharpening or filing

Use a taper saw file with the correct size. Fix the saw in a saw vice, handles to the right with the gullets about 4mm above the jaw. Place the file in the gullet in the left of the first tooth bent towards you. Swing the handle of the file to the left keeping the file at the same angle and parallel to the floor, file in each alternate gullet.

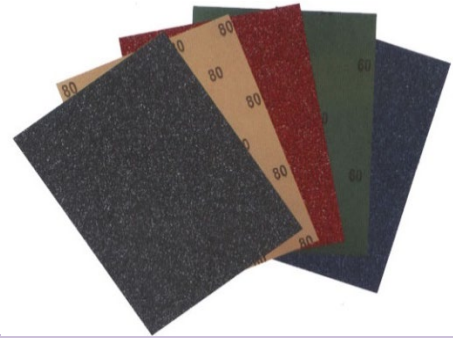


Activity

4. Why is oil used on the oilstone during the sharpening process?
5. What is the convenient angle for sharpening chisels?
6. Name the tool used for holding saws while sharpening?
7. Why is setting of saw teeth necessary?
8. Explain the term kerf?

Abrasives

Most of the finishing work is carried out using abrasives. Abrasives are used before applying any finish. There are many types of abrasives available but in this chapter we will look at the three common types.



Garnet

It is for hand and machine sanding and is made from natural crushed garnet stone for sanding wood, glass, provides sharp cutting edges, cuts quick, long lasting and does not build up heat that may burn finishing's or the wood. Excellent for cabinet makers in both soft and hardwoods and those who require a good smooth finish. As the grade number of the garnet increases, the paper becomes coarser. Garnet paper is available in sheets and also in rolls.



Glass Paper

Glass paper grit is made from crust glass and bottles. They are first crust graded and are glued on paper or cloths backing. The '00' or 'Flour' grade is still preferred by many uses for cutting back through the layers of polish. 'Worn' particle paper i.e. slightly clogged up, used for the final finishes although two papers rubbed together to remove the bite was also used. As the grade number increases from '00', it becomes coarser.



Backing material

The backing material is usually cloth or paper. The cloth variety is more expensive but lasts longer under hardy conditions. Probably used extensively in the motor trades and engineering rather than woodwork.

Open/close coat

Open coat is used chiefly for machine sanding. The open pore helps to prevent clogging. The speed of rotation provides a smoothing action.

Closed coat is for hand sanding papers.

Wet and Dry Paper

Wet and dry paper is abrasive material used for sanding metal paint and other hand finishing jobs. It can be used either as wet or dry hence the name and can be supplied in different varieties including full sheets down or even fine grits. As the grade of the paper increases, it becomes finer



How to use wet sandpaper?

Fill a bucket with lukewarm water. Prepare the surface of the vehicle by wiping all dust and debris off the painted surface with a wiping cloth.

Wet sandpaper or glass paper is a heavy paper with abrasive material attached to its surface. Sandpaper is part of the coated abrasives family of abrasive products. It is used to remove paint and clean up metal before finishing. Mostly used in car industry.



Activity

1. What are garnet papers made from?
2. Which abrasive paper is used in cleaning metals?
3. Which paper is used in car industry?

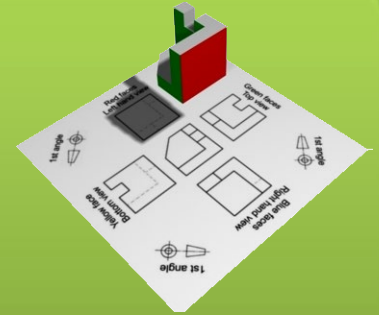
Chapter 5

STRAND: 5 GEOMETRICAL DRAWING

Outcome

After studying this chapter, students will:

- Recognize and develop skills in orthographic projection.
- Acquire the concepts of 3rd angle projection.
- Produce 3rd angle projections of simple solids.
- Recognize and develop skills in making simple pictorial drawings.
- Acquire the concepts of pictorial drawings.
- Convert orthographic projections into pictorial drawings.
- Identify the different types of prisms and pyramids (limited to 6 sides).
- Develop skills in geometrical drawing of prisms, pyramids, cylinder and cone.
- Produce the full surface developments of prisms, pyramids, cylinder and cone.



Introduction

Engineering drawings are used to indicate the shape and the size of an object. All objects have three dimensions, i.e. length, width and height. Usually an object is represented in a pictorial projection (isometric, oblique and perspective). Problems associated with conveying an impression of the object and associated dimensioning can be overcome with orthographic projection.

Four different methods are commonly used for representing drawings. The first and the most important of the four methods is the orthographic projection.

Orthographic projections enable us to see the objects in its real shape with all correct angles. Orthographic projection looks at the true shape of the object in each view. This means that each view is seen in its real form. Each view is also seen in line with the next either sideways or on top.

This is the international language of drawings used by engineers, architects, designers, draughtspersons and all others involved in the field of technology around the world. Pictorial views such as the **isometric**, **oblique** and **perspective** drawings are used to express the natural look of the solids, objects, etc., as they show the three different views in one.

Pictorial Drawings show the three views together. Orthographic Projection shows the same three views separately but linked together. The FRONT ELEVATION and the END ELEVATIONS are views of the VERTICAL PLANE. The PLAN is a view seen in the HORIZONTAL PLANE and is in line with the Front Elevation.

Solid geometry is concerned with showing the orthographic views of an object together with development of its surface showing how it was formed or the outline from which it was made.

ORTHOGRAPHIC PROJECTION

The term ORTHOGRAPHIC PROJECTION means true or correct shape. This is the main reason why orthographic projection has been adopted by the building, engineering and technology fields.

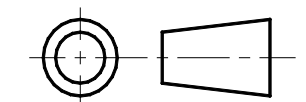
Since ORTHOGRAPHIC PROJECTION is the most important method used in Technical Drawing, it must be mastered thoroughly. Orthographic Projection has two methods which generally gives us the same information.

The two methods of Orthographic Projection are the FIRST ANGLE PROJECTION and the THIRD ANGLE PROJECTION. The First Angle Projection was developed by the British while the Third Angle Projection was developed by the Americans and has now developed to be the standard method used throughout the world.

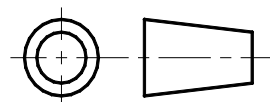
During the Second World War when the Americans were producing weapons for the Europe and the British, engineering drawings were being exchanged. There was a clash with regards to the two standards being used. It was realised back then that it did not really matter which standard was to be adopted as long as the person concerned understood them. That is why it is called the Universal Language.

The above led to the formation of the two conventions. The first and the third angle projections are known as the conventions. Conventions in drawings are used so that anyone could read and understand it.

It must be understood that both the methods do the same job. Each of the convention was given a symbol. Below are the two symbols that are being used by all to give clarity to their drawings.

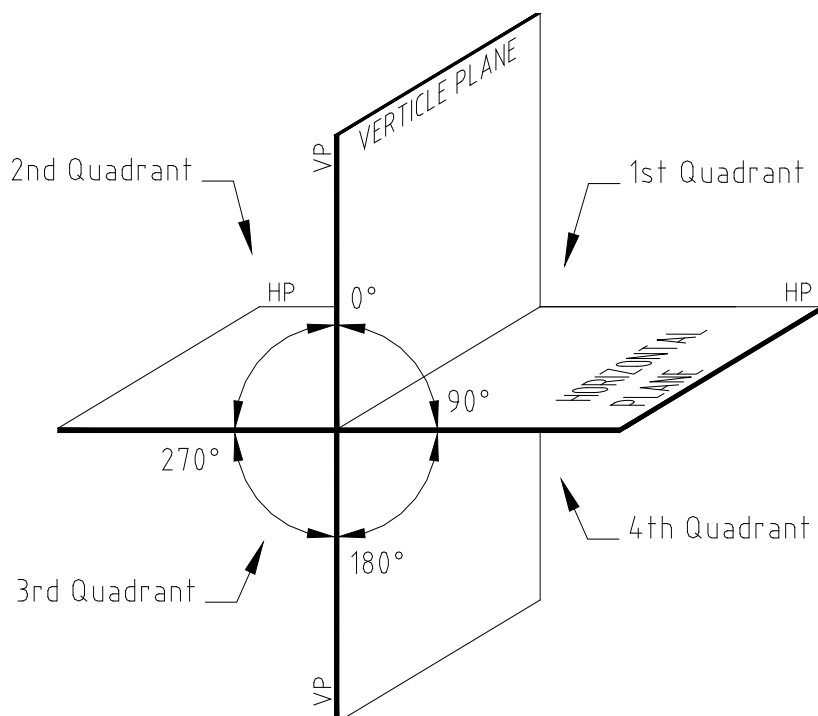


THIRD ANGLE PROJECTION



FIRST ANGLE PROJECTION

ORTHOGRAPHIC PROJECTIONS BOX



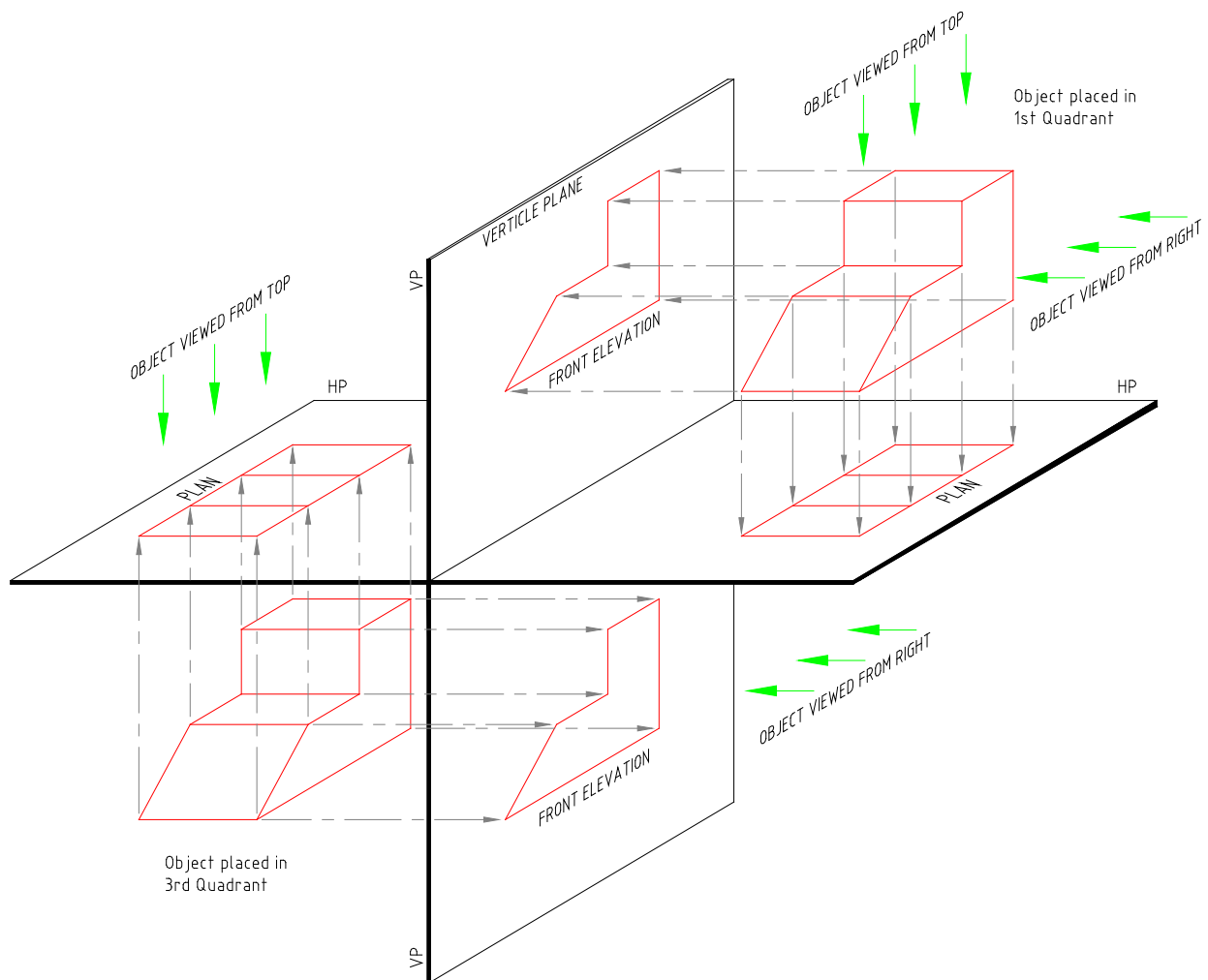
The 1st Quadrant 0° - 90° is the First Angle. The 3rd Quadrant 180° - 270° is the Third Angle.

The common question asked at this stage is "What about the 2nd and 4th Quadrants?"

If the above drawing is studied carefully, then it is noticed that for the 2nd Quadrant, the Front Elevation will be the same as that for the 1st Quadrant and the Plan would be the same as that for the 3rd Quadrant. It is the opposite in the 4th Quadrant. Therefore the 2nd and the 4th Quadrants are not used.

In the 1st Angle Projection, the object to be viewed is placed in the 1st Quadrant and viewed from the right and the top. The Front Elevation is seen on the V.P. and the Plan is on the H.P.

In the 3rd Angle Projection, the object to be viewed is placed in the 3rd Quadrant and again viewed from the right and the top. The Front Elevation is seen on the V.P. and the Plan is on the H.P.

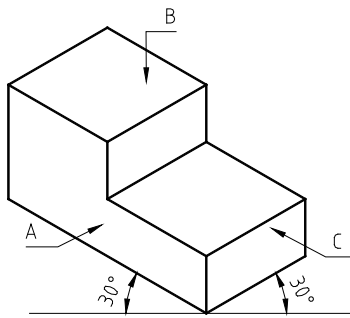


Note that in both the cases, the object is viewed from the right and the top. In the first angle projection, the object is viewed from one side and drawn on the other side as is the case with X-rays in the hospitals. While in the third angle projection, the object is viewed and drawn on the same side as is the case with the photos taken from a camera.

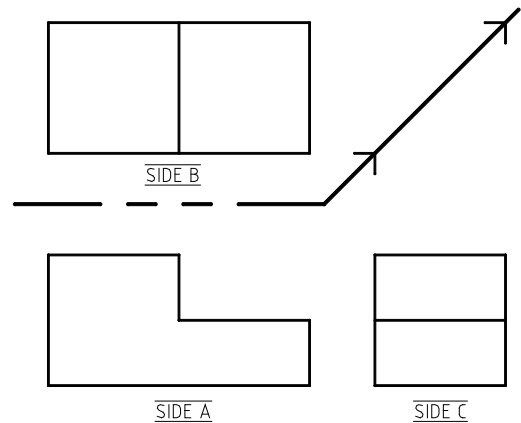
3rd ANGLE ORTHOGRAPHIC PROJECTION

Orthographic projection is most commonly used for representing drawings.

The simple shaped block shown below will be used in understanding orthographic projection.



Shown below are three drawings that represent the three views of the simple shaped block but seen from three different positions. Each view is seen perpendicular to or at 90° to each of the surfaces. The three views are seen from the top, the front and the side.

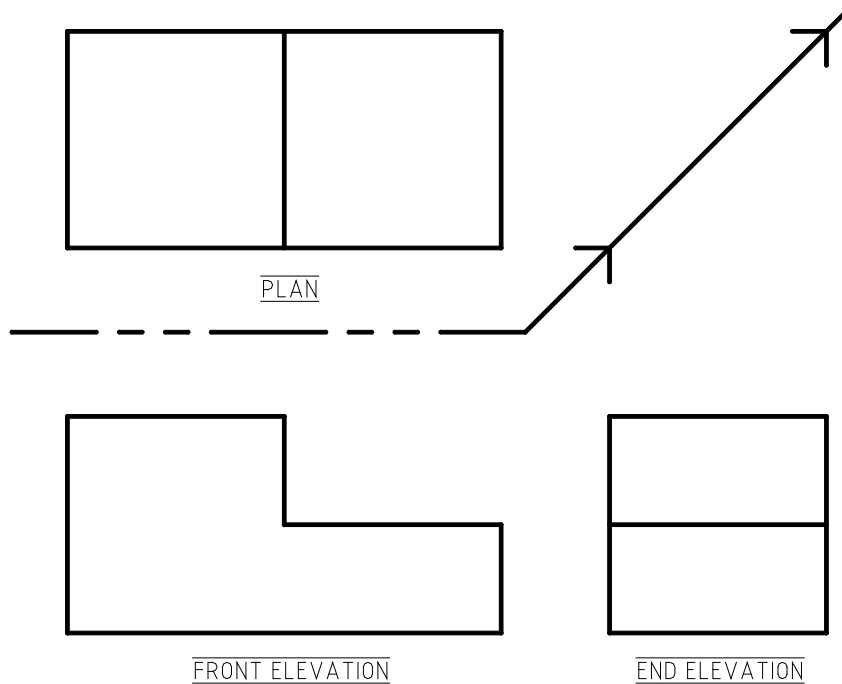


Orthographic projection looks at the real form or the true shape of the simple shaped block in each view. Each view is in line with the next, either sideways or on the top.

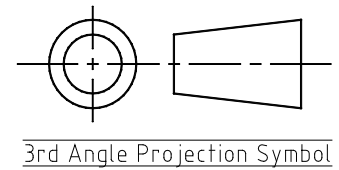
Orthographic Drawing is the international language of drawings used by engineers, architects, designers, draughtspersons and all others involved in the field of technology around the world.

The orthographic projection shown below shows the correct names of each of the views. View A is known as the **front elevation**. It looks at the drawing from the front and gives us the most information. It also gives us the length and the height of the shaped block. View B is known as the **plan**. It looks at the drawing from the top and gives us the length and the width of the shaped block. View C is known as the **end elevation**. It looks at the drawing from the end of the shaped block and gives us the width and the height of the solid.

Each solid has two end elevations but we will draw only one of the end elevations.



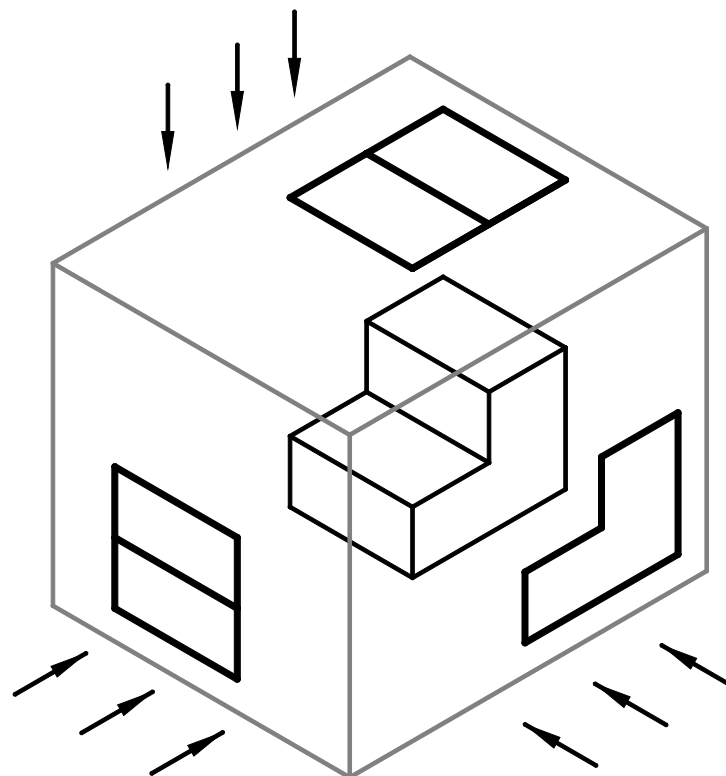
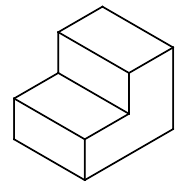
The symbol shown below is used to represent 3rd Angle Orthographic Projections and Drawings.



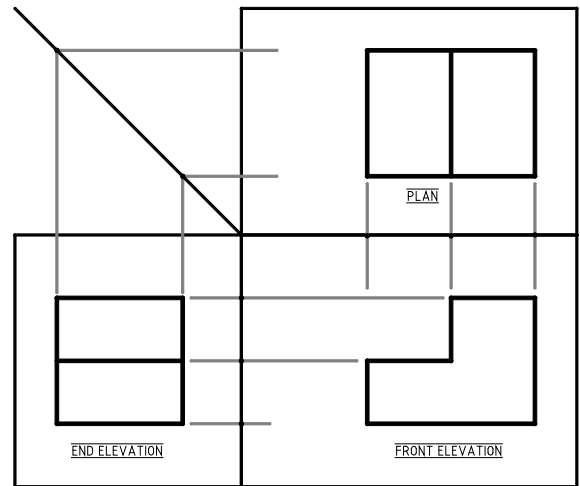
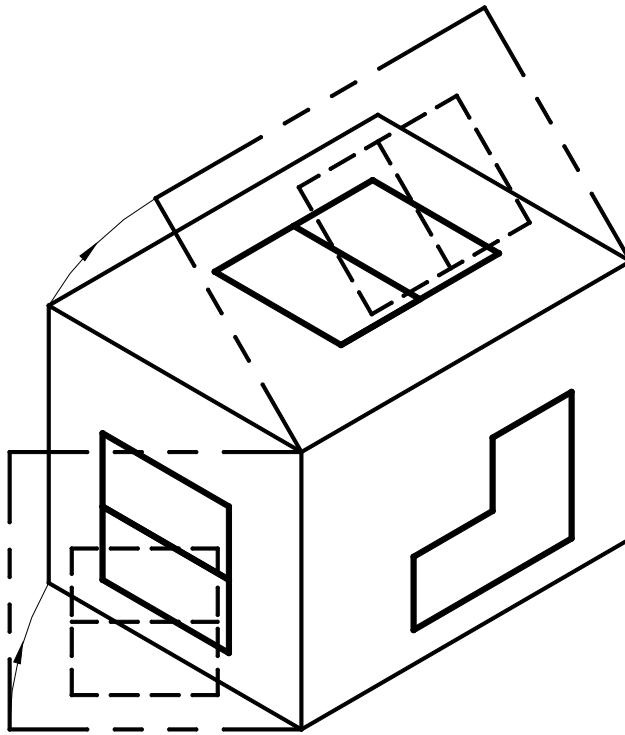
3rd ANGLE ORTHOGRAPHIC PROJECTION

An L - shaped block is used for understanding 3rd angle projection in which the object is viewed and drawn on the same side.

Note: In orthographic projections, the longer side (length) of any object will be taken in the front elevation while the shorter side (width) will be taken in the end elevation.



Note: When the box is opened, the plan and the end elevation are in line with the front elevation. The plan is drawn in line & above the front elevation and the end elevation drawn in line but on the side of the front elevation where it is viewed from.



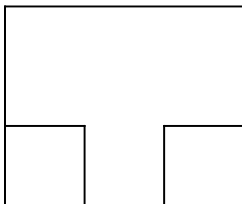
Shown above is the 3rd Angle Orthographic projection of the shaped block.

Activity

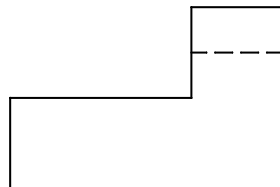
Match the elevations drawn below with its corresponding pictorial views on the right. Write the answers in the boxes provided next to each of the pictorial view. No: 2 has been done as an example.

Orthographic Projections of Simple Shaped Blocks

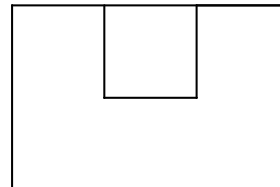
1



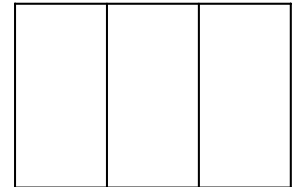
2



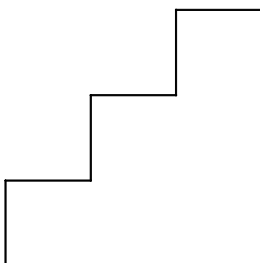
3



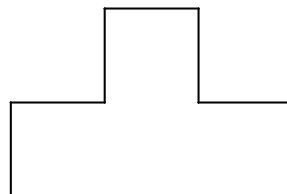
4



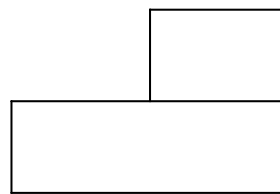
5



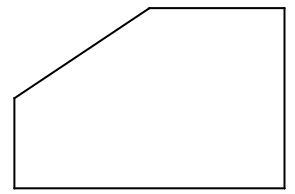
6



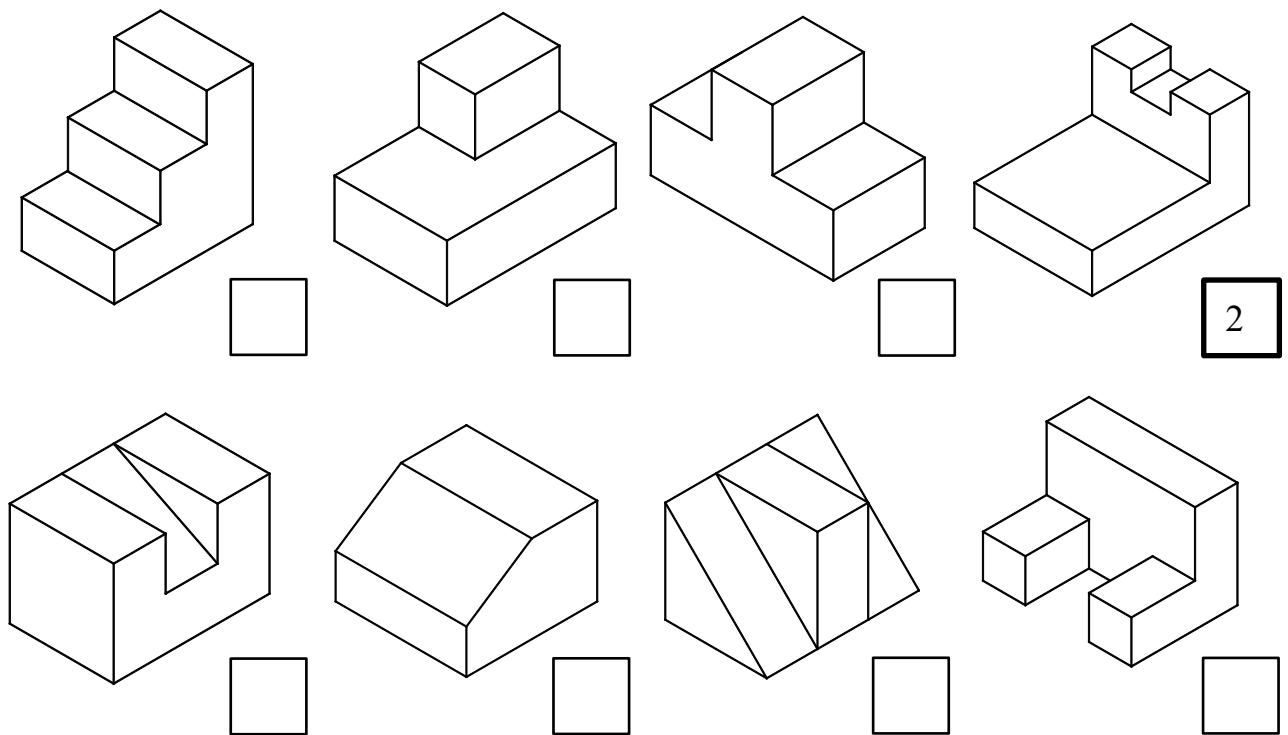
7



8



Pictorial Views of Simple Shaped Blocks

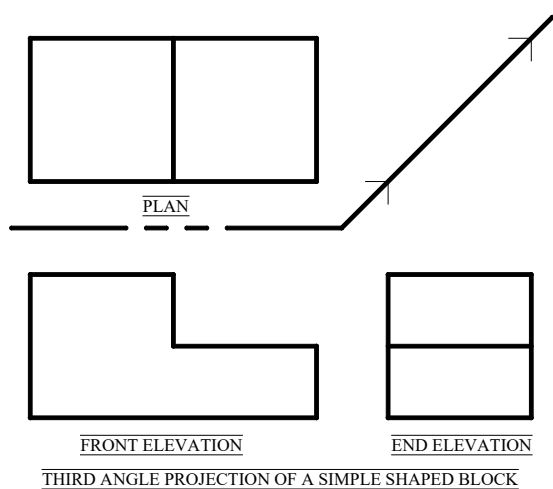


ISOMETRIC VIEWS AND DRAWINGS

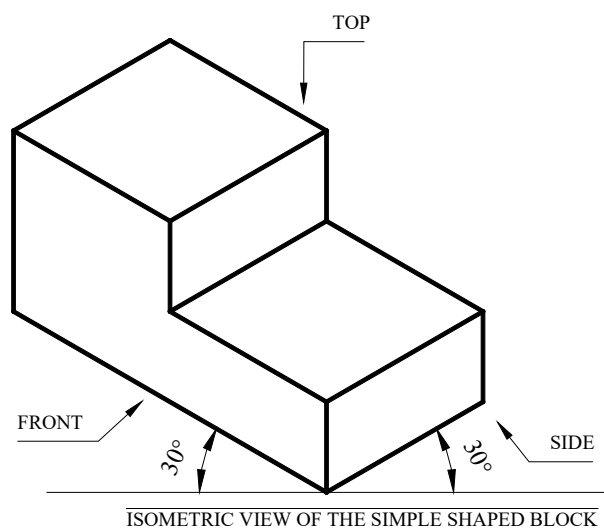
ISOMETRIC DRAWING

Is the most natural looking drawing and it is easy to produce.

The orthographic projection shown below is of a simple shaped block used to understand and draw an isometric view.



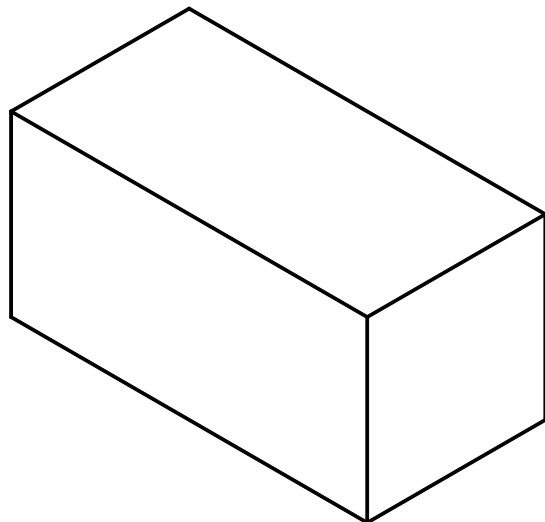
The isometric view shown below is drawn from the orthographic views of the simple shaped block.



Isometric drawing helps in understanding the shape of a shaped block. Isometric drawings are drawn using the 30°/60° set square.

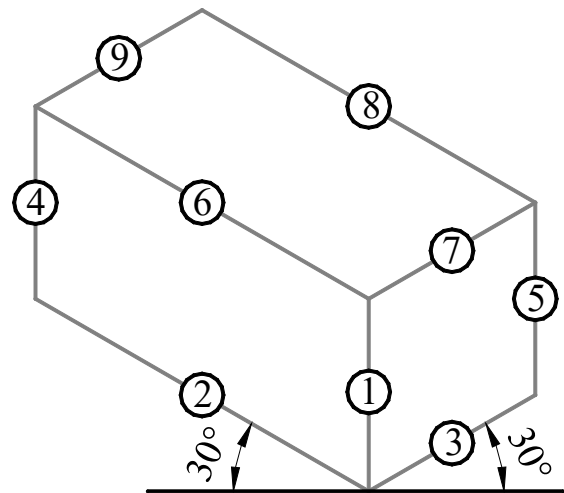
The length and width on the isometric drawing is drawn at 30° and the height is drawn vertically at 90° to the horizontal.

The isometric crate shown below is used as a guide to draw the actual shape. The isometric crate is drawn using construction lines.



ISOMETRIC

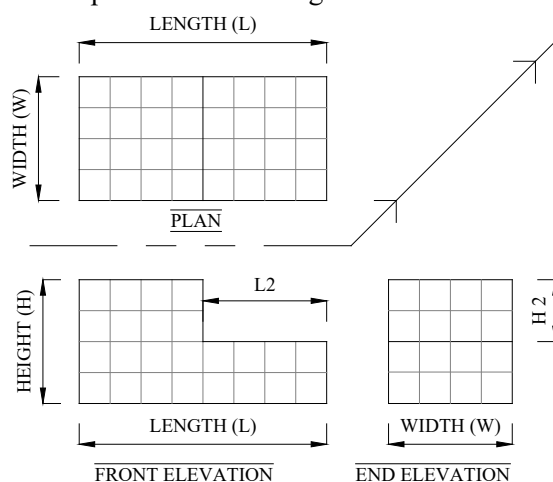
The isometric box below illustrates the order in which the 9 lines are drawn. Note that three of the lines represent the length, three represent the width and three represent the height of the shaped block.



ISOMETRIC CRATE - CONSTRUCTION ORDER

CONVERTING ORTHOGRAPHIC DRAWING TO ISOMETRIC VIEW USING GRIDS

The front elevation of an orthographic drawing shows the length & height of the shaped block. The end elevation shows the width & height and the plan shows the length & width.

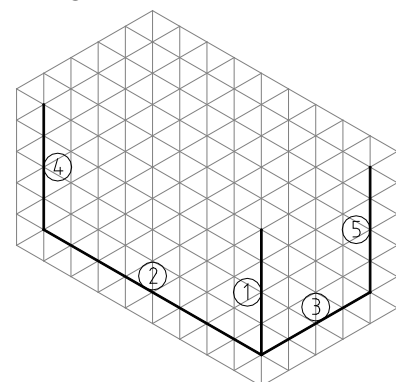


3RD ANGLE ORTHOGRAPHIC VIEW OF A SIMPLE SHAPED BLOCK

Construction of the Isometric Crate

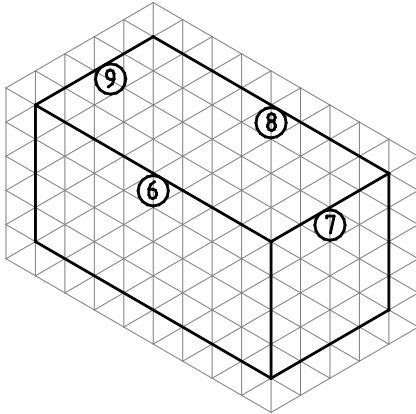
First the height line is drawn vertically. Then the length and width of the isometric crate are drawn.

The other 2 height lines are added as well.



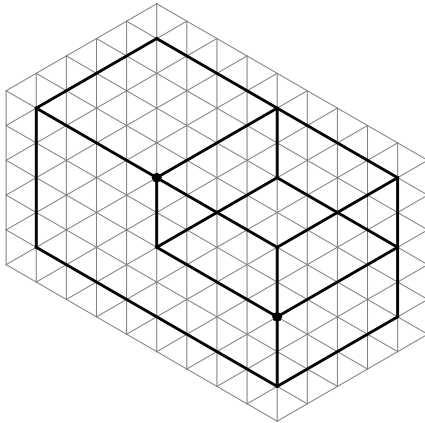
Construction of the Isometric Crate

The other length and width lines are then drawn to complete the isometric crate.



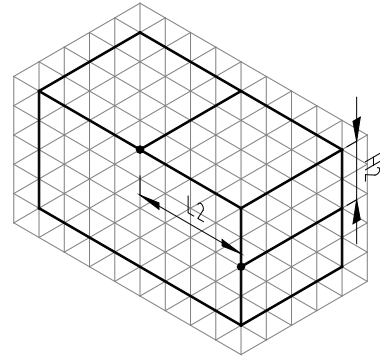
Completing the shape

The remaining lines are drawn to complete the shape.



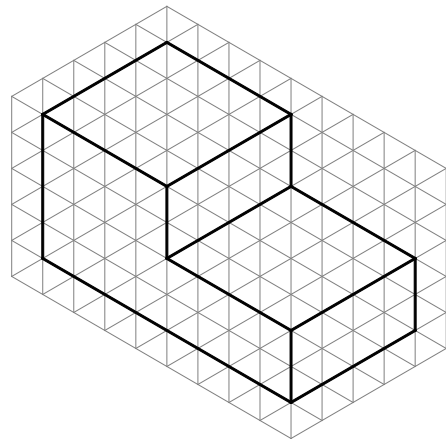
Completing the shape

To get the shape of the object, the length (L2) and height (H2) are taken from the front & end elevation and drawn by counting the number of grids.



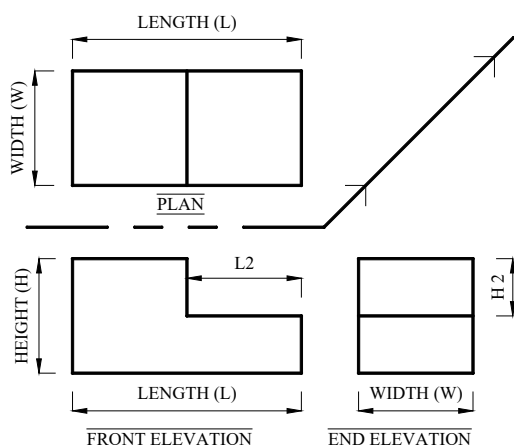
Completing the shape

The shape is completed by drawing the OUTLINES.



CONVERTING ORTHOGRAPHIC DRAWING TO ISOMETRIC VIEW

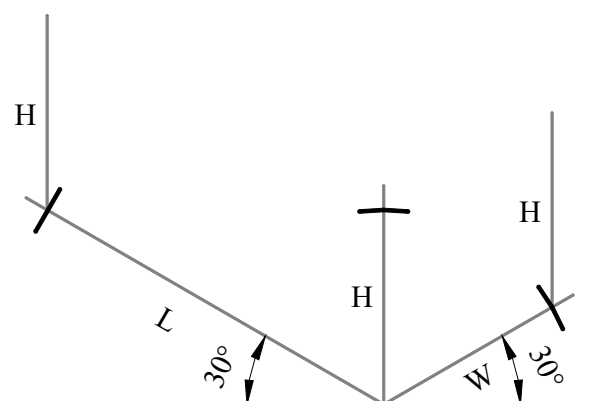
The front elevation of an orthographic drawing shows the length & height of the shaped block, the end elevation shows the width & height and the plan shows the length & width.



3RD ANGLE ORTHOGRAPHIC VIEW OF A SIMPLE SHAPED BLOCK

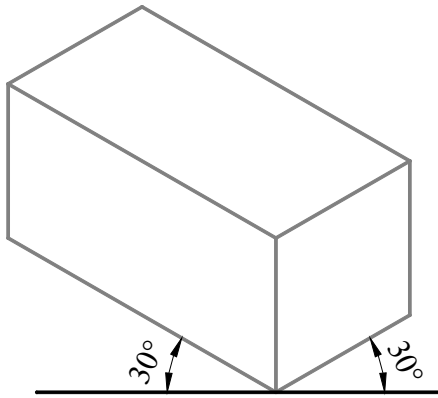
Construction of the Isometric Crate

Draw the isometric axis by drawing the height at 90° from the base line, the length and width at 30° to the base line. Mark the length, width and height.



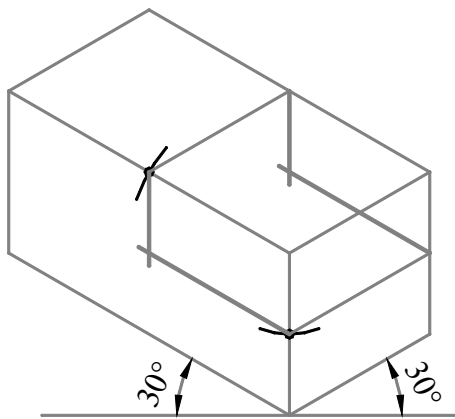
Construction of the Isometric Crate

Complete the isometric crate by drawing lines parallel to the length, width and height.



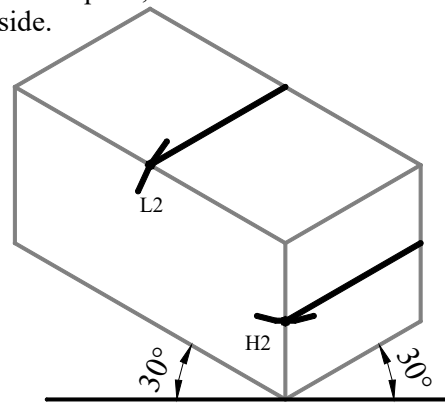
Completing the shape

Using the 30°/60° set square, draw the remaining lines parallel to the length, width and slightly longer than L2 and H2. Complete the shape.



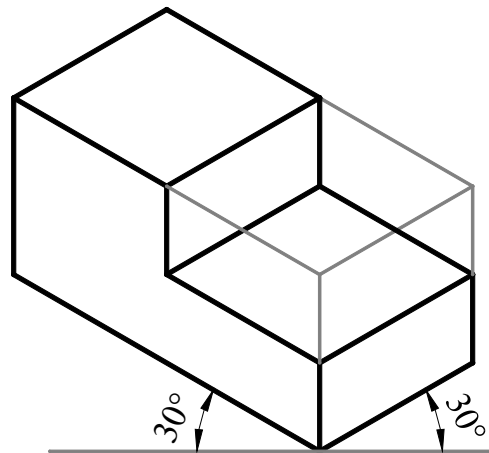
Completing the shape

Using a compass, measure length (L2) and height (H2) and transfer to the isometric box. Using the 30°/60° set square, draw 30° lines to touch the other side.



Completing the shape

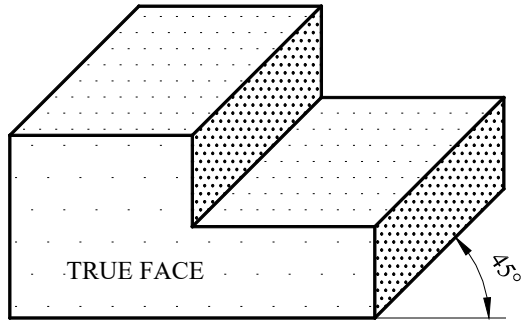
The shape is completed by drawing the OUTLINES.



OBLIQUE VIEWS AND DRAWING

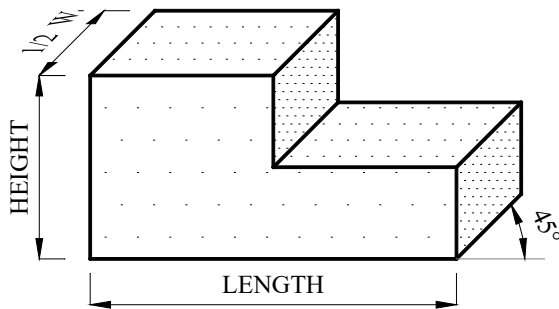
OBLIQUE drawing is another form of natural - looking drawing. Oblique drawings are also used in understanding the shape of a shaped block. The front view of an oblique drawing is the same as the front elevation in the orthographic drawing.

A shaped block shown below is used in understanding and drawing oblique views.

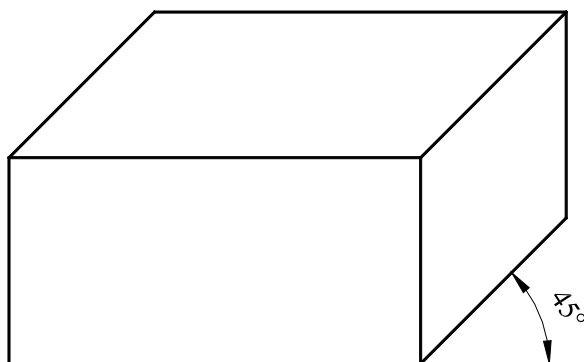


OBLIQUE DRAWING

To overcome the problem of shapes being distorted, CABINET oblique drawing shown below, drawn with the receding lines reduced by half.



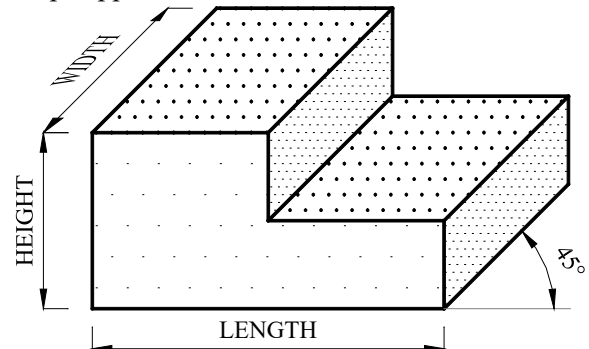
CABINET OBLIQUE



OBLIQUE BOX

Types of oblique drawings

CAVALIER oblique drawing shown below, is drawn using the full scale for all axes but the shape appears distorted.

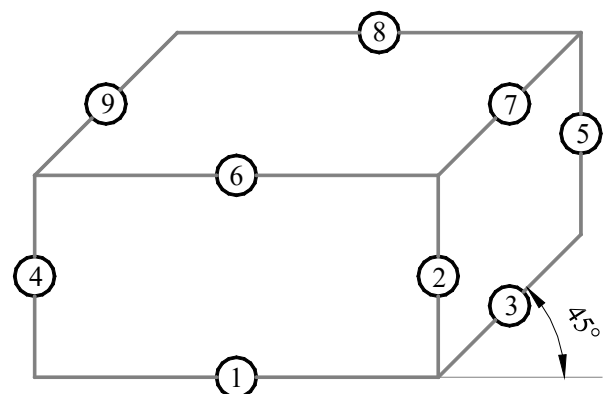


CAVALIER OBLIQUE

The length is drawn parallel, the height vertically or at 90° and the width on the receding lines at an angle of 45° to the horizontal.

The 45° set square is used for drawing oblique views.

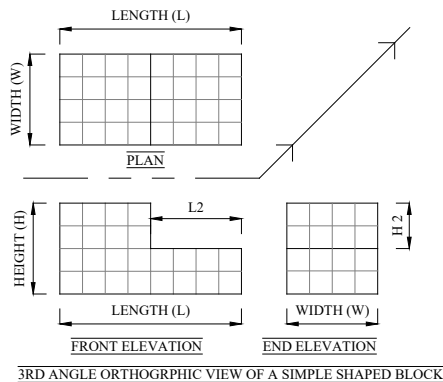
The oblique box shown below is used as a guide to draw the actual shape. It is drawn using construction lines and it also illustrates the order in which the 9 lines are drawn. Note that three of the lines are representing the length, three representing the width and three representing the height of the shaped block.



CONSTRUCTION STEPS

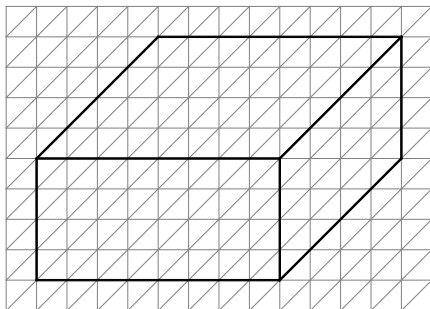
CONVERTING ORTHOGRAPHIC DRAWING TO OBLIQUE VIEW USING GRIDS

The front elevation of an orthographic drawing shows the length & height of the shaped block, the end elevation shows the width & height and the plan shows the length & width.



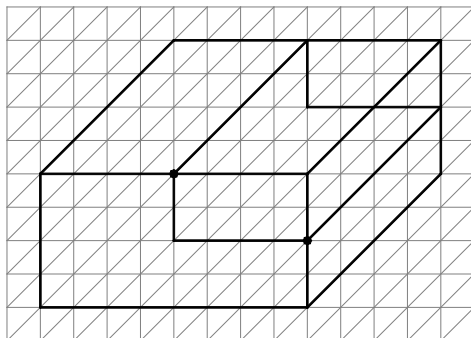
Construction of the OBLIQUE Box

The other length and width lines are then drawn to complete the oblique box.



Completing the shape

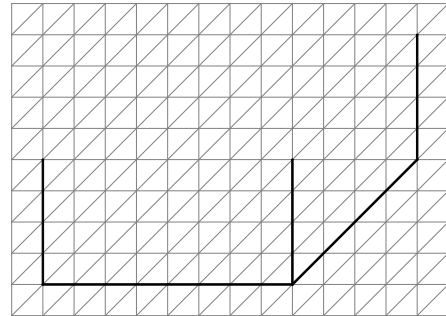
The remaining lines are drawn to complete the shape.



Construction of the OBLIQUE Box

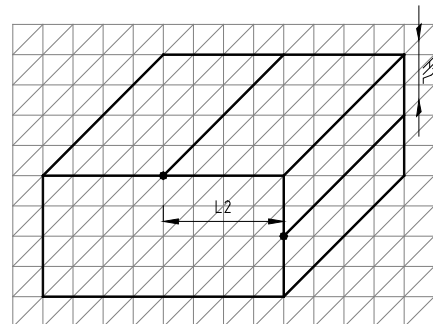
First, the height line is drawn vertically then the length horizontally and the width receding at an angle of 45° to the horizontal.

The other 2 height lines are added as well.



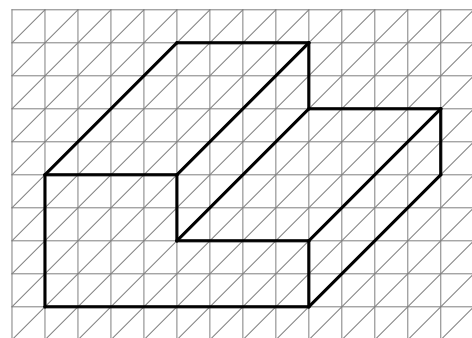
Completing the shape

To get the shape of the object the length (L2) and height (H2) are taken from the front and end elevation and drawn by counting the number of grids.



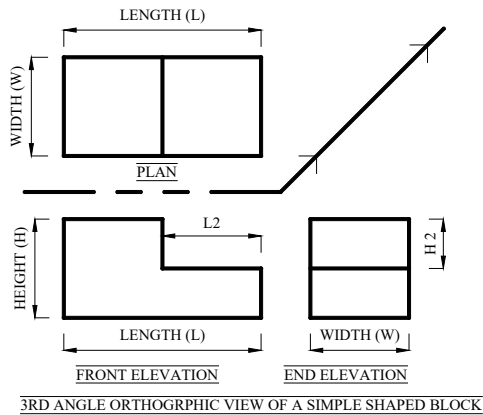
Completing the shape

The shape is completed by drawing the OUTLINES.



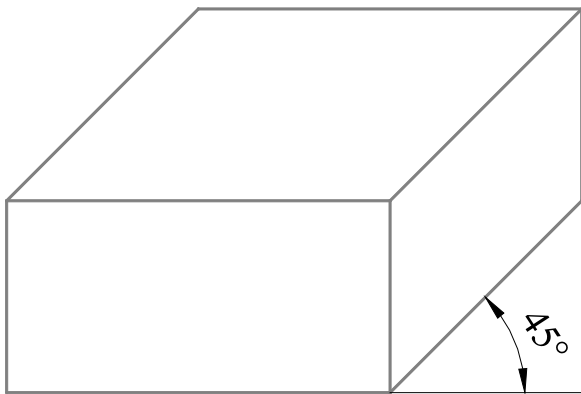
CONVERTING ORTHOGRAPHIC DRAWING TO OBLIQUE VIEW

The front elevation of an orthographic drawing shows the length & height of the shaped block, the end elevation shows the width & height and the plan shows the length & width.



Construction of the OBLIQUE Box

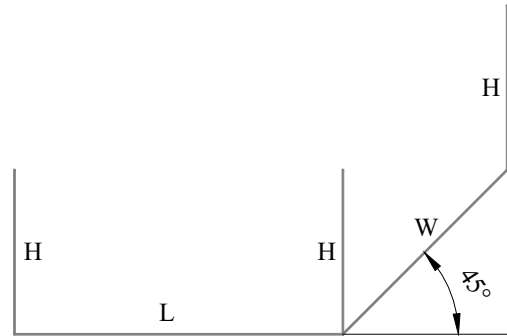
Complete the oblique box by drawing lines parallel to the length, width and height.



Construction of the OBLIQUE Box

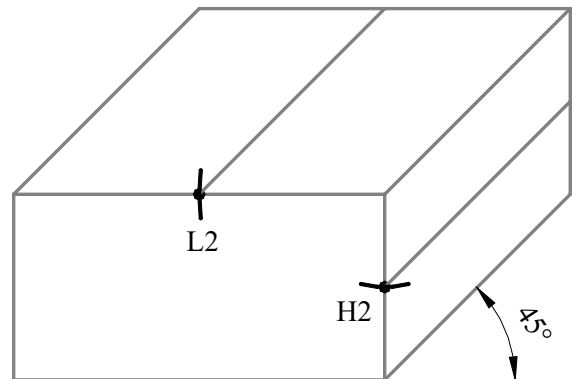
First, the height line is drawn vertically then the length drawn horizontally and the width receding at an angle of 45° to the horizontal.

The other 2 height lines are added as well.



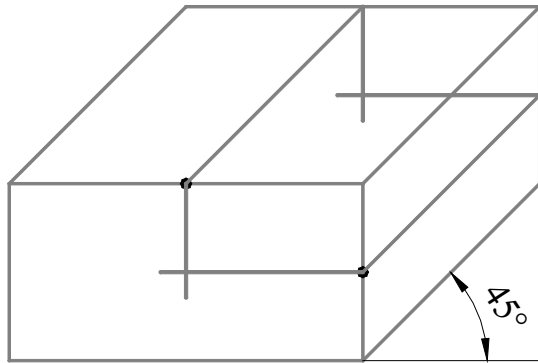
Completing the shape

Using a compass, measure length (L2) and height (H2) and transfer to the oblique box. Using the 45° set square, draw 45° lines to touch the other side.



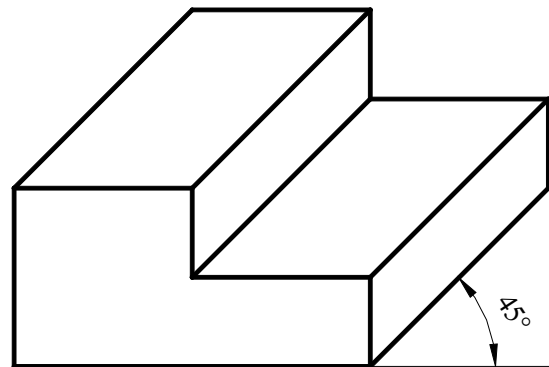
Completing the shape

Using the set squares, draw the remaining lines parallel to the length and width slightly longer than L2 and H2 and complete the shape.



Completing the shape

The shape is completed by drawing the OUTLINES.

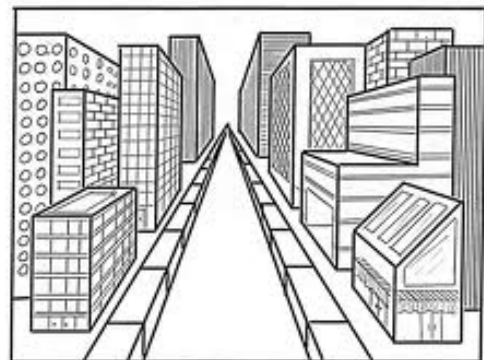


PERSPECTIVE VIEWS AND DRAWINGS

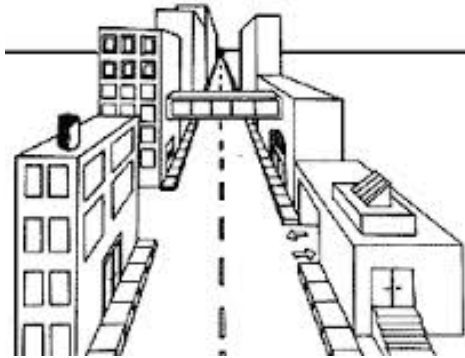
Perspective views or drawings are used to represent objects as they would look like to the viewer when standing at a particular position. One point perspective drawing is a form of natural - looking pictorial drawing.



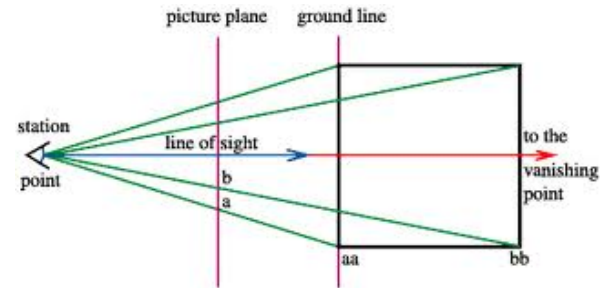
Perspective views or drawings can be used with detailed drawings where a more accurate illustration or creative effect is required than that achieved by means of working drawing.



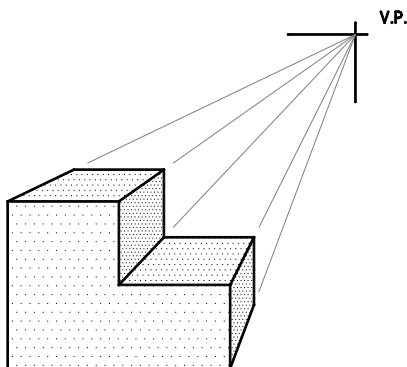
Lines appear to converge to a point on the horizon known as the vanishing point (VP). All parallel lines which are on the ground or parallel to the ground will have the same vanishing point on the horizon.



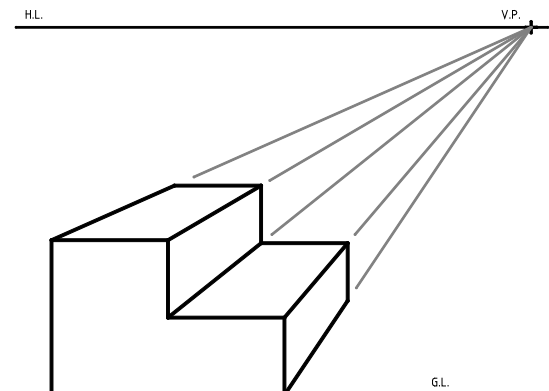
Objects drawn in perspective appear to recede towards the back and converge or become closer together.



Note that the object drawn in perspective shows its front view in full size. The other views will converge and appear to vanish towards a point known as the VP.



When drawing an object in the perspective view, it is good practice to first draw the front elevation.

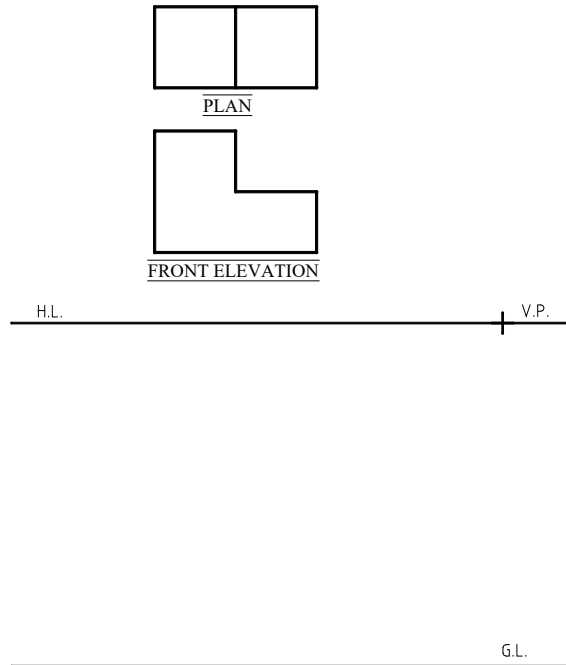


CONVERTING ORTHOGRAPHIC DRAWING TO PERSPECTIVE DRAWING

Example 1

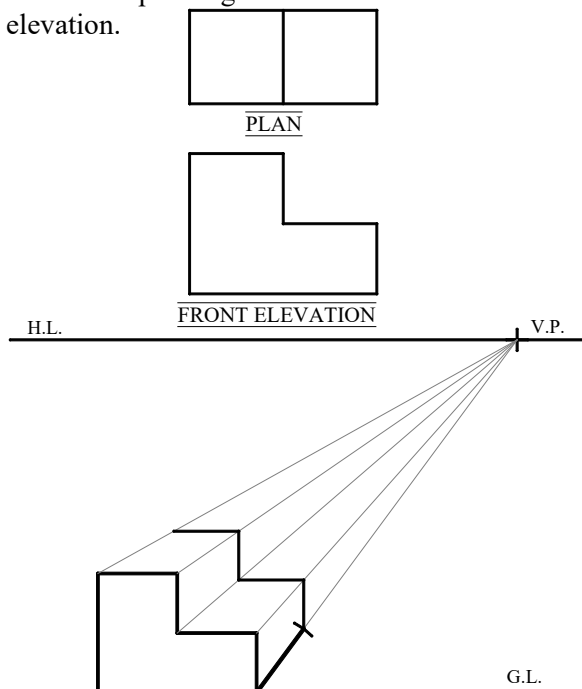
Given: The plan and front elevation of a shaped block, Ground Line (GL), Horizon line (HL) and a Vanishing Point (VP).

Required: Draw the shaped block in one point perspective view



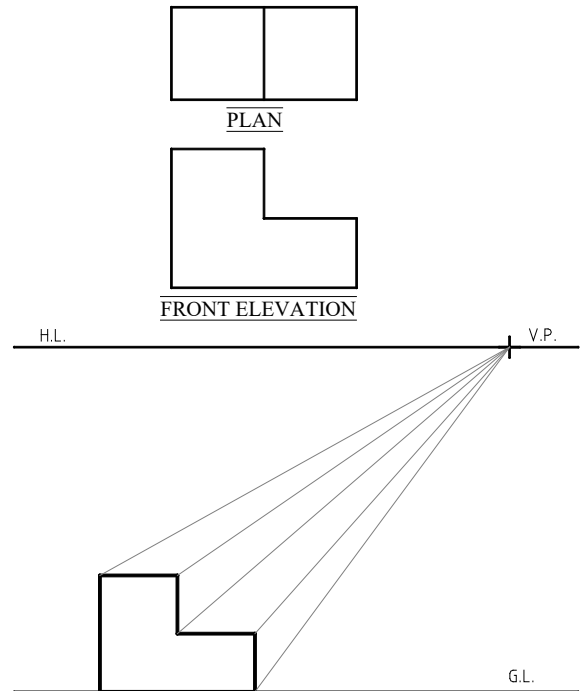
Step 3: Measure the width of the object and mark it on the visual ray drawn from the bottom of the elevation

Step 4: Using this point draw lines parallel to their corresponding sides from the front elevation.

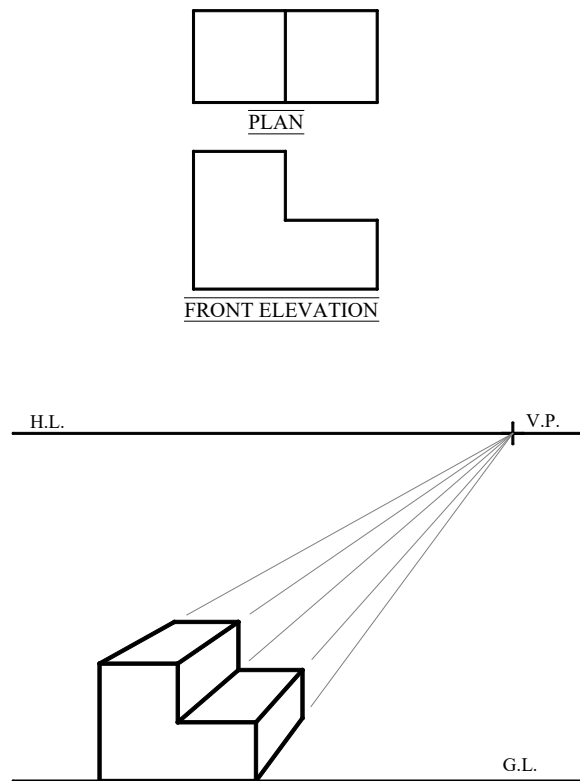


Step 1: Draw the front elevation on the opposite side of the vanishing point

Step 2: Draw visual rays from the corners of the front elevation to the vanishing point



Step 5: Complete the shape by drawing the OUTLINES.

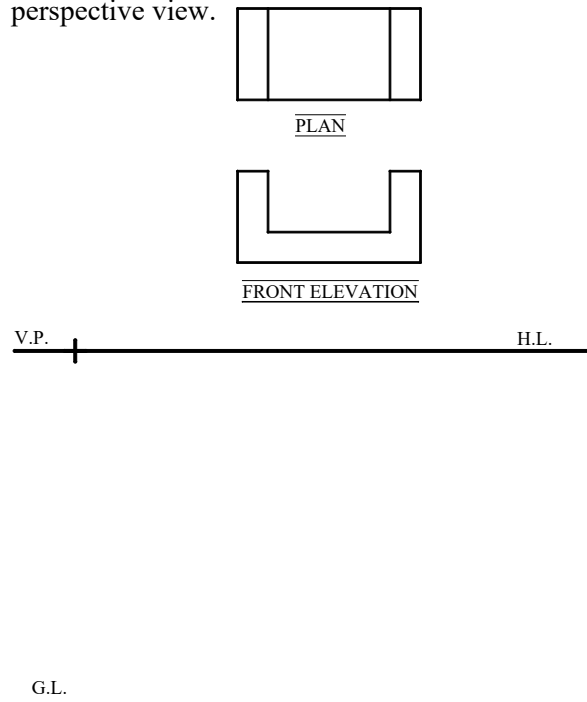


CONVERTING ORTHOGRAPHIC DRAWING TO PERSPECTIVE DRAWING

Example 2

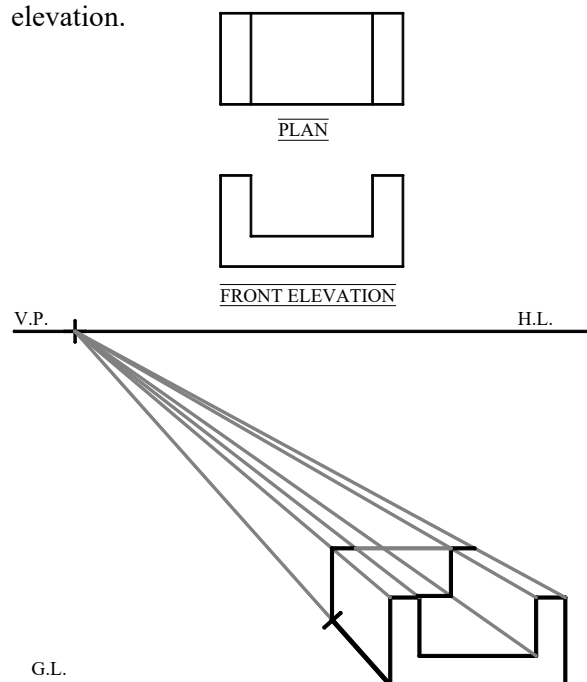
Given: The plan and front elevation of a shaped block, Ground Line (GL), Horizon line (HL) and a Vanishing Point (VP).

Required: Draw the shaped block in one point perspective view.



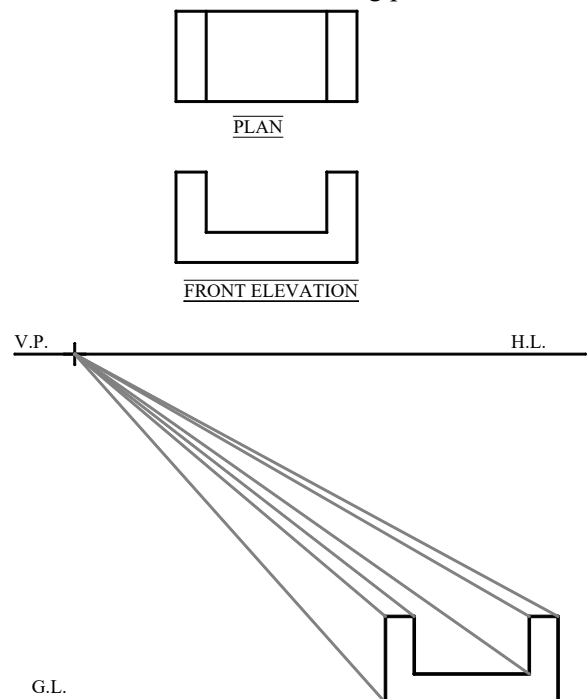
Step 3: Measure the width of the object and mark it on the visual ray drawn from the bottom of the elevation.

Step 4: Using this point, draw lines parallel to their corresponding sides from the front elevation.

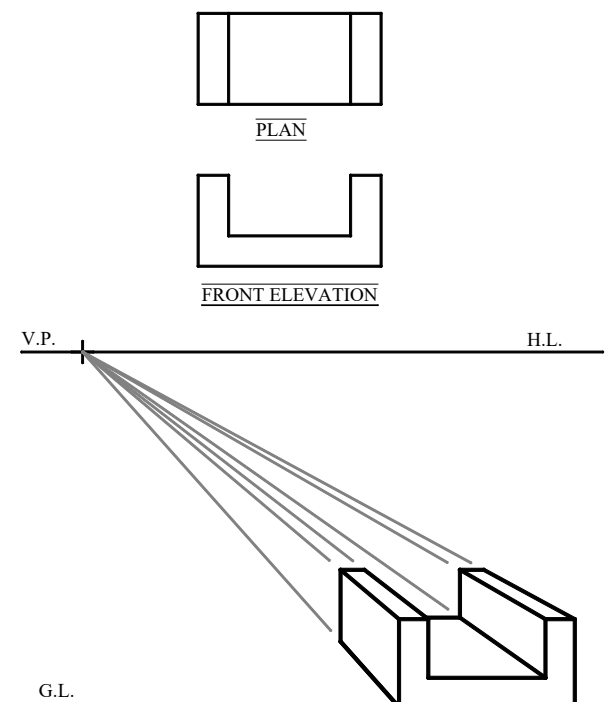


Step 1: Draw the front elevation on the opposite side of the vanishing point.

Step 2: Draw visual rays from the corners of the front elevation to the vanishing point.



Step 5: Complete the shape by drawing the OUTLINES.



Solid Geometry

Solid Geometry is the geometry of three-dimensional objects or solids.

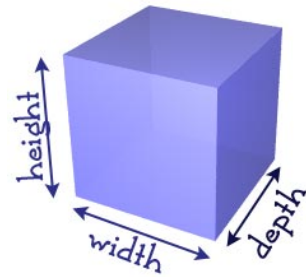
It is a basic part of technical drawing. In order to be able to apply it in solving problems, it is necessary to understand geometrical solids and their constructions.



Three Dimensions

It is called **three-dimensional**, or **3D** because there are three dimensions on the solids:

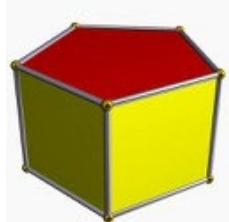
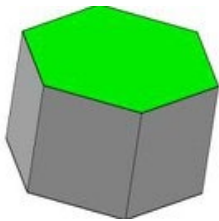
1. depth / length
2. width
3. height



Properties

Solids have *properties* (special things about them), such as:

- **Volume** (think of how much water it could hold).
- **Surface area** (think of the area that has to be painted).
- The number of vertices (corner points), faces and edges they have.



There are two types of commonly used solids. The first type has all its sides parallel to each other and is known as **Prisms**. The second type has its sides converging towards a point and they are known as **Pyramids**. Both the prisms and the pyramids are named from their bases and their sides.

Prisms

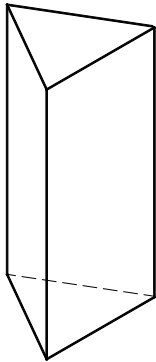
A right regular prism is a solid whose sides consist of equal rectangles and two equal ends with its axis perpendicular to its base.

Pyramids

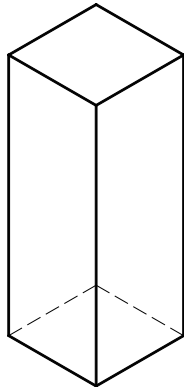
A right regular pyramid is a solid whose sides consist of equal isosceles triangles meeting at a point above the base called the apex. The axis of the pyramid is perpendicular to its base.

COMMON GEOMETRICAL SOLIDS

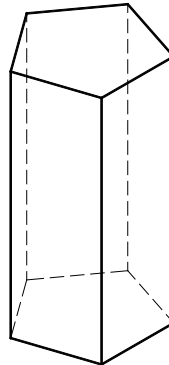
Prisms



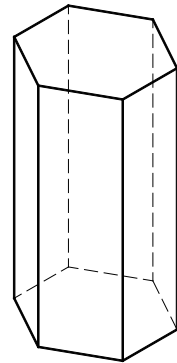
TRIANGULAR
PRISM



SQUARE
PRISM

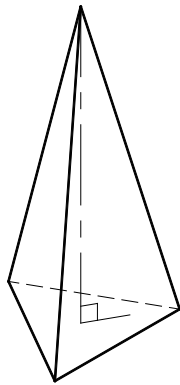


PENTAGONAL
PRISM

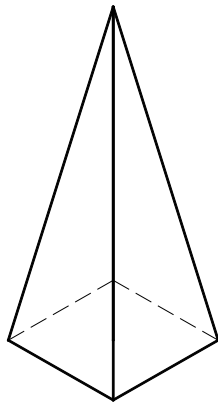


HEXAGONAL
PRISM

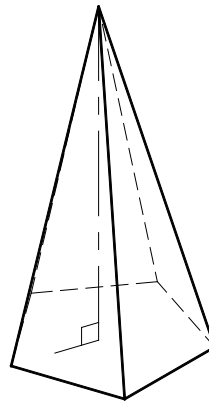
Pyramids



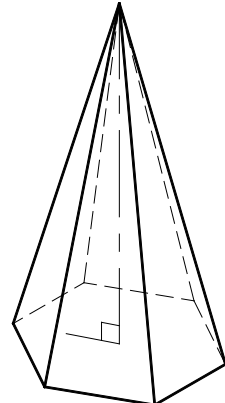
TRIANGULAR
PYRAMID



SQUARE
PYRAMID



PENTAGONAL
PYRAMID



HEXAGONAL
PYRAMID

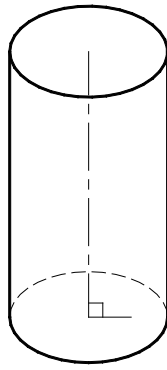
Cylinders and Cones - Solids with curved surfaces

Cylinders and cones are a special case of prisms and pyramids. Their sides are curved.

Cylinder

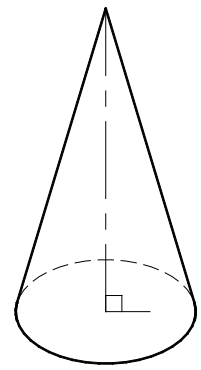
Cone

A right cylinder is a solid generated by rotating a rectangle about one of its sides. The fixed side becomes the axis. The right cylinder has two circular ends that are parallel to each other.



CYLINDER

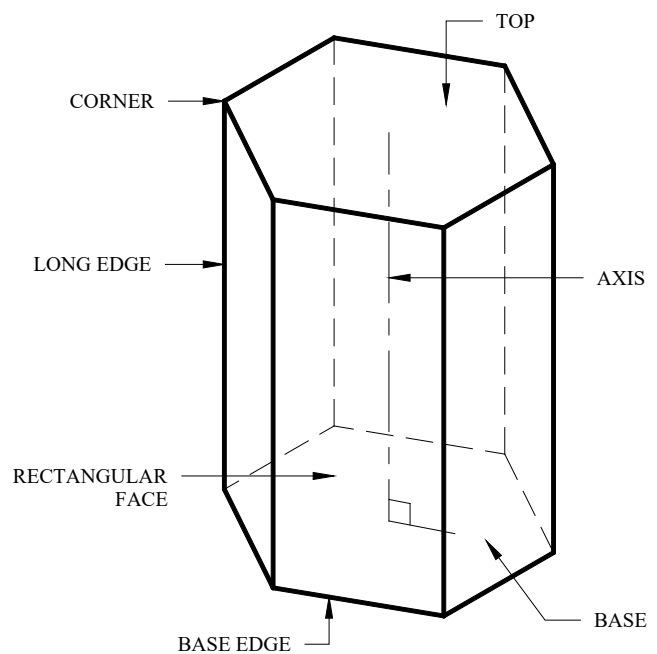
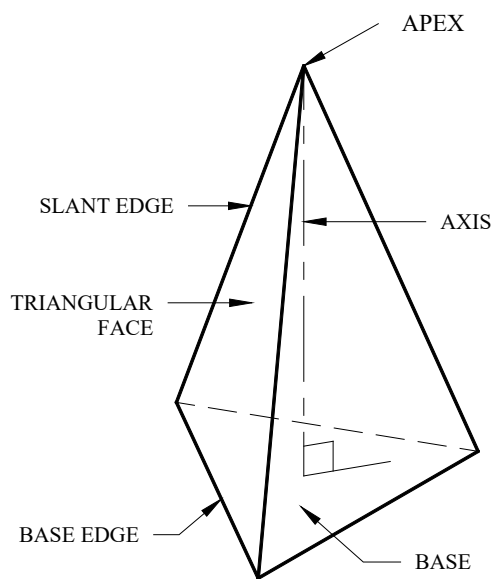
A right cone is a solid generated by rotating a right angled triangle about its perpendicular side. The right cone has a circular base at right angles to the axis. The point where the sides meet is called the apex.

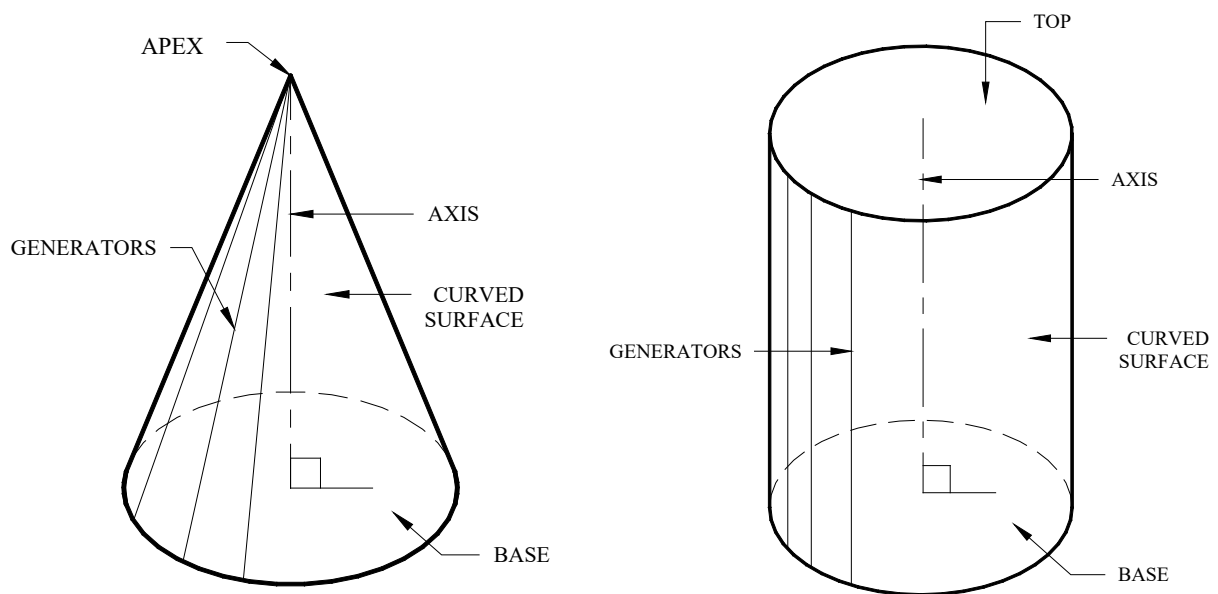


CONE

GEOMETRICAL TERMS

The following diagrams show the terms associated with the geometrical solids and their location or position on the solids.

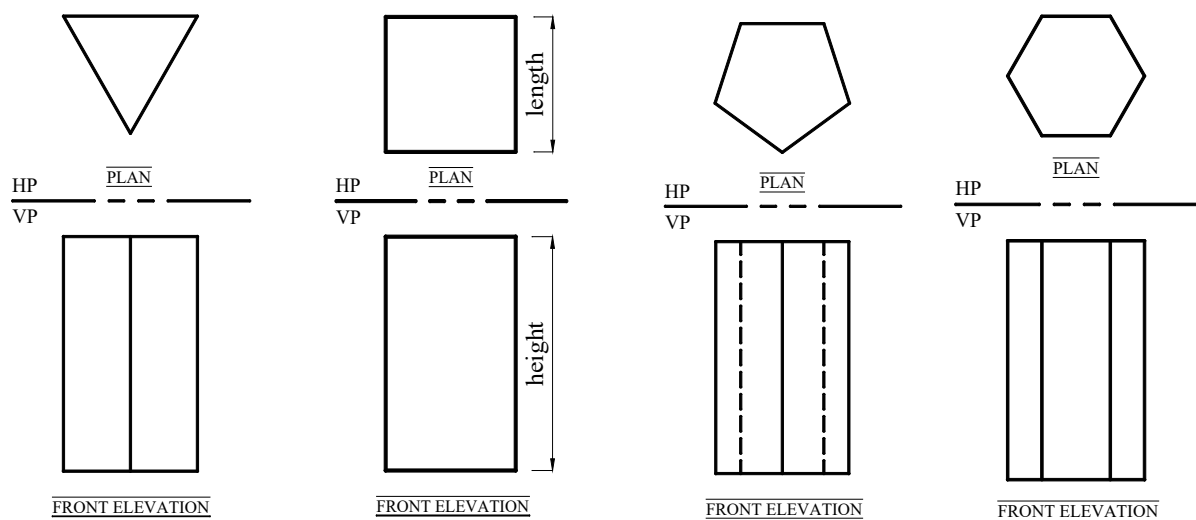




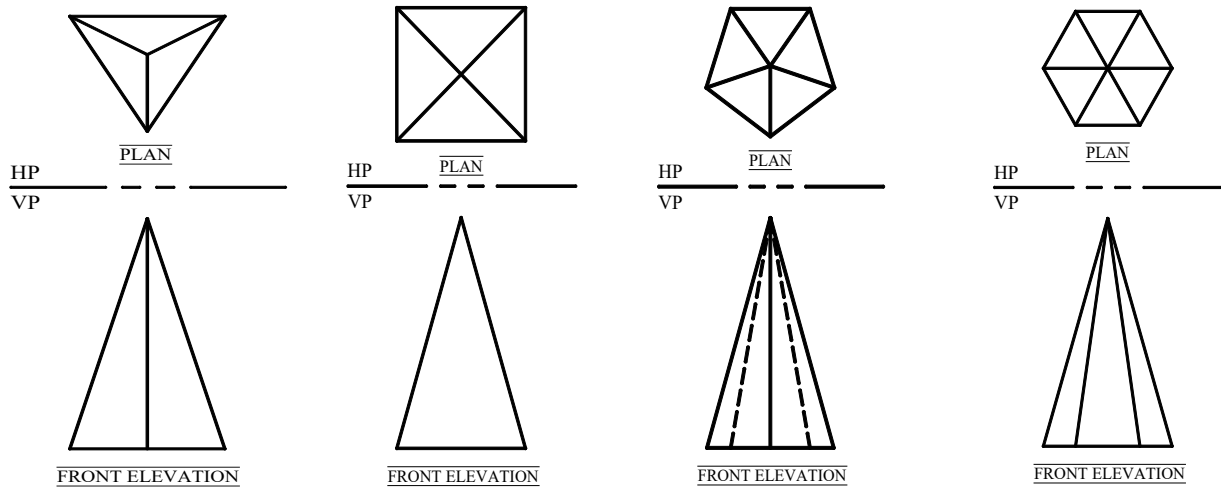
ORTHOGRAPHIC PROJECTION OF GEOMETRICAL SOLIDS

In order to view the details or find the exact dimensions of a 3-dimensional solid, it is necessary to draw the geometrical solids in orthographic projections. The orthographic drawings will show the position of the solids in respect of the location of its base and faces to the horizontal and vertical planes.

Projection of Prisms in 3rd angle

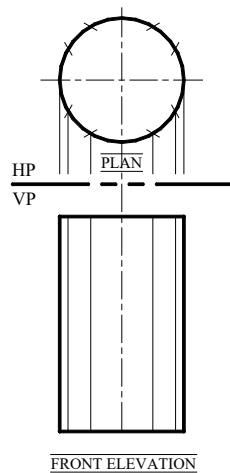


Projection of Pyramids in 3rd angle



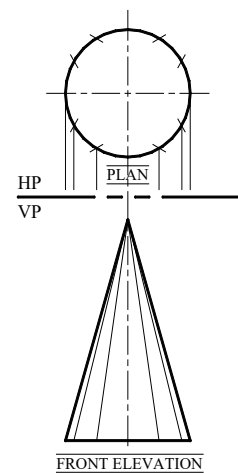
Projection of Cylinder In 3rd angle

The problem of projection of a curved surface is overcome by the use of ordinates or lines that allow for identifying points on the surface of the cylinder. The circular end is divided into twelve equal parts.



Projection of Cone in 3rd angle

The curved surface of a cone is treated much the same way as the cylinder. The base is divided into twelve equal parts and the points are sloping up to the vertex from the circumference.



SURFACE DEVELOPMENTS OF GEOMETRICAL SOLIDS

Surface Development of an object is the drawing of the actual surfaces of an object, in other words, it is the unwrapping of the object to show its full surface. The surfaces are drawn in such a way that when folded or bent along its fold lines will form the required object.

There are many examples of developments in everyday life e.g. the packaging of toothpaste, milk, etc.

It is very important to develop the surface of an object prior to it being made to avoid wastage of material.

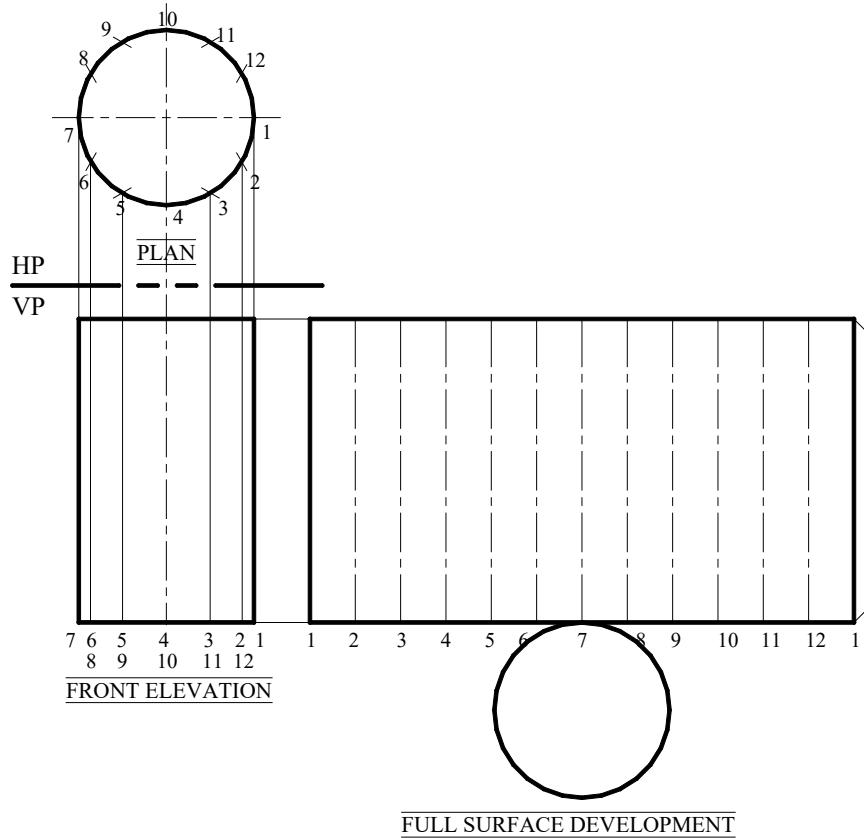


All prisms and the cylinder are developed using the **parallel line development** method as all the sides are parallel to each other. All the pyramids and the cone are developed using the **radial line development** method as all the sides are radiating outwards from a point.

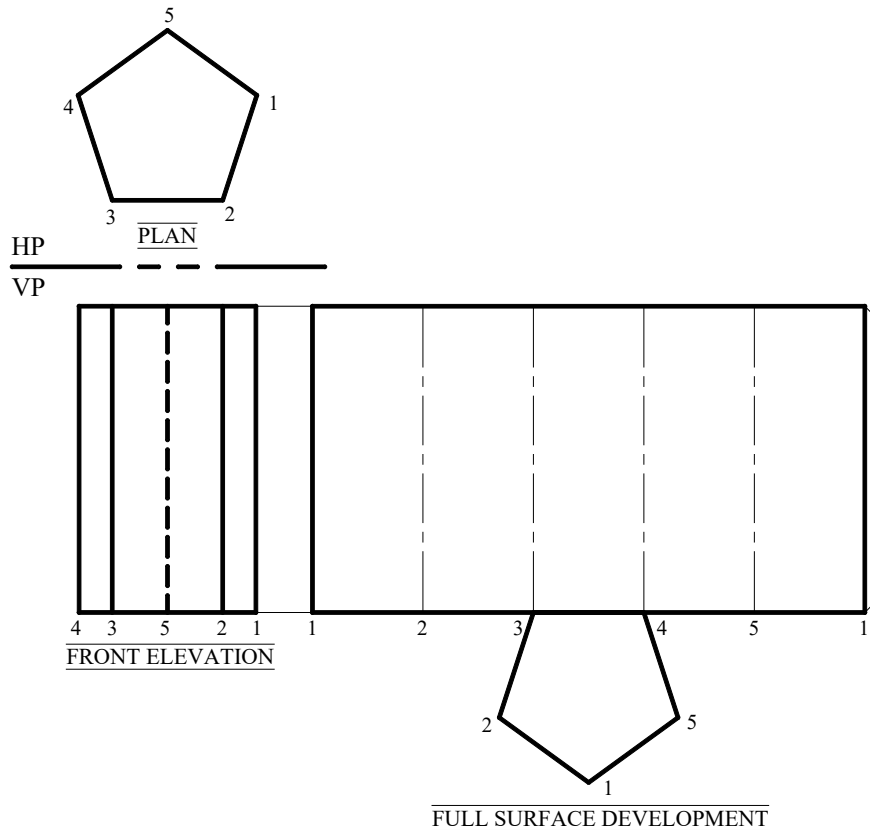
PARALLEL LINE DEVELOPMENT

The surfaces of the object are rolled out as all the sides are parallel to each other. In order to draw the surface development of the object, it is important to first draw its orthographic projection. The surface is developed from the elevation and the dimension for its base size is taken from the plan. The surfaces are always numbered to help in the construction of the development.

Surface development of a cylinder



Surface development of the pentagonal prism



RADIAL LINE DEVELOPMENT

The surfaces of pyramids and cone are developed using the radial line development. In order to draw the surface development of the object, it is important to first draw its orthographic projection. Pyramids and cone have sloping edges radiating from the vertex. The true length of its sloping side is taken as the radius of an arc and drawn from a centre with the base length taken from the plan.

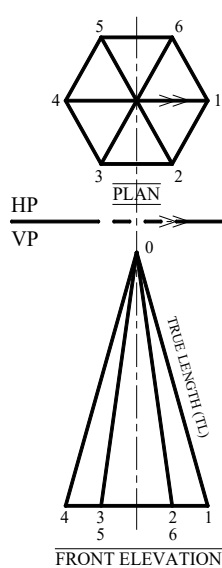
TRUE LENGTH LINES

Due to the rotation and position of the object in space, it is necessary to obtain the true length of sloping edges and generators to draw a radial line development.

The hexagonal pyramid shown on the diagram have two of its generators 01 and 04 seen parallel to the reference in the plan or the HP.

Therefore, the measurements of the distances 01 and 04 are true length lines as seen in the front elevation.

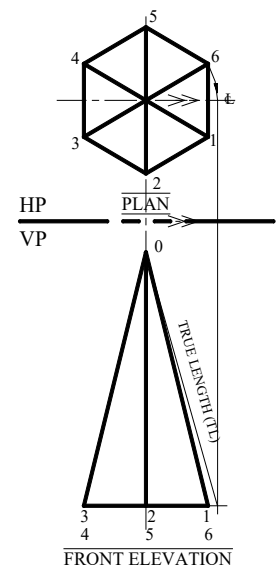
It is this true length line that is used for the full surface development of the hexagonal pyramid.



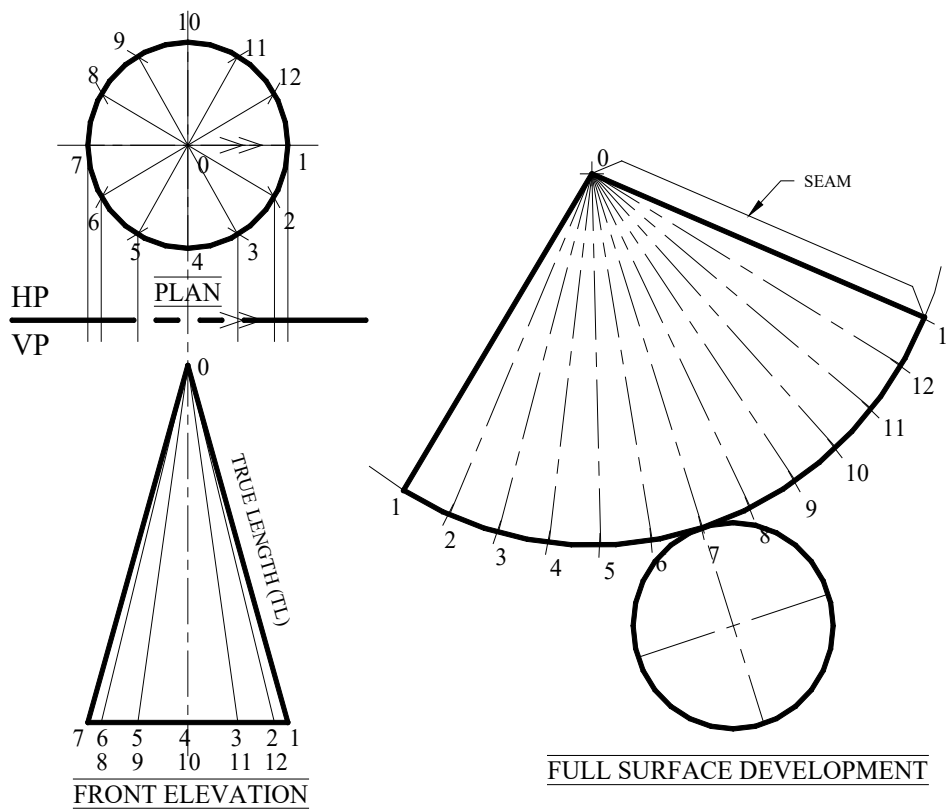
The hexagonal pyramid shown on the diagram does not have any of its generators (01 – 06) seen parallel to the reference line in the plan or the HP.

Therefore, to obtain the true length, the pyramid must be rotated about its centre 0 until any one of its edges are seen parallel to the reference line in the plan or the HP. The distance from the rotated edge is transferred to the front elevation and the true length found.

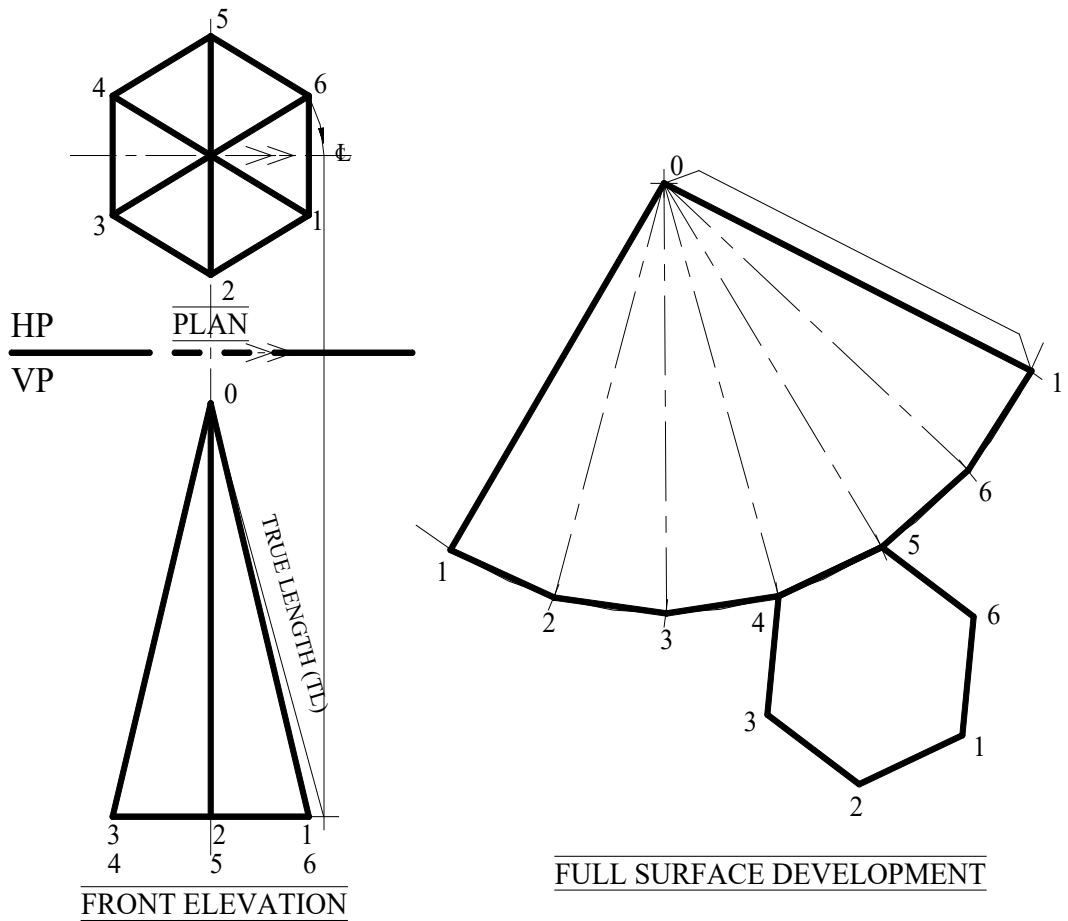
It is this true length line that is used to construct the full surface development.



Surface development of a cone



Surface development of the pentagonal prism



Chapter 6

STRAND: 6 JOINTS AND PROCESSES

Outcome

After studying this chapter, students will:

- a. Identify and state the use of simple woodworking joints
- b. Develop practical skill in construction of simple woodworking joints.
- c. Identify and state the use of simple metalworking joints.
- d. Develop practical skill in construction of simple metalworking joints

Introduction

Joinery is a part of woodworking that involves joining together pieces of wood, to produce more complex items. Some wood joints employ fasteners, bindings, or adhesives, while others use only wood elements. The characteristics of wooden joints in terms of properties are strength, flexibility, toughness, appearance, etc...



WOOD JOINTS

Successful woodwork depends mainly on the correct and accurate jointing of pieces of wood. It is not always realized just how many pieces of timber are often required to make up even a small article. The increasing use of woodworking machines has of course eliminated much of the tedious hand work involved in joint making. However, as the principles involved in both hand and machine made joints are basically the same, knowledge of the construction and uses of the more commonly used woodworking joints is necessary before attempting to make them by hand or machine.

Angle joint

Angle joint is a joint where the sides of the pieces (the wide surfaces) meet at right angles to each other. Angle joints are used for box-like constructions such as small boxes, tool boxes etc. In this chapter we will consider the most common types of angle joints and their construction.



Nailed butt joint

The simplest angle joint is the nailed butt joint. The end of one piece of wood is cut square, and then butted against the face of the other piece. It is held in place with nails, or both nail and glue (Fig. 1).



Fig. 1

Plain Mitred Joint

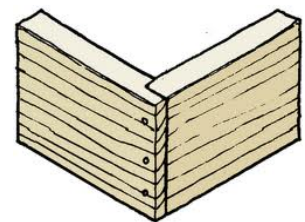
The ends of the pieces are mitred (cut at 45 degrees) across the thickness. The mitred ends are butted together and held in place with glue and nails (Fig. 2). This is a weak type of joint although it is stronger than the butt joint because it is nailed from two sides. Its advantage is that the end grain is not exposed to damage from water or Insects, and it has a neater appearance.



Rebated butt joint

The end of one piece fits into a rebate at the end of the other piece. This joint is strong because two surfaces are available for nailing, and because the shoulder of the rebate supports and helps to hold the other piece (Fig. 3). The lap is the section of wood which is left projecting after the rebate is cut (Fig. 3). The lap is usually one-half of the thickness of the board. This lap will be important later when we are figuring out the length of our pieces for making a box.

The rebated butt joint is simple to construct. In the following sequence of operations we will describe how to make a simple box with this joint while also





practicing some techniques discussed earlier, like making a cutting list and preparing timber to size.

The tools required here will be the same ones we used for the preparation of timber with the additions of a firmer chisel, a smoothing plane, and a backsaw. Make sure you know what these tools are and how to use them before you go on.

CONSTRUCTION OF REBATED BUTT JOINT

Step 1. Preparation of timber

Make a cutting list. The end pieces can be cut to the required length, which is the outside width of the box minus the width of the two laps. Allow 3 mm extra at each end of the side pieces for planning off after assembly.

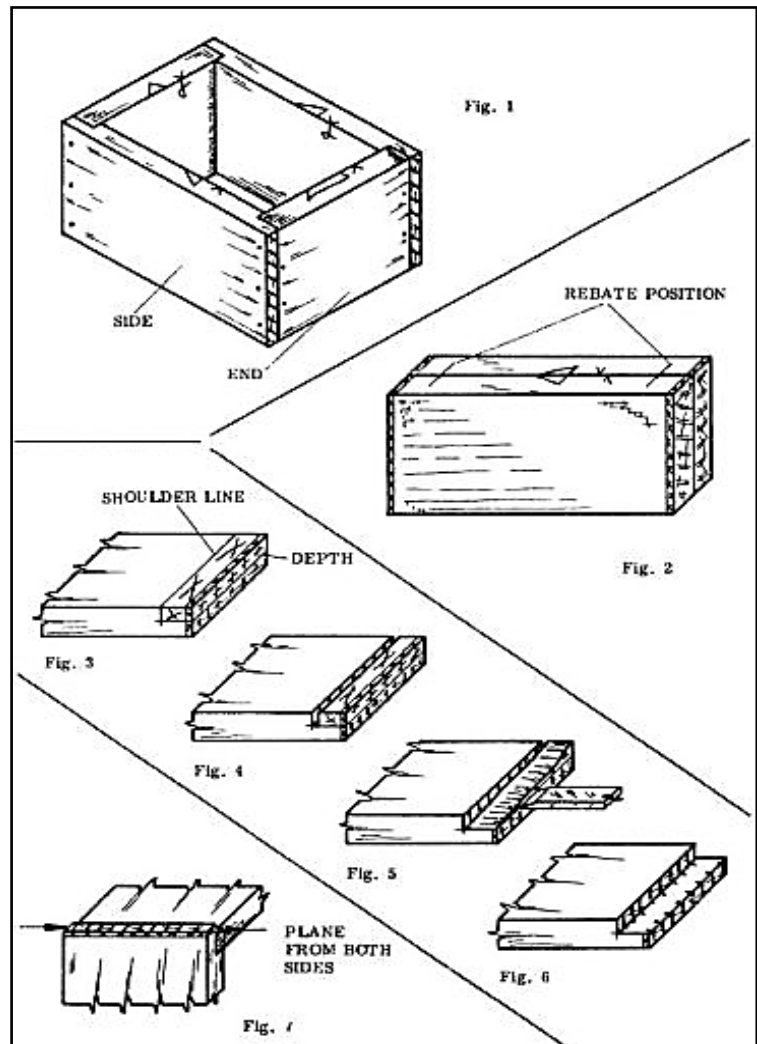
Step 2. Marking out

Mark the sides and ends as shown in Fig. 1, on the face edges (Marking of frames). All further marking will be done from the sides with these marks.

Place the two sides together and mark the position of the rebates, squaring with the try square (Fig. 2).

Mark the shoulder lines of the rebate on the inside face of the piece, using the try square (Fig. 3).

Mark the depth of the rebate on the end grain and the edge, using a marking gauge (Fig. 3). Mark the waste with crosses.



Step 3. Cutting the rebate

Saw the shoulders down to the gauge line. Cut on the waste side of the line. If the piece is very wide, nail or clamp a guide over the line to guide the saw. Use a backsaw (Fig. 4).

Remove the waste carefully to the gauge line with a firmer chisel (Fig. 5). Find out the direction of the grain by chiselling out small pieces first, so that you don't accidentally chisel too deep.

Step 4. Assembling

Clean up the inside faces with a smoothing plane (Fig. 6).

Assemble the box with glue and nails.

Measure the diagonals to check for squareness.

Clean up the face and bottom edges with a smoothing plane.

Plane off the waste from the sides with the smoothing plane. Prevent splintering by working inwards from the ends (Fig. 7).

THE HOUSING JOINT

These joints are another type of angle joint, also used in box-like constructions. Housing consists of sinking the end of one piece into a trench which is cut into the face of another piece (Fig. 1). *The tools required for making this type of joint are the same ones used to make the rebated butt joint.*

CONSTRUCTION OF HOUSING JOINT

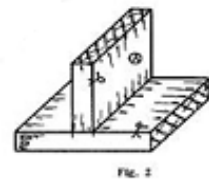
Step 1. Preparation of the timber

Make a cutting list.

Prepare the timber (see Preparation of Timber section).

Step 2. Marking out

Mark one edge of the trench with a try square and the other edge by using piece A as a guide (Fig. 2). (Smoothen piece A before using it to mark the trench).



Gauge the depth of the trench at each edge (Fig. 3).

Mark the waste with small crosses (Fig. 3).

Step 3. Cutting the trench

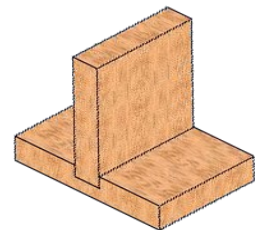
Saw the sides of the trench (on the waste side of the lines) down to the gauge lines (Fig. 4).

Chisel out the waste from the trench (Fig. 5).

Step 4. Assembling

Assemble the two parts with nails and glue.

Clean up the edges with a smoothing plane.



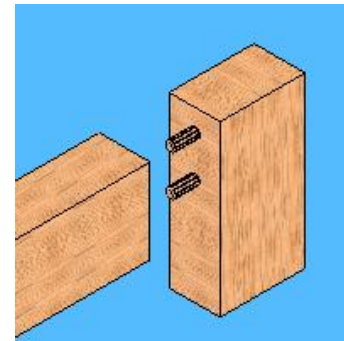
Dowel joint

A dowel joint is a butt joint reinforced with wooden pegs. A well-made dowel joint is as strong as a mortise and tenon joint. It is often used instead of the mortise and tenon joint if cost is a consideration in the assembly of the work. Dowel holes must be a perfect 90 degrees to the face of the work or the dowel will twist when it is assembled. There are different types of dowel joints.



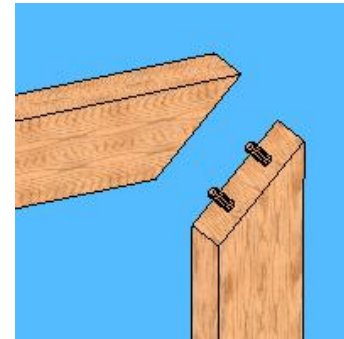
Dowelled butt joint

This style of butt joint is the simplest of all methods of joining two pieces of wood together. However the strength of the joint relies entirely on the glue and any additional fasteners such as screws, pins, nails, tongues, biscuits, corner blocks or in this case, the dowels used to strengthen, reinforce and hold the wood together.



Dowelled mitred butt joint

A mitre joint is a form of decorative butt joint which no end grain is visible. A mitre halves the angle between the parts being joined. It is normally cut at 45° but different angles can be used for non square frames. The joint is usually reinforced with pins, nails, or in this case, dowels to improve its strength.



Application of woodworking joints.

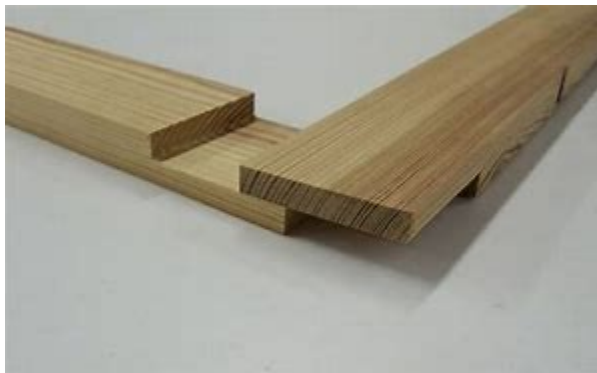
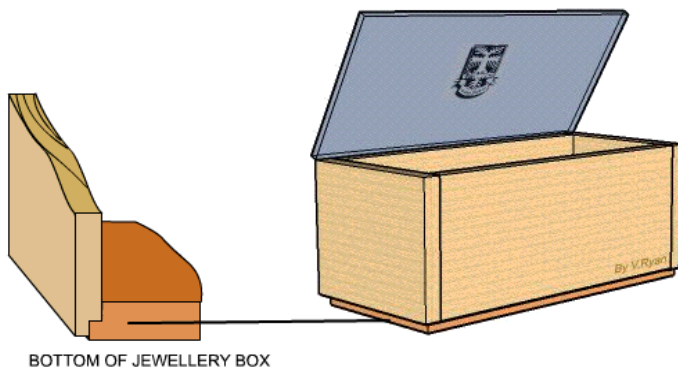
NAIL BUTT JOINT

The simplest and most easy joint to learn or create is the butt joint. This is the first joint you'll probably learn how to make, and it can be used in a variety of different ways. Butt joints are often used during the framing of a house, for example.

To create a butt joint, simply place the end of a piece of wood against another piece and fasten either using a metal clip, a nail or a screw. Butt joints may not be the fanciest or nicest looking joints, but they are extremely stable and can hold up fairly heavy loads.



REBATED BUTT JOINT



DOWELLED JOINT



Application of Timber Joint Projects



1. Sketch the following joints

- a. Mitre joints
- b. Simple butt joints
- c. Rebated butt
- d. Dowel joints



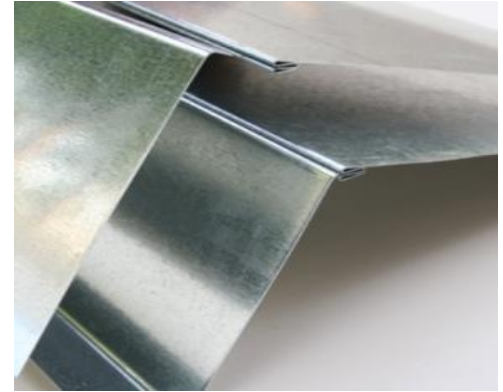
2. Write down the uses of the joint stated above?

METAL WORK JOINT

Sheet metal is simply metal formed into thin and flat pieces. It is one of the fundamental forms used in metalworking and can be cut and bent into a variety of different shapes.

The thickness of the sheet metal is called its gauge. The gauge of sheet metal ranges from 30 gauge to about 8 gauge. The higher the gauge, the thinner the metal is.

There are many different metals that can be made into sheet metal, such as aluminum, brass, copper, steel, tin, nickel and titanium. For decorative uses, important sheet metals include silver, gold and platinum.



Sheet metal

Sheet metal is available in flat pieces or as a coiled strip. Sheet metal has uses in car bodies, airplane wings, medical tables, roofs for buildings and many other things.

Types of joints in metalwork

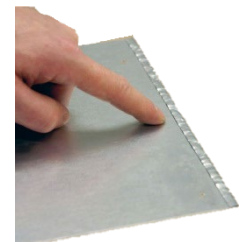
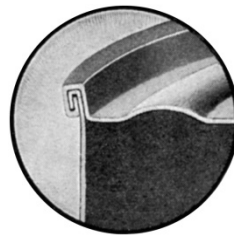
Sheet metal is frequently used in all levels of construction, be it home, public or commercial. The most useful way to permanently join two pieces of metal together is to weld them. However, the use of fasteners, rivets, screws and solders are also very widely used in sheet metal industry.

FABRICATION OF EDGES, JOINTS, SEAMS, AND NOTCHES

There are numerous types of edges, joints, seams, and notches used to join sheet-metal work. We will discuss those that are most often used.

Edges

Edges are formed to enhance the appearance of the work, to strengthen the piece and to eliminate the cutting hazard of the raw edge. The kind of edge that you use on any job will be determined by the purpose, size, and by the strength of the edge needed.



Single-Hem Edge

This edge is shown in. This edge can be made in any width. In general, the heavier the metal, the wider the hem is made. The allowance for the hem is equal to its width (W in fig. 2-54).

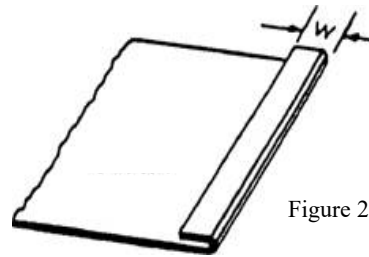


Figure 2-54.-Single-hem edge

Double-Hem Edge (fig. 2-55)

It is used when added strength is needed and when a smooth edge is required inside as well as outside. The allowance for the double-hem edge is twice the width of the hem.

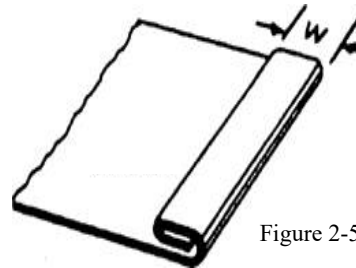


Figure 2-55.Double-hem edge

WIRE EDGE

It is often specified in the plans of objects such as ice-cube trays, funnels, garbage pails, and other articles formed from sheet metal and fabricated with wire edges to strengthen and stiffen the jobs to eliminate sharp edges. The allowance for a wire edge is $2\frac{1}{2}$ times the diameter of the wire used. For example: using a wire that has a diameter of $\frac{1}{8}$ inch. Multiply $\frac{1}{8}$ by $2\frac{1}{2}$ and your answer will be $\frac{5}{16}$ inch which you will allow when laying out sheet metal for making the wire edge.

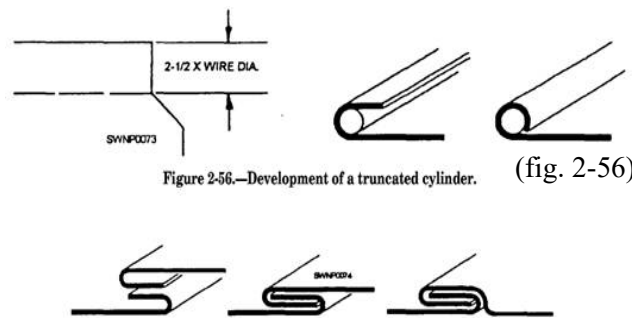


Figure 2-56.—Development of a truncated cylinder.

Grooved Seam Joint

It is one of the most widely used methods for joining light- and medium-gauge sheet metal. It consists of two folded edges that are locked together with a HAND GROOVER (fig. 2-58).

When making a grooved seam on a cylinder, you fit the piece over a stake and lock it with the hand groover (fig. 2-59). The hand groover should be approximately 1/16 inch wider than the seam. Lock the seam by making prick punch indentions about 1/2 inch in from each end of the seam.

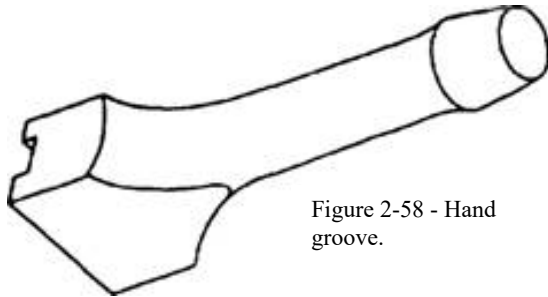


Figure 2-58 - Hand groover.

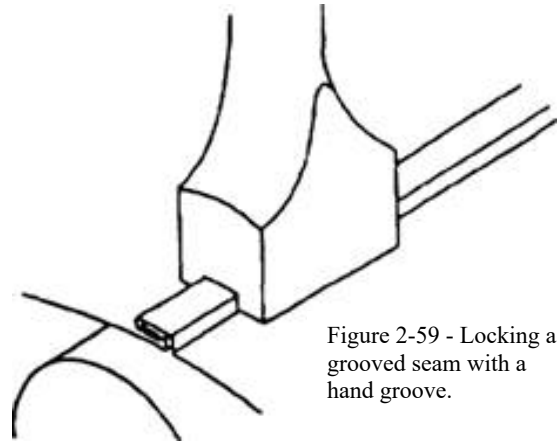


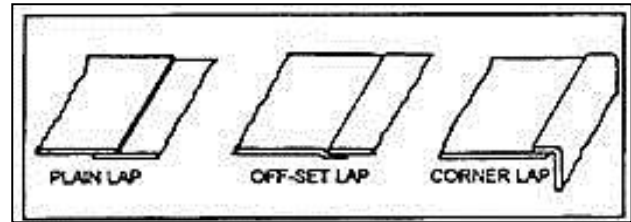
Figure 2-59 - Locking a grooved seam with a hand groover.

SEAMS

Many kinds of seams are used to join sheet-metal sections. Several of the commonly used seams are shown in figure 2-65. When developing the pattern, ensure you add adequate material to the basic dimensions to make the seams. The folds can be made by hand, however, they are made much more easily on a bar folder or brake. The joints can be finished by soldering and/or riveting.

When developing sheet-metal patterns, ensure you add sufficient material to the base dimensions to make the seams. Several types of seams that are used to join sheet-metal sections are discussed in this section. There are three types of lap seams: PLAIN LAP seam, OFFSET LAP seam and CORNER LAP seam.

Lap seams can be joined by drilling and riveting, by soldering or by both riveting and soldering. To figure the allowance for a lap seam, you must first know the diameter of the rivet that you plan to use. The centre of the rivet must be set in from the edge at a distance of $2\frac{1}{2}$ times its diameter.

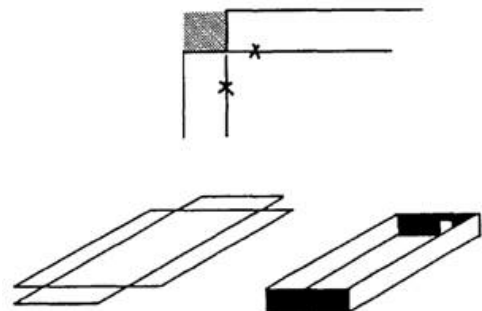


Notches

Notching is the last but not the least important step to be considered when you are getting ready to lay out a job. Before you can mark a notch, you will have to lay out the pattern and add the seams, the laps, or the stiffening edges. If the patterns are not properly notched, you will have trouble when you start forming, assembling and finishing the job.

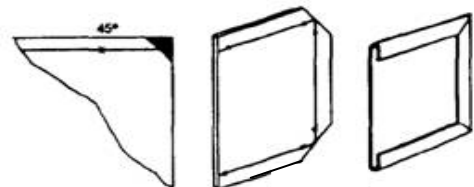
Square Notch

It is the kind you make in your layout of a box or drip pan and is used to eliminate surplus material. This type of notch will result in butt comers. Take a look around the shop to see just how many different kinds of notches you can see in the sheet-metal shapes.



Slant notch (Figure 2-77)

Is a cut at a 45° angle across the corner when a single hem is to meet at a 90 -degree angle. Figure 2-77 shows the steps in forming a slant notch.



V NOTCH

Is used for seaming ends of boxes. You will also use a full V notch when you have to construct a bracket with a toed-in flange or for similar construction. The full V is shown in figure 2-78.

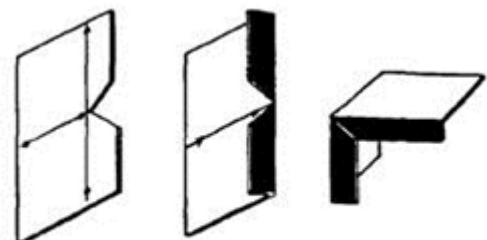
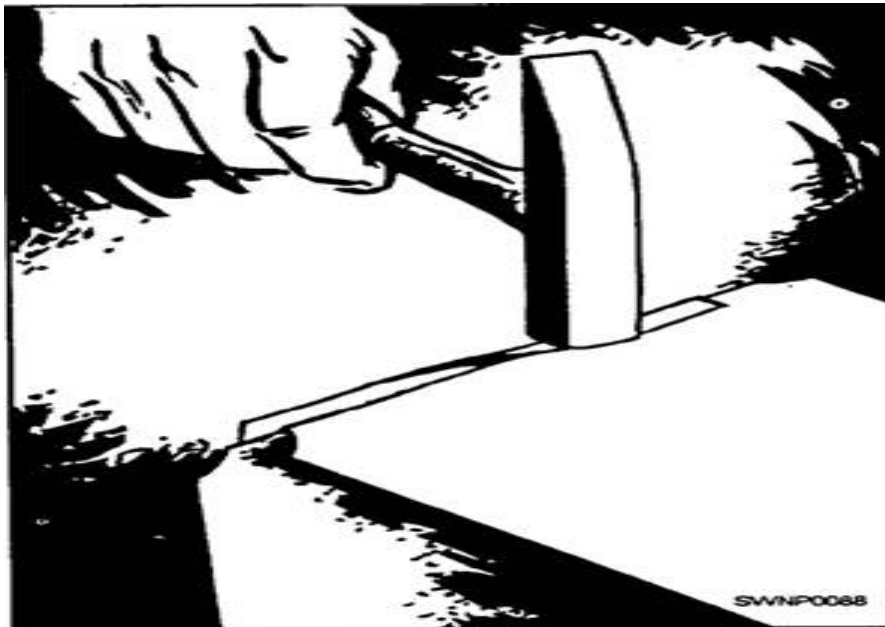
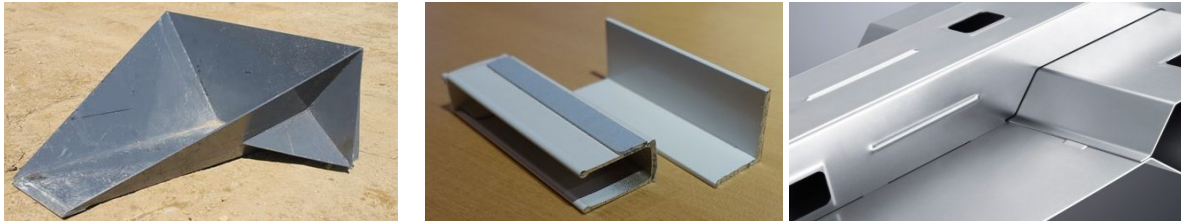
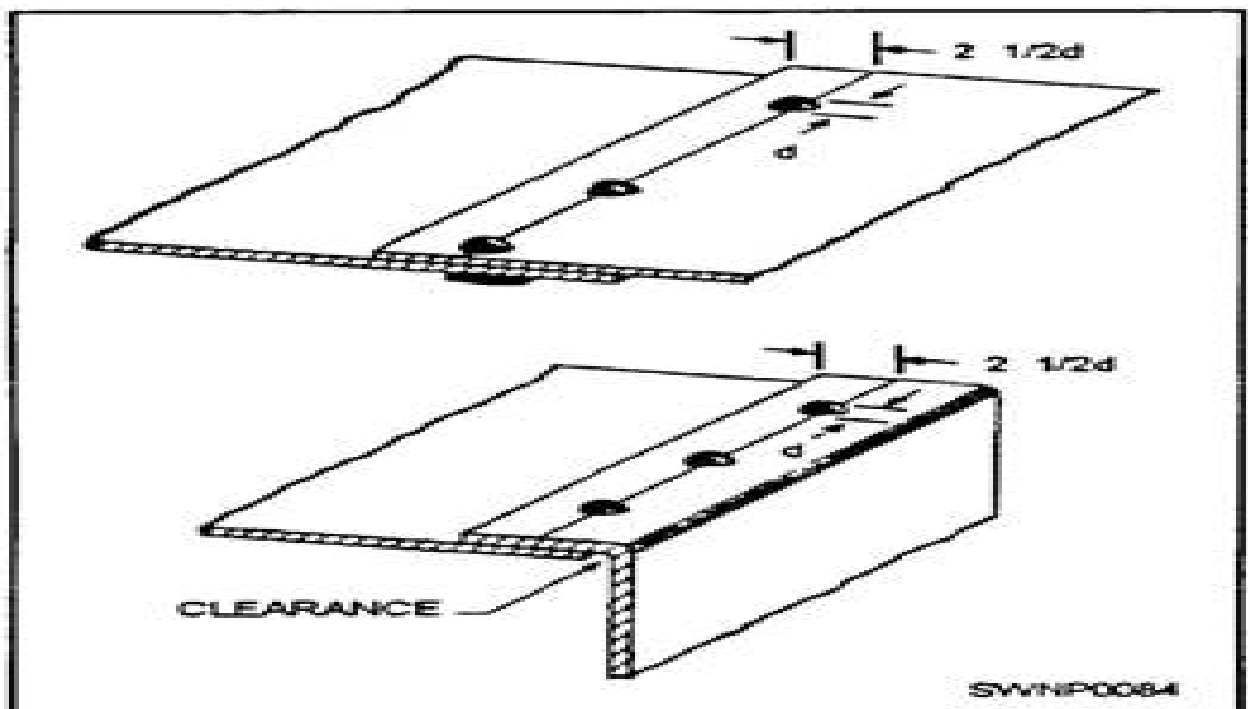
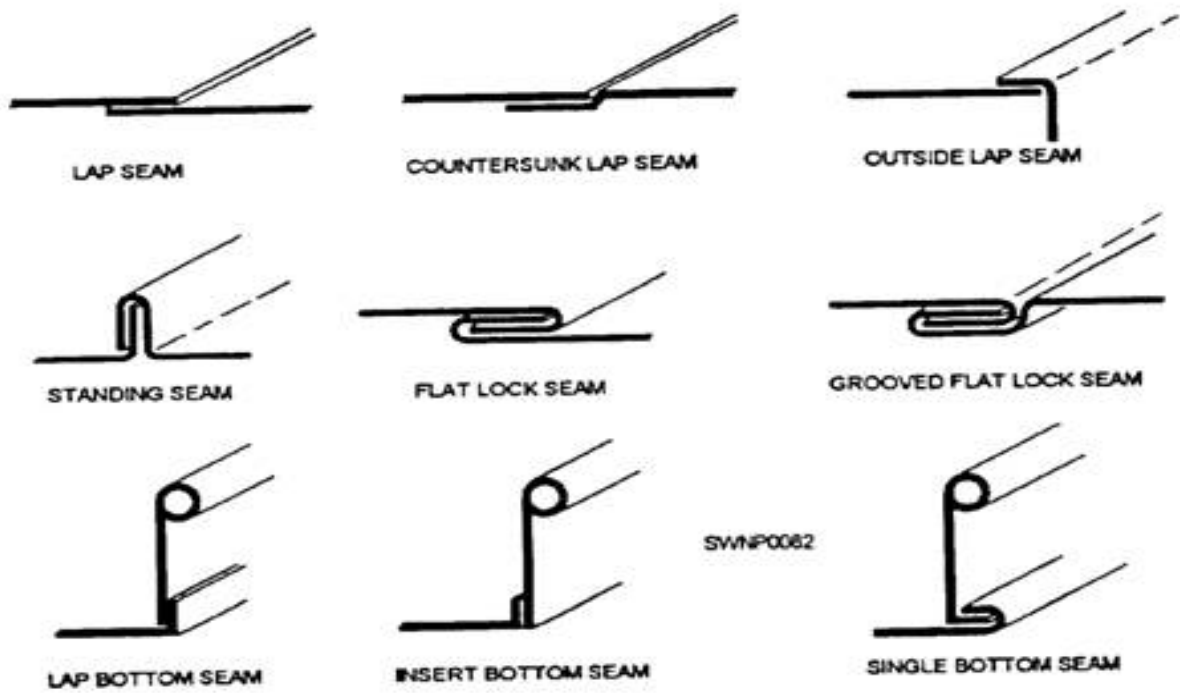


Figure 2-78.—V notch.

APPLICATION OF METAL WORKING JOINT



Sheet metal is metal formed by an industrial process into thin, flat pieces. Sheet metal is one of the fundamental forms used in metalworking and it can be cut and bent into a variety of shapes. Countless everyday objects are fabricated from sheet metal. Thicknesses can vary significantly; extremely thin sheets are considered foil or leaf, and pieces thicker than 6 mm are considered plate. Sheet metal is available in flat pieces or coiled strips. The coils are formed by running a continuous sheet of metal through a roll slitter.



1. Sketch and name six types of seam.
2. What is a sheet metal?
3. Name three types of sheet metals?
4. What is the name given to the thickness of a sheet metal?
5. Sketch any six type of metal working joint.



